

1. Results

1.1 Climate and Well Network Characterization

1.1.1 Climate Record

The RAF Valley climate station provides a continuous monthly record from December 1930 to February 2026 (Figure 4a) (Section 3.1.2). Over the well-record period (April 2005 to February 2026), mean annual precipitation was 890 mm and mean annual Thornthwaite PET was 652 mm, giving a mean P/PET ratio of 1.36 (Table 1). The wettest year on record within this period was 2008 (1,040 mm), followed by 2012 (1,022 mm); the driest complete year was 2010 (736 mm). Four years recorded a P/PET ratio below 1.2 — indicating near-balance between precipitation and evaporative demand — namely 2007, 2010, 2011 and 2022, with 2022 recording the lowest ratio of the monitoring period at 1.12. There is no meaningful difference in mean annual precipitation between the pre-felling (2006–2017: 887 mm) and post-felling (2018–2025: 893 mm) periods, confirming that the post-2018 hydrological signal detected in the BACI analysis (Section 4.6) is not driven by a concurrent change in rainfall amount.

PET shows strong seasonality but negligible inter-annual trend across the record, with the 12-month rolling mean remaining stable at approximately 50–55 mm/month throughout (Figure 4b). Consequently, inter-annual variability in the water balance is driven almost entirely by precipitation rather than evaporative demand.

The levelled cumulative water balance series (Figure 4c), expressed relative to the study-period mean of 19.46 mm/month, oscillates with a clear seasonal signal — rising sharply each winter as precipitation exceeds PET and falling each summer — while tracking the multi-year rhythm of wetter and drier phases. No sustained multi-year drift away from the baseline is evident, confirming that the site maintains a consistent long-run surplus throughout the monitoring period. The water table clearly follows this movement (Figure 4d). Linear regression of the cumulative balance against the unsmoothed network mean water level at zero lag yielded $R^2 = 0.514$, confirming that approximately half the variance in network mean water table depth is explained by antecedent climate forcing alone. The residual variance reflects the spatial heterogeneity of the network, local boundary subsidies, and land cover effects explored in subsequent sections. This close correspondence between cumulative climate forcing and water table behaviour provided the empirical motivation for the state-space modelling framework described in Section 3.4.

The seasonal structure of the climate record has direct implications for aquifer behaviour. Mean monthly rainfall peaks in October–December (around 101–109 mm/month) and falls to its minimum in April (around 46 mm/month; Figure 4a), whilst Thornthwaite (1957) PET peaks in July (around 107 mm/month) and is negligible from November to February (Figure

4b). The seasonal opposition between peak PET and minimum rainfall concentrates atmospheric evaporative demand on the aquifer during the period of lowest recharge input, driving summer water table drawdown that must be overcome by the following winter's recharge before any slack flooding can occur. This seasonal asymmetry — not the annual rainfall total — is the primary control on dune slack hydrology and is reflected throughout the subsequent analyses. Over the monitoring period, mean annual winter rainfall (October–March) was 516 mm and mean annual summer rainfall (April–September) was 373 mm, together

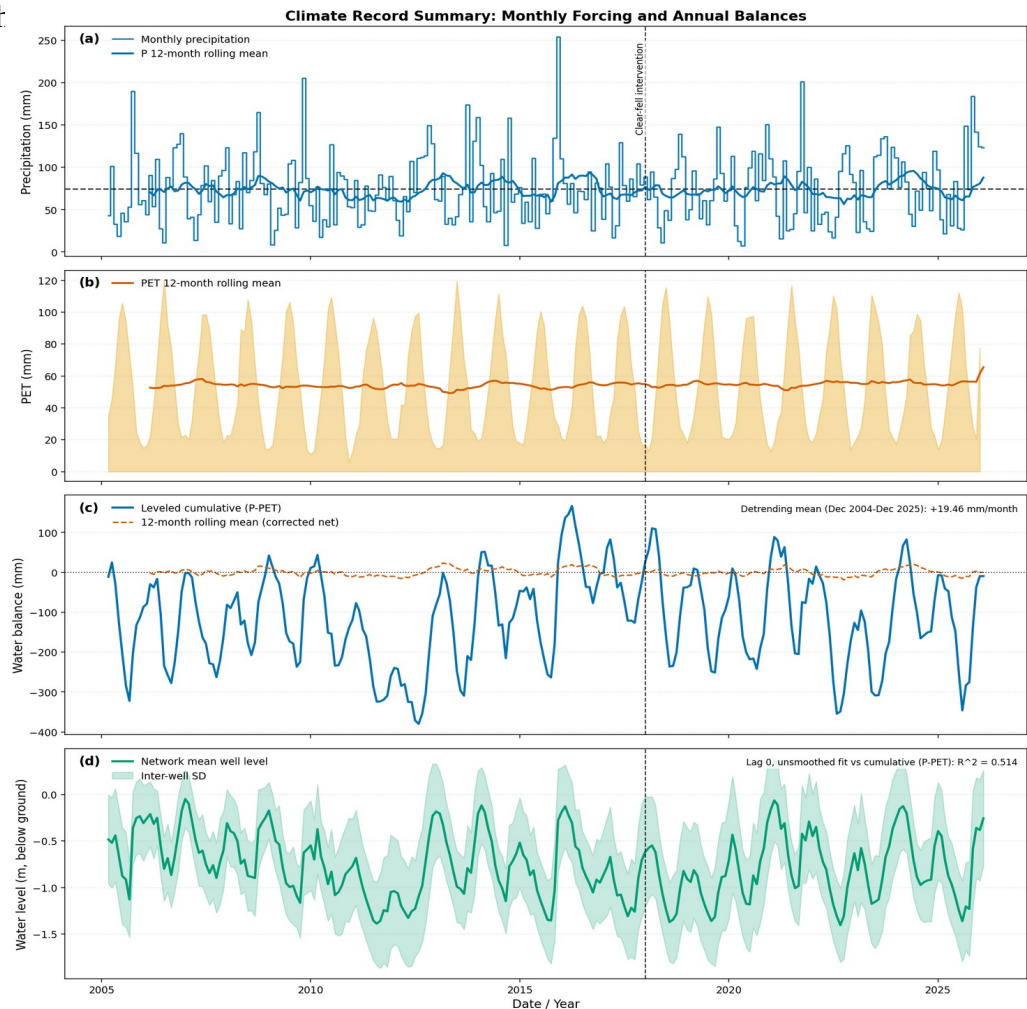


Figure 1.1: Climate record summary for the well-monitoring period (April 2005 – February 2026), derived from RAF Valley Met Office station. Panel (a): monthly precipitation (mm) with long-term monthly mean indicated by dashed line and 12-month rolling mean as solid line. Panel (b): monthly Thornthwaite PET (mm) as filled area with 12-month rolling mean overlaid. Panel (c): levelled cumulative water balance ($P - PET$, mm), corrected by subtracting the long-term mean monthly net balance to centre the series on zero; the dashed orange line shows the 12-month rolling mean of the corrected net balance. The intercept correction value is shown inset. Panel (d): network mean monthly water level (m below pipe top) across the 66-well reference network, with inter-well standard deviation as shaded envelope; the R^2 inset gives the coefficient of determination from linear regression of the cumulative balance against the unsmoothed network mean at zero. (Source: 00_01_climate_timeseries_short.png)

Year	Annual P mm	Annual PET mm	Months complete	P PET ratio
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2005	693.1	611.9	10	—
2006	954.0	655.4	12	1.46
2007	773.7	653.7	12	1.18
2008	1039.7	648.7	12	1.60
2009	871.3	647.0	12	1.35
2010	736.2	628.8	12	1.17
2011	760.6	647.5	12	1.17
2012	1021.7	637.0	12	1.60
2013	844.0	633.6	12	1.33
2014	943.0	662.5	12	1.42
2015	974.0	631.2	12	1.54
2016	858.4	652.0	12	1.32
2017	871.0	659.4	12	1.32
2018	873.2	653.0	12	1.34
2019	832.0	654.4	12	1.27
2020	944.2	656.6	12	1.44
2021	906.6	649.0	12	1.40
2022	747.2	668.2	12	1.12
2023	986.2	672.9	12	1.47
2024	911.2	657.8	12	1.39
2025	943.0	675.5	12	1.40
2026	247.2	77.4	1	—
Long-term mean	889.6	652.2		1.36

Table 1.1: Annual climate summary for RAF Valley station over the well-monitoring period. Annual precipitation (P) and Thornthwaite PET are in mm. P/PET ratio below 1.0 indicates evaporative demand exceeds precipitation. Months_complete gives the number of months with valid data in that year. Partial years (2005: April–December only; 2026: January–February only) are included for completeness but no annual P/PET ratio is calculated for these years (shown as —). Long-term mean row is calculated from complete years 2006–2025 only. (Source: 00_01_annual_climate_summary_short.csv).

The RAF Valley record also shows a statistically significant warming trend in mean summer maximum temperature over the full 95-year record (slope $+0.014^{\circ}\text{C yr}^{-1}$, $R^2 = 0.142$, $p < 0.001$; Figure 5). The post-2013 period has run 0.94°C above the pre-2013 baseline mean of 17.89°C , with six of the thirteen summers between 2013 and 2025 exceeding this baseline by more than 1°C (2013, 2018, 2021, 2022, 2023 and 2025). The most extreme anomalies occurred in 2022 ($+1.77^{\circ}\text{C}$) and 2023 ($+1.64^{\circ}\text{C}$). The year 2022 combined a $+1.77^{\circ}\text{C}$ temperature anomaly with only 747 mm annual rainfall — the most severe dual-stress year in the monitoring period. Annual rainfall shows no significant trend over the full record ($p = 0.498$).

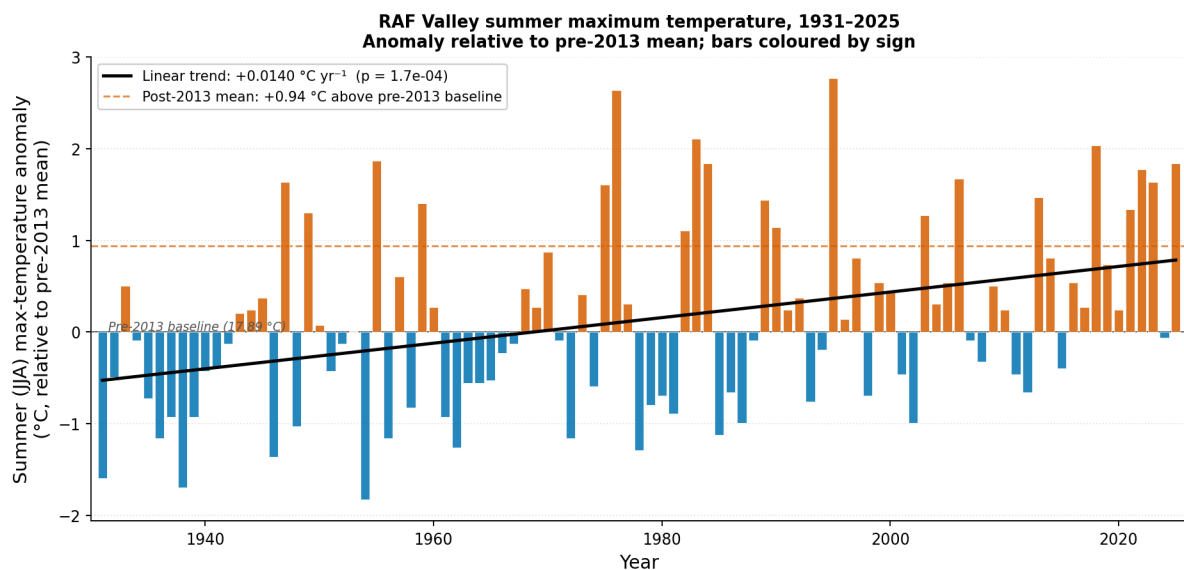


Figure 1.2: Summer maximum temperature anomaly at RAF Valley, 1931–2025, expressed as departure from the pre-2013 mean (17.89°C). Bars coloured by sign: orange = above baseline, blue = below. Solid black line: OLS linear trend ($+0.014^{\circ}\text{C yr}^{-1}$, $p < 0.001$). Dashed orange line: post-2013 mean anomaly ($+0.94^{\circ}\text{C}$ above baseline). (Source: 00_03_summer_warming_trend.png)

4.1.2 Well Network

The 66-well reference network (Section 3.1.1) spans a median record length of 191 months (15.9 years), with individual cleaned series ranging from 139 to 250 months (Figure 6, upper left); all wells satisfy the 100-month minimum-observation threshold for admission to the reference network. The network mean water level across all wells and months was 0.74 m below pipe top, with individual well means ranging from 0.26 m above pipe top to 2.07 m below, reflecting the considerable topographic and hydrogeological diversity of the site (Figure 6, upper right). The median seasonal amplitude — defined as the difference between mean August and mean February water levels — was 0.69 m, with the water table deepest in late summer and shallowest in late winter, consistent with the expected seasonal recharge cycle for a maritime dune system (Figure 6, lower right). The network mean time series (Figure 6, lower left) shows a clear seasonal cycle throughout the record, with the inter-well standard deviation envelope widening visibly after the December 2017 clearfell intervention,

reflecting the divergence of the impact zone wells from the broader network that is explored in detail in Section 4.6.

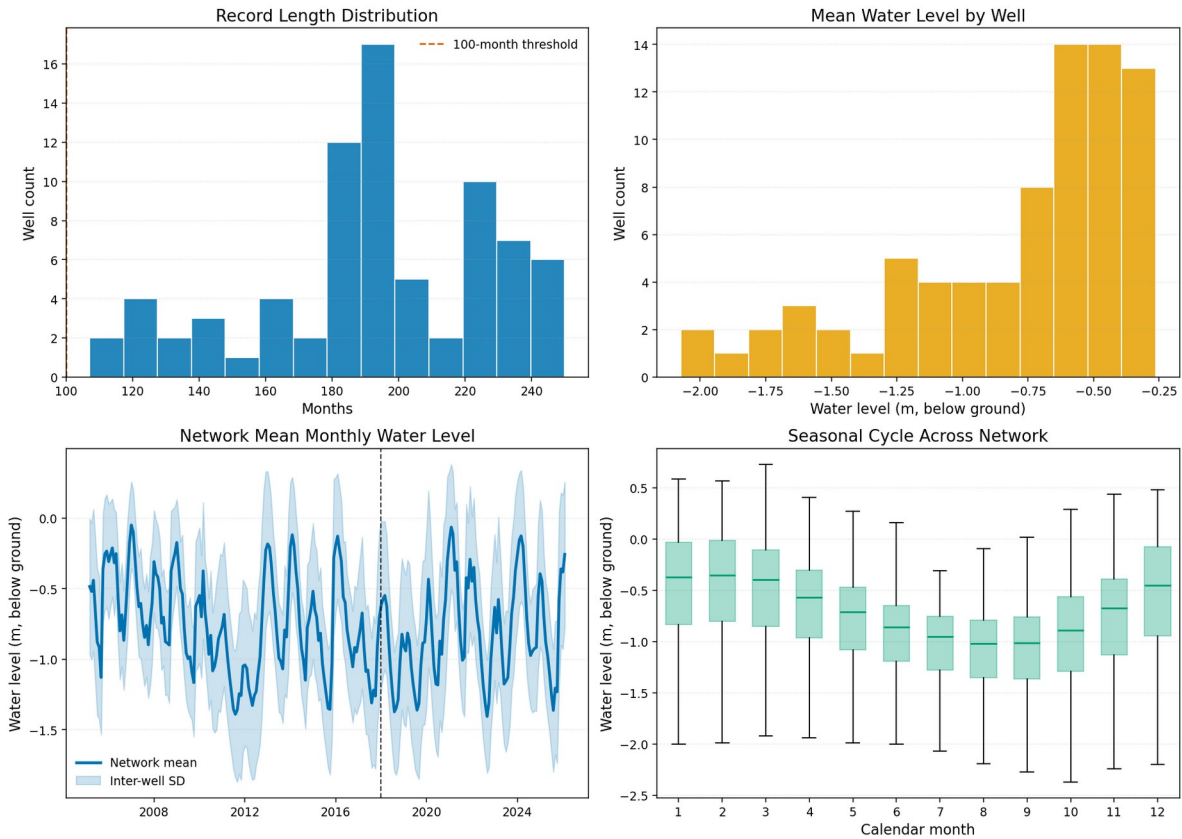


Figure 1.3: Well network characterization for the 66-well reference network. Upper left: frequency distribution of record lengths (months), with the 100-month minimum threshold indicated by dashed line. Upper right: frequency distribution of mean water level by well (m below pipe top; negative values indicate water standing above pipe top). Lower left: network mean monthly water level (m below pipe top) with inter-well standard deviation as shaded envelope; vertical dashed line marks the December 2017 clearfell intervention. Lower right: network-wide seasonal cycle shown as monthly box plots pooling all well-month observations; boxes show interquartile range, horizontal line shows median, whiskers extend to $1.5 \times \text{IQR}$, outliers suppressed. (Source: 00_02_well_network_summary_short.png)

Mapped against the eco-hydrological thresholds of Curreli et al. (2013), the network-wide seasonal extremes reveal a system already under considerable stress. The mean annual summer minimum across the reference network was -1.18 m below pipe top, ranging from -0.71 m to -2.32 m across individual wells. No well maintains a mean summer minimum shallower than the wet slack viability threshold of -0.61 m; 28 wells fall within the dry slack zone (-0.61 to -0.98 m); and 38 wells record mean summer minima deeper than the dry slack threshold of -0.98 m, of which 20 lie below the SD16 recovery limit of -1.20 m — placing them beyond the reach of a single scraping event under average conditions. For winter maxima, 25 wells exceed the SD15b wet slack flooding threshold of -0.10 m in an average year and 38 exceed the SD16 dry slack threshold of -0.25 m, though only 11 record

mean winter maxima above pipe top. These network-level baselines provide the reference against which cluster-specific dynamics, intervention effects, and climate trajectory are assessed in subsequent sections.

1.2 Network Clustering and Spatial Architecture

The clustering analysis (Section 3.2) identified $k = 5$ as the preferred cluster solution (silhouette coefficient = 0.40; bootstrap stability analysis confirming four of five clusters with median per-well co-assignment ≥ 0.93). The five clusters are spatially coherent and hydrogeologically interpretable (Table 3; Figures 7–9); their mechanistic characterization including LCSC values is presented in Section 4.2.2.

The dendrogram confirms a primary binary division separating eastern wells (C1, C2) from western and forest wells (C3, C4, C5) at a Ward linkage distance of approximately 1.0. This top-level split reflects the division between the till-and-estuary-influenced eastern dune aquifer and the deeper, more uniform western sand block. The forest clusters (C4 Main Forest, C5 Coastal Forest) separate from adjacent C3 wells despite geographical proximity, confirming that plantation land cover modifies groundwater behaviour independently of the broader dune aquifer geology. The separation also reflects the position of the C4 wells in the northern part of the study area, closest to the rock ridge, at the highest topographic positions in the dune system. C5 (Coastal Forest) occupies the coastal edge of the plantation where canopy and coastal influences combine.

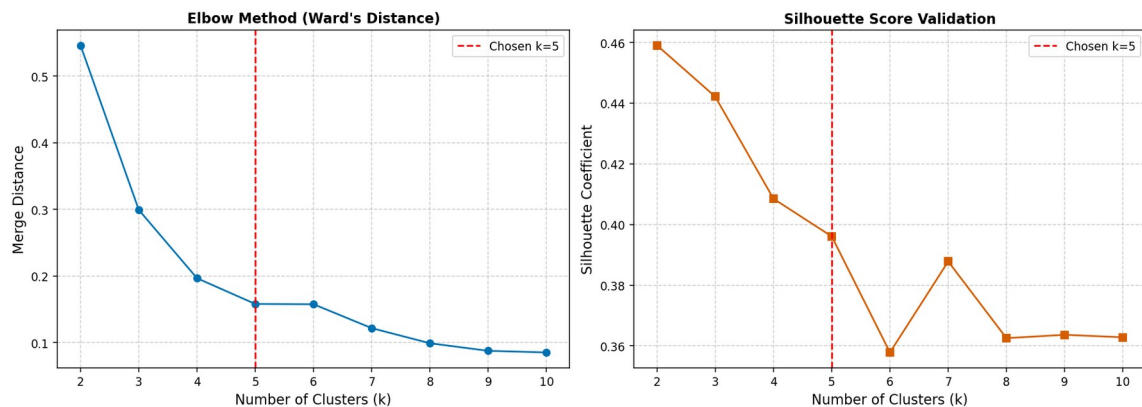


Figure 1.4: Cluster validation plots: silhouette coefficient, Calinski–Harabasz index, and Ward's merge distance for $k = 2$ – 10 , with chosen $k = 5$ indicated. The silhouette coefficient peaks at $k = 2$ reflecting the fundamental binary contrast between eastern and western aquifer domains; $k = 5$ was selected as the finest partition at which bootstrap stability analysis confirms robust, reproducible cluster assignments. (Source: 02_02_validation_plots.png).

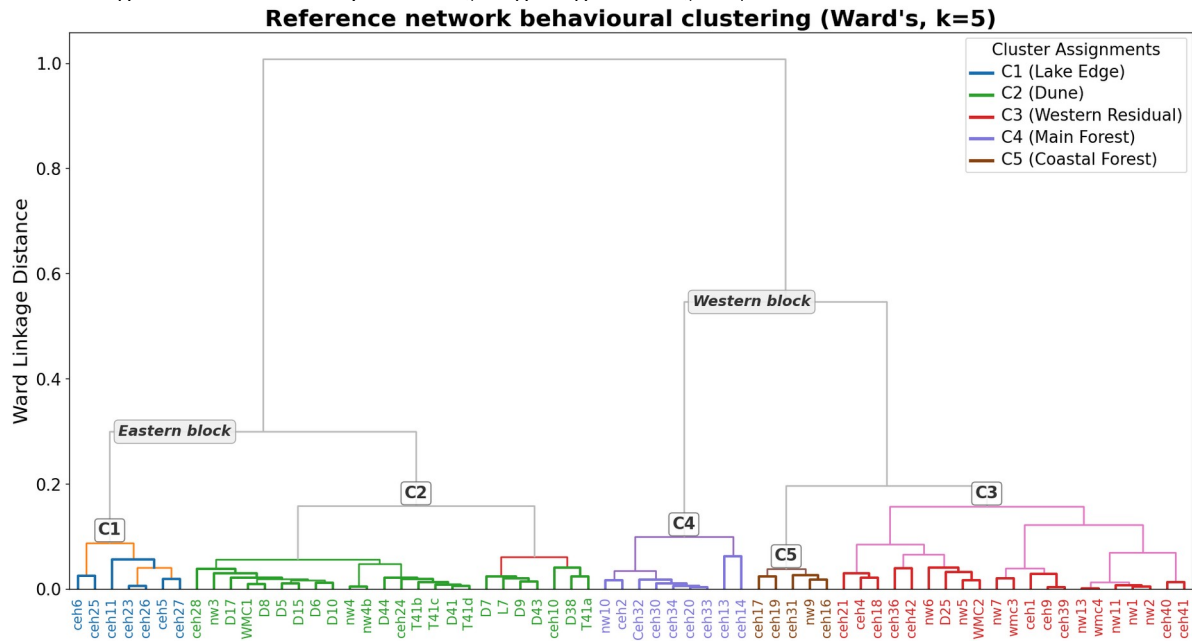


Figure 1.5: Clustering results: Ward's hierarchical clustering dendrogram for the 66-well reference network. Wells are labelled by identifier and coloured by assigned cluster. The primary binary split at a Ward linkage distance of approximately 1.0 separates the eastern till-influenced clusters (C1, C2) from the western sand and forest clusters (C3, C4, C5). The Forest cluster (C4) separates from the adjacent Western Residual (C3) at a lower linkage distance, confirming that plantation land cover imposes a distinct hydrogeological signature independent of the shared sandy substrate. (Source: 02_01_dendrogram.png).

1.2.1 Cluster Hydrographs and Seasonal Amplitudes

The five cluster average hydrographs (Figure 9b) reveal a clear stratification of groundwater behaviour that reflects the hydrogeological architecture identified by the clustering analysis. C1 (Lake Edge) shows the shallowest mean depth (−0.45 m), consistent with its low storage capacity and hydraulic connectivity to Llyn Rhos-Ddu. C2 (Dune) tracks C1 closely but sits systematically deeper (mean −0.60 m), reflecting greater distance from the lake boundary. The fundamental geological contrast between the Eastern and Western Blocks is visible as a persistent vertical offset between C1/C2 and C3 (Western Residual), with C3 running approximately 0.3 m deeper on average (mean −0.89 m) despite occupying a broadly similar topographic position — a contrast that reflects the deeper sand accumulation and higher storage capacity of the Western Residual aquifer. C4 (Main Forest) is the most distinctive cluster, sitting 0.6 m deeper than adjacent C3 wells throughout the record (mean −1.47 m). This persistent depression relative to the surrounding dune aquifer is consistent with the canopy interception interpretation: the Corsican pine plantation intercepts a substantial fraction of gross precipitation before it reaches the water table, suppressing recharge and depressing the water table below that of the adjacent unforested dune. The C4 cluster mean hydrograph should be interpreted cautiously prior to 2010, when only three of the nine cluster wells were operational (CEH2, CEH13 and NW10); these wells are disproportionately located at higher elevations on the ridge margin so the apparent high mean heads in 2006–2008 reflect an unrepresentative spatial sample rather than a genuine network-wide signal. Full nine-well coverage was not achieved until mid-2010. C5 (Coastal

Forest) sits at comparable depth to C4 (mean -1.25 m) but with a damped seasonal signal, consistent with its position at the coastal edge of the plantation where both canopy interception and proximity to the coastal discharge boundary attenuate the seasonal response. Per-well seasonal amplitudes and their temporal evolution are analysed below.

The cumulative water balance panel (Figure 9a) confirms that the cluster separation is not an artefact of differential climate forcing — the 12-month rolling mean of the corrected net balance remains close to zero throughout the record, indicating that all clusters are responding to the same climatic regime. The structural differences between clusters therefore reflect hydrogeological and land cover controls rather than spatially variable precipitation or evapotranspiration.

The network mean water table elevation was 7.60 m AOD across the 66-well reference network, ranging from 3.12 m AOD (CEH11, C1 Lake Edge) to 13.30 m AOD at the ridge flank (CEH14, C4 Main Forest). CEH11's assignment to C1 is discussed in Section 3.2.3: its low elevation and coastal position place it near the Menai Strait tidal boundary rather than Llyn Rhos-Ddu, and it represents the low-elevation extreme of the reference network rather than a typical lake-edge well. Excluding CEH11, the C1 cluster-mean elevation rises from 8.57 m AOD to 9.44 m AOD, and the network minimum shifts to CEH42 (C2, 4.47 m AOD). The Main Forest cluster (C4) occupies the highest mean water table elevation in the network at 9.52 m AOD, compared with 8.57 m AOD in the Lake Edge cluster (C1), 7.57 m AOD in the Dune cluster (C2), and 7.18 m AOD in the Western Residual (C3) — a contrast that reflects the topographic position of the plantation on the northern ridge flank rather than any difference in recharge behaviour. The Coastal Forest cluster (C5) sits at 4.41 m AOD, the lowest cluster-mean elevation in the network, consistent with its coastal-margin position.

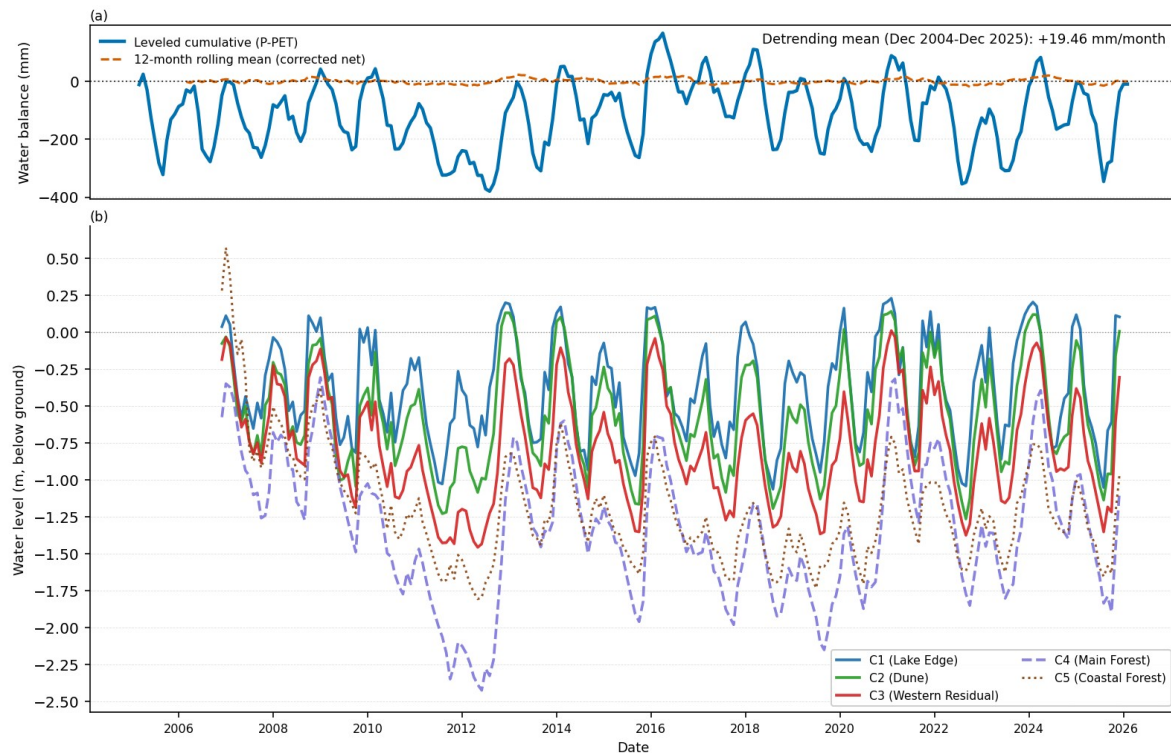


Figure 1.6: Cluster-mean relative water level hydrographs for the five hydrogeological clusters, April 2005 to February 2026. (a) Levelled cumulative water balance ($P - PET$), detrended at the study-period mean of $+19.46$ mm/month (December 2004 to December 2025); the 12-month rolling mean of the corrected net balance (orange dashed) confirms no long-run directional trend in climate forcing over the monitoring period. (b) Monthly cluster-mean water level relative to pipe-top datum (m), where zero approximates ground surface, computed from all wells assigned to each cluster in the 66-well reference network. Cluster labels and colours follow those used in Figure 8. (Source: 02_03_cluster_hydrographs_wb.png)

Separating the reference wells into one panel per cluster (Figure 10) shows the within-cluster spread that the cluster means in Figure 9 necessarily collapse. On the shared vertical axis the between-cluster stratification is evident: the two forested clusters (C4 Main Forest, C5 Coastal Forest) sit at consistently lower relative water levels than the lake-edge and dune clusters, and all clusters share the same seasonal winter-recharge and summer-recession cycle. C1 (Lake Edge) is the most internally coherent, while C2 (Dune) contains a small number of wells sitting persistently below the cluster mean, consistent with its greater within-cluster heterogeneity; the individual traces nonetheless broadly track their cluster's seasonal cycle throughout the record.

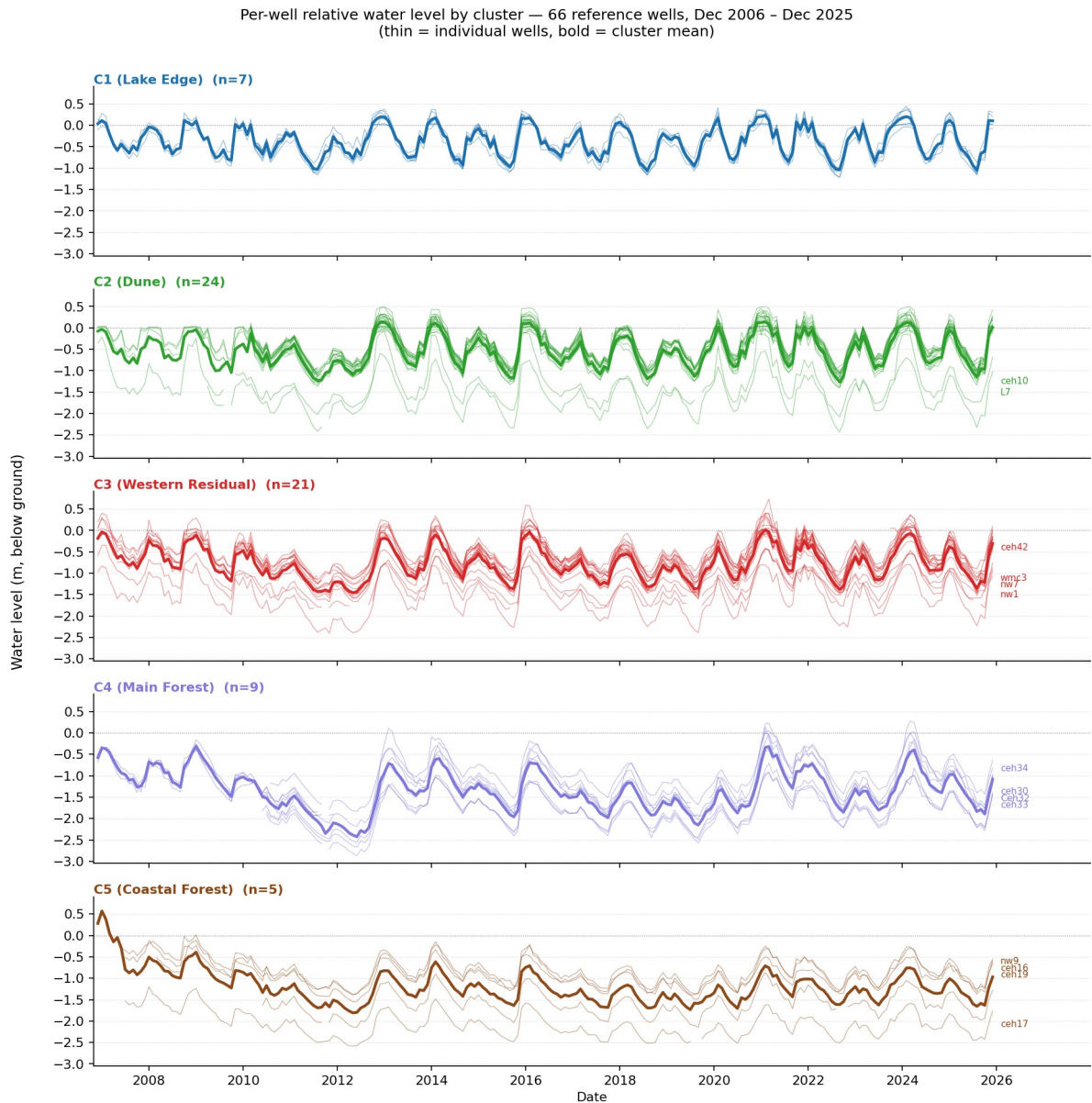


Figure 1.7: Per-well relative water level for the 66 reference wells, shown as one stacked panel per hydrogeological cluster, December 2006 to December 2025. Within each panel the thin lines are individual wells and the bold line is the cluster mean. All panels share a common vertical axis: monthly water level relative to pipe-top datum (m), where zero approximates ground surface, as in Figure 9b. Cluster labels and colours follow Figure 9; well counts per cluster (n) are given in each panel title. (Source: 02_03b_cluster_spaghetti.png)

Per-Well Seasonal Amplitude. Per-well inter-decile amplitudes (p90 – p10) computed across three windows (full record, pre-2018, post-2018) confirm and refine the cluster-mean amplitude readings (Section 4.2.1; Figure 9). Cluster medians of per-well amplitudes are summarized in Table 2.

Cluster	n	Pre-2018 (m)	Post-2018 (m)	Δ pre → post (%)	Climate-normalized Δ (%)
C1 Lake Edge	7	0.83	1.00	+20%	+16%
C2 Dune	24	1.00	1.02	+2%	–7%
C3 Western Residual	21	1.04	1.06	+1%	–7%
C4 Main Forest	9	1.16	1.09	–6%	–6%
C5 Coastal Forest	5	0.89	0.73	–18%	–21%

Table 1.2: Negative values indicate amplitude reduction (damping); positive values indicate amplitude amplification. Climate-normalized values exclude empirically-identified drought summers (2005, 2018, 2022) where Jun–Sep rainfall fell $\geq 1\sigma$ below the 1931–2017 baseline. (Source: 02_09_cluster_amplitude_summary.csv)

Three patterns emerge. First, the open dune clusters (C2, C3) show stable raw amplitude but modest climate-normalized damping ($\approx 7\%$), consistent with the post-2018 drought summers temporarily inflating apparent variability while the underlying seasonal range has slightly contracted. Second, **C5 Coastal Forest shows a pronounced amplitude collapse** of 18% (raw) to 21% (climate-normalized), the largest reduction in the network. Combined with C5's anomalously fast summer minimum decline (Section 4.8), this indicates that the C5 water table is becoming both lower in absolute terms and less seasonally responsive — recharge peaks no longer reach their pre-2018 heights even after correcting for drought summers, suggesting either progressive interception increase or a structural change in the recharge pathway at the coastal margin. Third, **C1 Lake Edge shows the opposite pattern:** amplitude has *increased* by 16–20%, driven primarily by deeper summer minima rather than higher winter peaks (median post-2018 summer minimum -1.04 m vs -1.42 m at C3; Table 2). The C3 cluster shows the highest within-cluster spread (post-2018 ratio max:min ≈ 2.0), approaching the amplitude-heterogeneity flagging threshold (Section 3.2.4) and consistent with the geological transition between deep aeolian sand and the western dune-till margin.

The amplitude analysis confirms that the cluster-mean coefficients reported in Section 4.2.2 represent stable averages across wells whose individual amplitude responses differ within most clusters by less than a factor of two, but with sufficient within-cluster variation in C3 and C4 to motivate the per-well coefficient mapping presented in Section 4.9.

1.2.2 Cluster Mechanistic Characterization

The five clusters are characterized mechanistically in Table 3 using a state-space model fitted to the average hydrograph for each cluster (Section 3.4.1). β_1 is the recharge sensitivity

— the water table rise (mm) per mm of rainfall; β_2 is the atmospheric draw — the water table decline (mm) per mm of PET; and $-\beta_3$ is the proportional drainage decay rate — the fraction of antecedent displacement above the drainage datum (Section 3.4) dissipated by lateral drainage each month. All five cluster-centroid coefficient sets reported in Table 3 are statistically significant ($p < 0.01$); within-cluster, individual per-well coefficients vary in significance, most notably the near-zero C4 drainage coefficient and the small C1 atmospheric-draw term, as discussed below and in Section 4.9.

The recharge sensitivity coefficient β_1 is highest in the Lake Edge cluster (C1: 4.58), indicating that each millimetre of rainfall drives a proportionally larger water table rise in the shallow-till aquifer than in the Dune cluster (C2: 3.97) or Western Residual (C3: 3.57). The two forested clusters return near-identical and substantially lower recharge sensitivities (C4: 2.49, C5: 2.42), consistent with canopy interception suppressing the rainfall signal reaching the water table. The atmospheric draw coefficient β_2 is highest at C4 (2.58), followed by C3 (1.83) and C2 (1.76), with C5 (1.31) and C1 (0.96) lowest. The C4 value of β_2 slightly exceeds its β_1 (2.58 against 2.49), meaning that the Main Forest cluster loses as much water table per mm of PET as it gains per mm of rainfall — the plantation suppresses recharge (low β_1) while simultaneously amplifying the atmospheric draw signal relative to unforested clusters, producing a symmetrically sensitive water balance. Because β coefficients fold together vegetation effects, specific yield and soil moisture dynamics, they cannot be decomposed into crop coefficients without independent S_y estimates; this decomposition is developed in Section 5.6.1. That C5 (Coastal Forest) has low β_2 despite also carrying Corsican pine canopy suggests that coastal proximity moderates the atmospheric draw signal, possibly through higher ambient humidity or higher specific yield in the sandier coastal substrate. C1's low β_2 (0.96, the lowest in the network) reflects its shallow water table and rapid lateral drainage to the lake — the head-dependent drainage term captures the dominant loss mechanism, with the depth-dependent PET diagnostic (Section 3.7.1) confirming that C1 has the strongest residual depth-coupling ($\lambda = 1.90 \text{ m}^{-1}$). The drainage decay coefficient $-\beta_3$ is highest at C1 (0.090) where rapid lateral exchange with Llyn Rhos-Ddu drives fast recession, similar at C2 (0.066) and C3 (0.058), lower at C5 (0.045), and lowest at C4 (0.020). The C4 value is weakly identified, however — the forest-interior drainage coefficient is degeneracy-affected (Section 5.2.4), and a triangulation-constrained estimate places it nearer 0.058 month^{-1} . The LCSC ($= 100/\beta_1$) expresses these contrasts as a directly interpretable management metric: 22–25 mm of rainfall is required to raise Lake Edge and Dune water tables by 10 cm, compared with 28 mm in the Western Residual and 40–41 mm beneath the plantation.

The C1 cluster-mean coefficients in Table 3 include CEH11, which is geographically separated from the other six C1 members (Section 3.2.3). CEH11 sits at 3.6 m AOD on the south-eastern coast, approximately 1.6 km from the nearest lake-edge well, and its damped hydrograph likely reflects proximity to the Menai Strait tidal boundary rather than to Llyn Rhos-Ddu. Its β_2 value of 0.77 is consistent with the C1 cluster mean (0.96) but may reflect a

coastal constant-head mechanism rather than the lake-buffering process that governs the remaining members. CEH11's inclusion extends the C1 elevation range from 8.6–10.9 m down to 3.6 m; if excluded, the six remaining wells (five excluding CEH6, whose negative β_2 is discussed below) occupy a tight 9.1–10.9 m elevation band. The cluster-mean coefficients are not materially affected by CEH11's inclusion — its β_1 (4.46) and β_3 (0.092) are close to the C1 mean — but per-well spatial analyses within C1 should be interpreted with awareness that CEH11 responds to a different boundary condition.

An extended form of the state-space model incorporating a depth-dependent evapotranspiration coefficient (Section 3.7.1) was evaluated at cluster level; as the results of this analysis bear primarily on the mechanistic interpretation of the Forest cluster β_2 coefficient, they are discussed in Section 5.6.1 rather than reported here. The within-cluster coefficient variation is substantial, particularly in the forest zone where per-well β_2 ranges from 0.873 to 3.723 across the 14 C4 and C5 wells. Whether this variation tracks topographic position (a proxy for the uncored substrate) or canopy differences is examined through per-well spatial regression in Section 4.9.5.

Cluster	Label	n	β_1	β_2	$-\beta_3$	LCSC (%)	R ²
C1	Lake Edge	7	4.576	0.956	0.088	21.85	0.73
C2	Dune	24	3.974	1.762	0.064	25.17	0.75
C3	Western Residual	21	3.573	1.833	0.057	27.98	0.81
C4	Main Forest	9	2.487	2.583	0.018	40.20	0.73
C5	Coastal Forest	5	2.423	1.307	0.044	41.27	0.69

Table 1.3: Cluster mechanistic characterization derived from the displacement-formulation SSM (Section 3.4). β_1 = recharge sensitivity (mm water table rise per mm rainfall); β_2 = atmospheric draw (mm water table decline per mm PET); $-\beta_3$ = proportional drainage decay rate (fraction of displacement above the 3.7 m drainage datum dissipated per month). LCSC = $100/\beta_1$; values express the millimetres of rainfall required to raise the cluster average water table by 10 cm. R² is one-step diagnostic fit. All coefficients significant at $p < 0.01$ or better. (Source: 03_03_cluster_mechanistic_coefficients.csv).

1.2.3 Recharge and Losses: A Cluster-Level Water Balance

Water balance decomposition applied to all five clusters (Section 3.7.2) reveals markedly different recharge sensitivities across clusters receiving identical climate forcing. The head-space decomposition is presented in Table 4a and Figure 11; an indicative volumetric conversion is presented in Table 4b and Figure 11b. The derivation and defence of the specific yield values is addressed in Section 4.2.4.

Recharge into the water table ($\beta_1 \cdot \text{PI}$) declines systematically from the Lake Edge cluster (C1: 0.340 m/month) through the Dune (C2: 0.296 m/month) and Western Residual (C3: 0.265 m/month) to the forested clusters (C5: 0.180 m/month; C4: 0.185 m/month). Total losses (atmospheric draw plus drainage) track recharge closely, ranging from 0.186 m/month at C4 to 0.345 m/month at C1. Residuals are less than 2.3% of total losses in all clusters, confirming that the three-term displacement-formulation SSM captures the principal water balance fluxes without requiring an unmeasured boundary contribution. The drainage

partition varies markedly across clusters: drainage accounts for 85% of total losses at C1 (reflecting rapid lateral exchange with Llyn Rhos-Ddu) but only 24% at C4, where atmospheric draw dominates. C2 and C3 show drainage-dominated partitions (68%/32% and 62%/38% respectively), while C5 is intermediate (60%/40%).

Cluster	Label	LCSC (%)	Recharge (m/month)	Atm. Draw (m/month)	Drainage (m/month)	Total Loss (m/month)	Residual (m/month)
C1	Lake Edge	21.9	0.340	0.053	0.292	0.345	−0.004
C2	Dune	25.2	0.296	0.096	0.203	0.299	−0.003
C3	Western Residual	28.0	0.265	0.100	0.166	0.266	−0.000
C4	Main Forest	40.2	0.185	0.142	0.044	0.186	−0.001
C5	Coastal Forest	41.3	0.180	0.072	0.113	0.184	−0.004

Table 1.4a: Mean monthly head-space water balance decomposition by hydrogeological cluster, Newborough Warren 2005–2026. Recharge ($\beta_1 \cdot Pc$), atmospheric draw ($\beta_2 \cdot PETc$) and gravity drainage ($\beta_3 \cdot hc_disp$) are expressed as metres of water table change per month (m/month), where hc_disp is the mean displacement above the 3.7 m drainage datum (Section 3.4). Coefficients are cluster-centroid fits from the displacement-formulation SSM (Table 3). Residuals are less than 2.3% of total losses in all clusters, confirming that the three-term formulation closes the water balance without requiring an unmeasured boundary contribution. All clusters receive identical climate forcing: $Pc = 74.4$ mm/month, $PETc = 55.1$ mm/month. Drainage and ET percentages shown below. (Source: 16_water_bal_table.csv).

An indicative volumetric expression of the water balance is presented in Table 4b and Figure 11b. All clusters receive 893 mm/yr of rainfall; for the forested clusters (C4, C5) canopy interception removes 214 mm/yr (24%; Freeman, 2008), leaving net rainfall of 678 mm/yr reaching the ground surface. The ET/drainage partition is bracketed by two independent methods — the SSM headspace β_2/β_3 ratio and a seasonal recession curve analysis — with the mid-point reported in Table 4b. At steady state, total losses equal total inputs (P or P_{net} for forest clusters) and the hatched band in Figure 11b spans the partition uncertainty range between the two methods.

Cluster	Label	P (mm/yr)	I (mm/yr)	P_net (mm/yr)	ET mid (mm/yr)	Drainage mid (mm/yr)
C1	Lake Edge	893	0	893	246	647
C2	Dune	893	0	893	390	503
C3	Western Residual	893	0	893	337	556
C4	Main Forest	893	214	678	431	247
C5	Coastal Forest	893	214	678	221	457

Table 1.4b. Indicative annual volumetric water balance by cluster, 2005–2026. At steady state, total losses (ET + drainage) equal total inputs (P for open dune clusters; $P - I$ for forested clusters). I = canopy interception (24% of P ; Freeman, 2008), applied to C4 and C5 only. ET and drainage mid-point values represent the mean of two independent partitioning methods: the SSM headspace β_2/β_3 ratio and seasonal recession curve analysis. The partition uncertainty range is shown graphically in Figure 11b as a hatched band. Interception appears identically on both input and loss bars of Figure 11b and cancels in the net surplus. (Source: 16_water_bal_vol_table.csv)

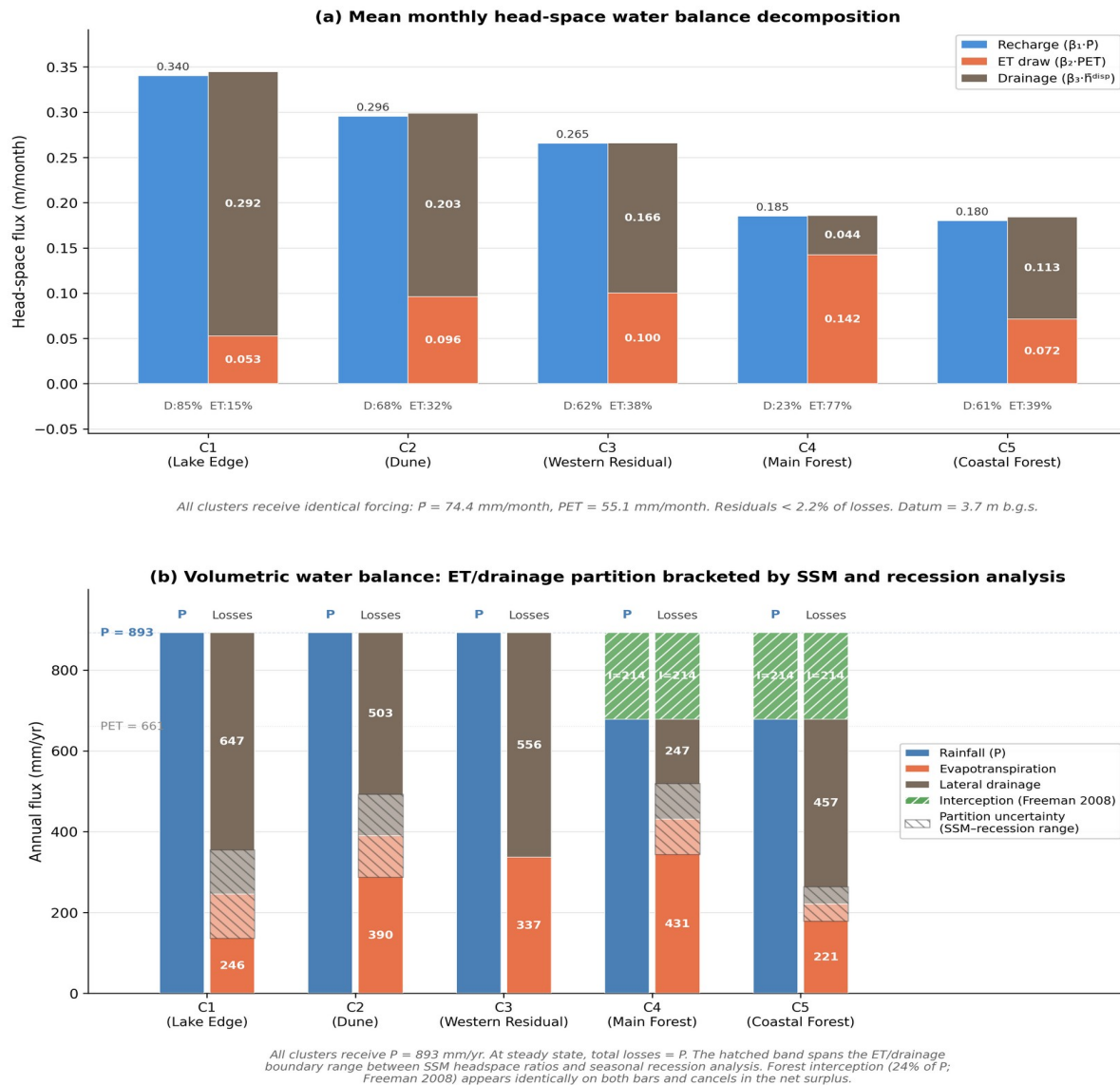


Figure 1.8: Water balance decomposition by hydrogeological cluster, Newborough Warren 2005–2026. (a) Mean monthly head-space decomposition: for each cluster, the left bar shows recharge ($\beta_1 \cdot \bar{P}$) and the right bar shows losses decomposed into gravity drainage ($\beta_3 \cdot \bar{h}_{disp}$) and atmospheric draw ($\beta_2 \cdot PET$). Drainage and ET percentages are shown below each cluster label. All clusters receive identical climate forcing: $\bar{P} = 74.4$ mm/month, $PET = 55.1$ mm/month. Drainage datum = 3.7 m b.g.s. Residuals < 2.3% of losses. (b) Indicative annual volumetric water balance. All clusters receive $P = 893$ mm/yr; for forested clusters (C4, C5) canopy interception ($I = 214$ mm/yr, 24% of P ; Freeman, 2008) appears identically on both input and loss bars and cancels in the net surplus. The hatched band spans the ET/drainage partition uncertainty between the SSM headspace ratio and seasonal recession analysis. (Source: 16_water_bal_bar_ms.png).

1.2.4 Specific Yield Estimation

WTF event-based median S_y estimates for all five clusters, derived from 34–64 qualifying monthly rising-limb events per cluster, are presented in Table 4c. Uncorrected median values range from 0.210 (C1 Lake Edge) to 0.356 (C5 Coastal Forest), with broad interquartile ranges reflecting the inherent noise in monthly cluster-mean data. The two forested clusters return uncorrected medians (C4: 0.315, C5: 0.356) that sit at or above the open dune range (0.210–0.325), because gross rainfall overstates the recharge flux reaching the water table beneath the canopy — the WTF method attributes the full rainfall to recharge when in reality 24% is intercepted, inflating the apparent S_y . After applying the interception correction of 24% of P (Freeman, 2008), the forested cluster medians fall to 0.259 (C4) and 0.321 (C5): C4 falls cleanly within the open dune range (0.210–0.328), while C5 sits at its upper edge. The C5 estimate is the least reliable in the network — a majority of its rising-limb events return event-level S_y at or above the 0.50 plausibility ceiling (Section 3.7.3), so its median is constrained by that limit rather than freely estimated, and its modest apparent response to the interception correction is partly an artefact of the clip; it is therefore reported as a weak corroborator. The C4 result is consistent with the attenuated water table response beneath the plantation reflecting the surface boundary condition (canopy interception) rather than a fundamentally different substrate. Within this corrected dataset, a secondary pattern is visible: the two clusters with the lowest corrected S_y — C1 (Lake Edge; 0.210) and C4 (Main Forest; 0.259) — return lower values than C3 (Western Residual; 0.328) and C5 (Coastal Forest; 0.321), with C2 (Dune; 0.281) intermediate. C1's low S_y is consistent with the shallow clay/till base and short drainage path to the lake seen in the borehole logs; C4 overlies the bedrock ridge, which outcrops within the forest, so an irregular, low-storage rock base is the natural reading of its low S_y and its within-cluster heterogeneity in β_3 — though the extent and depth of that base beneath C4, and hence the precise buried-ridge geometry, are unconstrained without coring. The pattern recurs in the per-well β_2 regression (Section 4.9.5) where elevation — reflecting both vadose zone thickness and proximity to buried bedrock — explains 96% of within-forest β_2 variance.

An independent cross-check on these estimates is available from a second WTF method: an ordinary-least-squares regression of winter water-table rise on net recharge, which returns a per-cluster S_y from the regression slope (Table 4c, OLS-winter column). The OLS estimates run higher than the event-medians at four of the five clusters (mean ratio 1.18), consistent with the documented tendency of WTF methods to bias high on absolute magnitude; the OLS values, like the event-medians, should be read as upper bounds rather than point estimates. The two methods agree on the principal spatial pattern — both place the deep-sand Western Residual (C3) and Coastal Forest (C5) clusters at the high-storage end of the network — which corroborates the substrate-driven interpretation above. They diverge at two clusters, and the divergence is itself informative. At C1 (Lake Edge), the OLS regression returns the weakest fit of the five clusters ($R^2 = 0.62$, with C2 next-weakest at 0.67 and C3–C5 at 0.84–0.88), so the OLS estimate is least reliable precisely where it disagrees; the

event-median's placement of C1 as the low-storage end-member, consistent with its constrained till-and-clay substrate, is retained. At C4 (Main Forest), the uncorrected OLS estimate ranks the cluster lowest in the network — but C4 is the cluster where canopy interception most distorts the raw WTF signal, which is exactly why the forest interpretation in this section rests on the interception-corrected event-median rather than any uncorrected estimate. Across both methods, the event-based median is retained as the primary specific-yield estimate for each cluster; the OLS regression serves as a corroborating check on the spatial pattern.

A third, independent WTF estimator — the rapid-recharge-event method of Crosbie et al. (2005; Section 3.7.3) — provides a further check that shares the assumptions of neither the event-median nor the winter-OLS estimate. Selecting only short, sharply-rising episodes in which drainage during the rise is negligible, it returns cluster median S_y values of 0.195 (C1, 95% CI [0.098, 0.230], $n = 20$), 0.261 (C2, [0.197, 0.336], $n = 21$), 0.334 (C3, [0.288, 0.437], $n = 17$), 0.274 (C4, interception-corrected, [0.229, 0.312], $n = 13$) and 0.298 (C5, interception-corrected, [0.269, 0.355], $n = 14$). Two features corroborate the interpretation above. At C3 (Western Residual) all three methods agree within 0.023 (0.328, 0.334 and 0.351), so the high-storage placement of the deep-sand cluster — the anchor for the drainage-reach interpretation of Section 4.9 — is robust to estimator choice. At C1 (Lake Edge) the rapid-event estimate (0.195) sits with the event-median (0.210) well below the winter-OLS value (0.341), independently reinforcing the reading that the drainage-corrected OLS over-corrects at this lake-buffered cluster, where the water table is decoupled from monthly rainfall; the low-storage placement of C1 is retained on the agreement of the two drainage-insensitive estimators. The coarse specific-yield gradient across the network is therefore consistent across three methodologically distinct estimators, while the fine ordering at the high-storage end (C3 versus C5) remains method-dependent — the winter-OLS placing C5 highest, the two event-based methods placing C3 highest — and all three, as noted, are upper bounds on effective storage rather than point estimates.

Individual well S_y values underlying the spatial mapping of Figure 12 are considered separately below. All five WTF medians exceed the assumed S_y values used in the water balance (C1: 0.08; C2–C5: 0.12) by a factor of two to three, consistent with the known tendency of monthly WTF to conflate gravity drainage with capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). The WTF estimates should therefore be interpreted as upper bounds on effective storage. Direct field measurement by slug test or pumping test at representative wells per cluster is identified as a priority for future work.

Cluster	n events	Sy assumed	WTF event-median	OLS-winter	Q25	Q75	Interception corrected?
C1 Lake Edge	59	0.08	0.210	0.341	0.129	0.259	No
C2 Dune	64	0.12	0.281	0.335	0.197	0.374	No
C3 Western Residual	57	0.12	0.328	0.351	0.283	0.410	No
C4 Main Forest	51	0.12	0.315	0.302	0.256	0.407	No
C5 Coastal Forest	34	0.12	0.356	0.419	0.319	0.431	No
C4 Main Forest (corrected)	62	0.12	0.259	—	0.178	0.330	Yes — Freeman (2008)
C5 Coastal Forest (corrected)	49	0.12	0.321	—	0.243	0.383	Yes — Freeman (2008)

Table 1.4c: Specific yield (Sy) estimates by hydrogeological cluster derived from the water table fluctuation (WTF) method (Healy and Cook, 2002). Event-based median Sy computed from monthly rising-limb events satisfying $\Delta h > 5$ mm and net recharge $P - PET > 10$ mm over the monitoring period April 2005 to February 2026. For the forested clusters (C4, C5), two estimates are reported: uncorrected (using gross rainfall) and corrected (net recharge adjusted for 24% canopy interception following Freeman, 2008). The corrected C4 median (0.259) falls within the range of the open dune clusters (0.210–0.328) and the C5 median (0.321) sits at its upper edge; the C5 estimate is constrained by the 0.50 plausibility clip and is treated as weakly corroborative. The C4 result, together with the continuous interpolated yield surface (Figure 12), is consistent with broadly uniform aquifer storage across the site. IQR = interquartile range; wide IQR reflects noise inherent in monthly cluster-mean data rather than genuine spatial heterogeneity in aquifer storage. Monthly WTF estimates are known to overestimate effective Sy through conflation of gravity drainage with capillary fringe release and should be interpreted as upper bounds rather than point estimates of long-term effective storage. A second, independent WTF estimate — the OLS winter-regression Sy (Sy OLS-winter) — is shown for the five uncorrected clusters; it is not computed on the interception-corrected forest series, shown as '—'. The event-median remains the primary estimate (see text). (Healy and Cook, 2002; Scanlon et al., 2002). (Source: 17_wtf_01_sy_estimates.csv).

Individual well WTF Sy estimates were derived from 27–72 qualifying rising-limb events per well (mean 51 across the 66-well reference network). Per-well cluster-mean values — C1: 0.209, C2: 0.254, C3: 0.312, C4: 0.246, C5: 0.309 — are consistent with the cluster-aggregate estimates of Table 4c, confirming that the spatial pattern is robust. These per-well means are by-products of the spatial interpolation: they describe the inputs to Figure 12 and preserve within-cluster variability, but each per-well Sy is noisier than the cluster-aggregate estimate of Table 4c, which fits a single median to the smoothed cluster-mean hydrograph and is therefore more robust as an estimate of cluster storage. The well-level analysis is presented here for its spatial content rather than as a competing storage estimate. The spatial distribution of Sy, shown in Figure 12, is broadly uniform across the site, with no abrupt discontinuities coinciding with the cluster margins, providing further confirmation that the cluster behavioural differences reflect surface hydrology and land cover rather than heterogeneity in aquifer storage architecture.

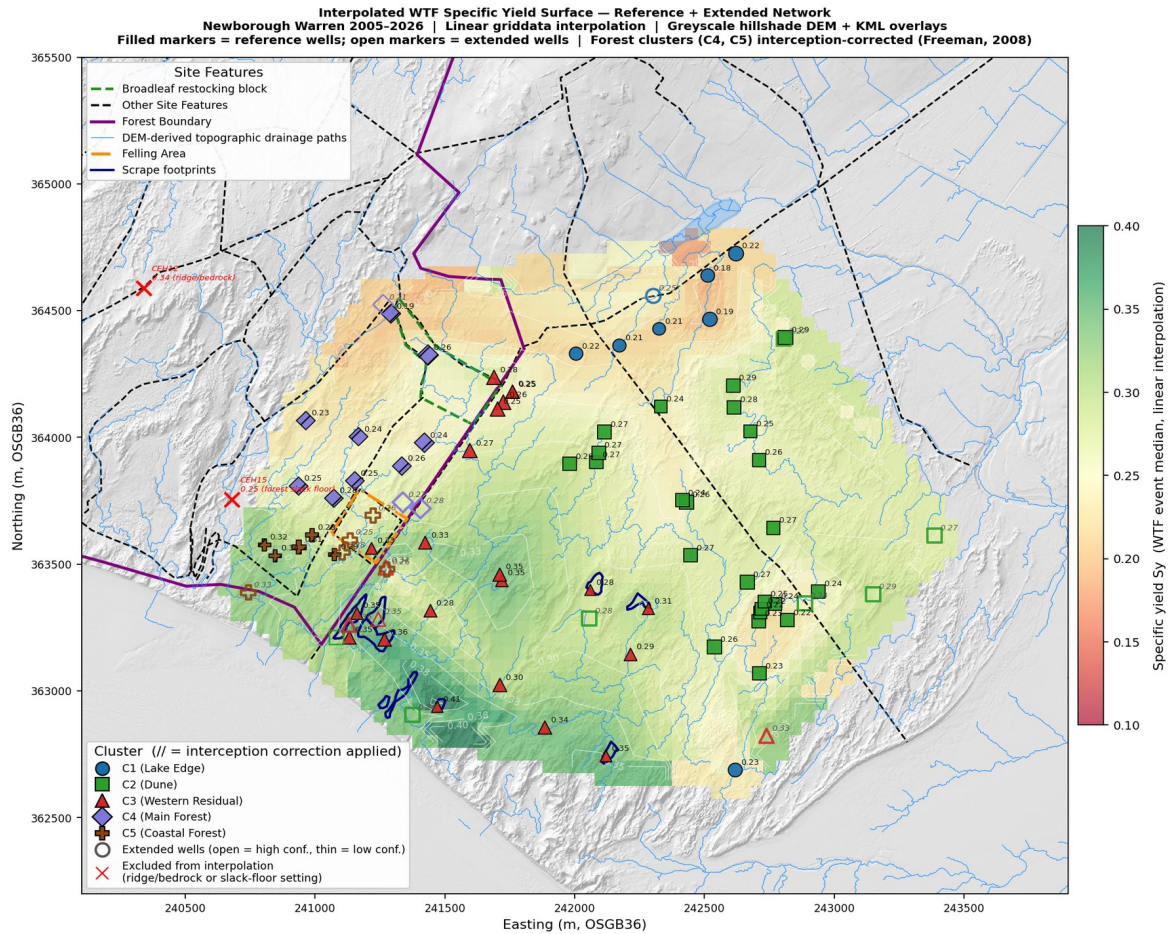


Figure 1.9: Interpolated WTF specific yield surface for the combined reference and extended networks, Newborough Warren 2005–2026. Per-well event-median S_y interpolated using IDW (power = 2) over 1 m LiDAR DEM hillshade. Filled markers denote reference-network wells; open markers denote extended-network wells classified by Pearson affinity (Section 4.3, best-match cluster). Forest cluster (C4, C5) wells are interception-corrected following Freeman (2008). Two wells are excluded from interpolation: CEH12 (ridge/bedrock setting) and CEH15 (forest slack floor), both of which produce anomalous S_y values inconsistent with the surrounding aquifer. Per-well cluster-mean S_y values — C1: 0.209, C2: 0.260, C3: 0.312, C4: 0.246, C5: 0.309 — are consistent with the cluster-aggregate estimates of Table 4c. The spatial distribution is broadly uniform across the open dune (S_y 0.22–0.31), with no abrupt discontinuities coinciding with cluster margins, providing further confirmation that the cluster behavioural differences reflect surface hydrology and land cover rather than heterogeneity in aquifer storage architecture. (Source: 18_wtf_04_sy_contour_extended.png) LiDAR data: © NRW & OS

1.3 Checking the Clusters Pearson Affinity Analysis

Of the 66 wells in the reference network, 63 confirmed their primary cluster membership through the Pearson Affinity Analysis (Section 3.3). Membership is characterized by three distinct tiers based on the correlation delta (Δr) between assigned and neighbouring cluster centroids: Core members (17 wells) with an unambiguous dominant affinity to their assigned cluster ($\Delta r > 0.05$), representing stable hydrogeological anchors within their designated geomorphological zone; Fuzzy members (46 wells) exhibiting closely competing affinities between their primary and secondary clusters ($\Delta r < 0.05$), identifying areas where the groundwater regime transitions smoothly between adjacent structural blocks; and three Spy classifications where the well's highest temporal affinity belongs to a different cluster than Ward's assignment. CEH21 and CEH4 (both assigned C3 Western Residual) show strongest affinity with C5 Coastal Forest, consistent with their coastal-margin position where the Western Residual and Coastal Forest signatures overlap; NW13 (assigned C3) shows strongest affinity with C2 Dune. All three Spy deltas are small ($|\Delta r| < 0.012$), confirming that these are borderline assignments at gradational boundaries rather than gross misclassifications.

Classification confidence was further characterized by the Multi-Cluster Affinity flag, raised where three or more cluster centroids achieved a correlation of $r > 0.90$. This criterion was met at 46 of 66 wells, reflecting the high mutual correlation among the five cluster centroids in a contiguous, highly transmissive sand aquifer. Of these, 21 are concentrated within the gradational C1/C2/C3 transitional zone of the mid-eastern dune sector, 14 lie along the C2/C3/C5 boundary where the Dune, Western Residual and Coastal Forest signatures converge, and 9 lie along the C3/C4/C5 forest–dune boundary, and 2 lie along the C2/C3/C4 boundary. The prevalence of MCA flags confirms the continuous nature of the sand matrix, where boundary wells respond to system-wide climatic forcing rather than localized topographic drivers, and identifies the precise locations of greatest structural ambiguity within the network (Figure 14a).

The extended network analysis (Section 3.3.3) classified all 22 extended wells against the reference centroids, yielding 3 Ext_Core and 19 Ext_Fuzzy assignments (Figure 14b). The high proportion of Fuzzy classifications reflects the shorter and earlier records of these wells rather than genuine hydrogeological ambiguity — with less temporal overlap against the centroid time series, affinity margins are naturally compressed.

Within the extended network, the clearfell treatment wells showed mixed affinities across the C3/C4/C5 boundary, with all wells carrying Multi-Cluster Affinity flags ($r > 0.90$ with three or more centroids). FE1 and LIS1 returned strongest affinity with C4 Main Forest ($r = 0.989$ and 0.973 respectively), while FE2 and FE4 showed marginally strongest affinity with C5 Coastal Forest ($r = 0.975$ and 0.972) with C4 within 0.006. FE3 showed strongest affinity with C5 ($r = 0.985$) followed by C3 ($r = 0.974$). WMC3, in the reference network, showed strongest affinity with C3 ($r = 0.965$) with C5 a close second ($r = 0.961$). The affinity

profiles for these and the other selected wells are plotted in Figure 13. The small deltas ($\Delta r = 0.003\text{--}0.037$) confirm that the clearfell zone sits at the gradational boundary between the Western Residual, Main Forest and Coastal Forest clusters rather than within any single hydrogeological unit. The persistence of forest-range affinities (C4 or C5) at these wells despite the December 2017 clearfell is consistent with either residual canopy influence from the surrounding intact plantation, incomplete hydrological recovery, or the high elevation and its inferred deep sandy substrate generating the forest signature independently of the canopy.

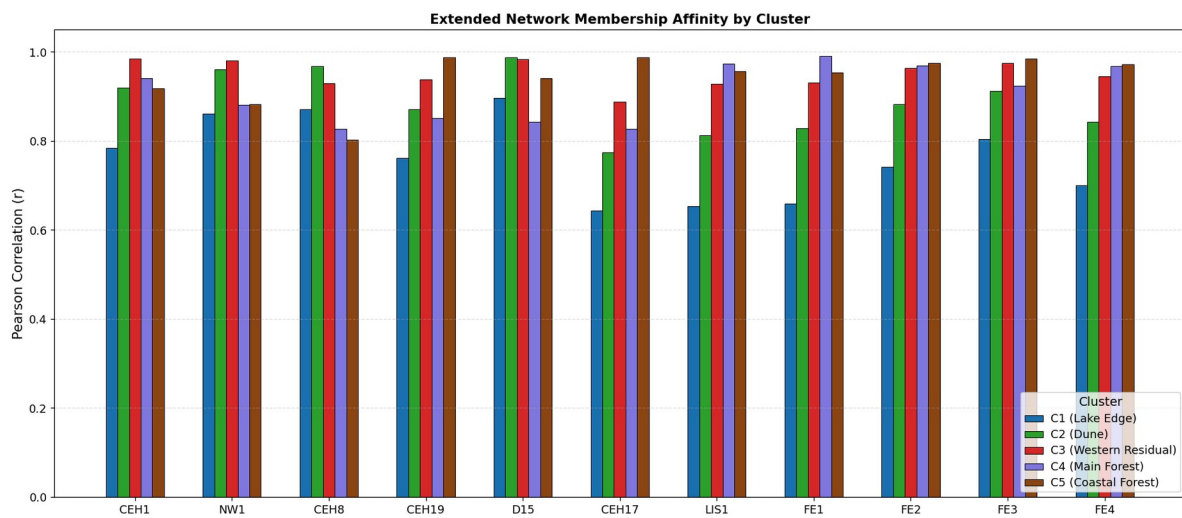


Figure 1.10: Pearson affinity analysis of selected reference and extended network wells, showing the correlation coefficient (r) between each well's z-scored time series and each of the five cluster centroids. Wells are selected to illustrate the range of affinity confidence from strong single-cluster dominance (Core members) to near-equal competing affinities (Fuzzy members), along with wells in or close to the clearfell experiment. (Source: 06_pear_01_affinity_chart_extended.png)

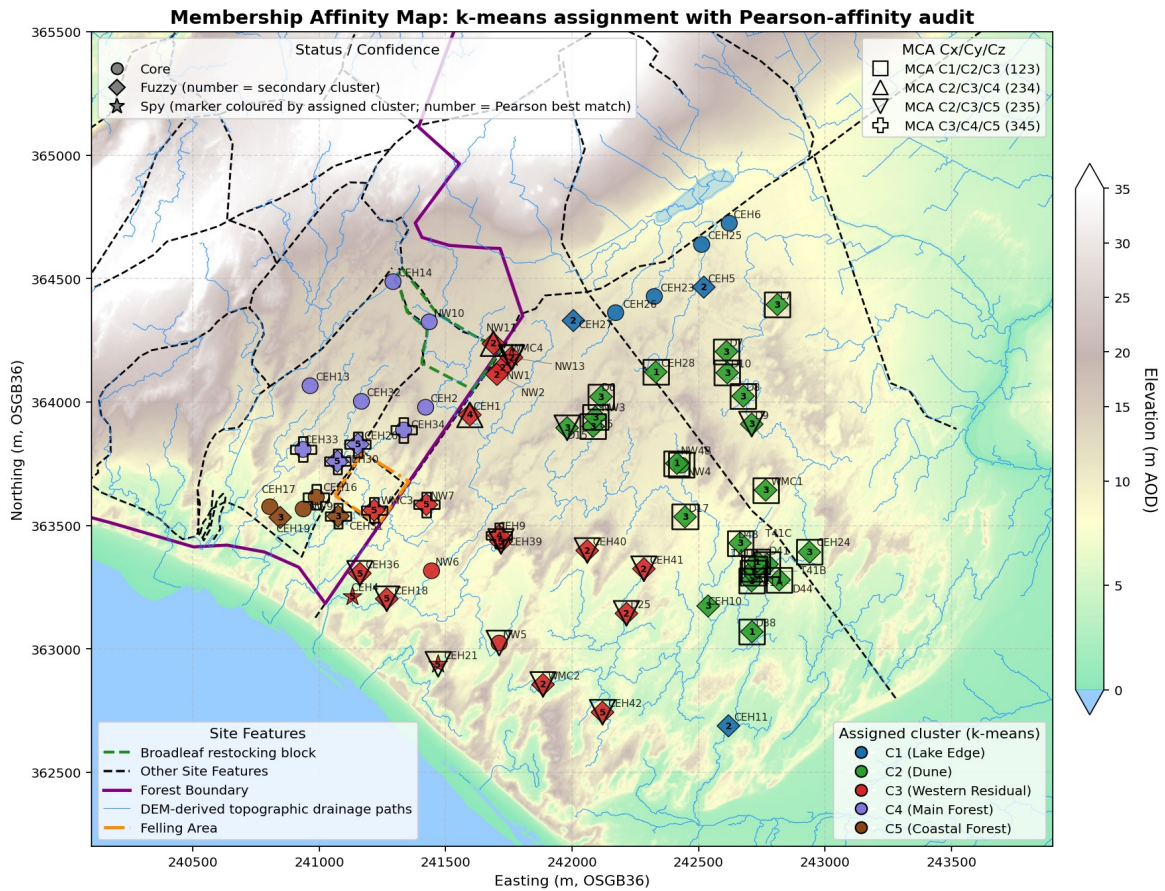


Figure 1.11a: Spatial distribution of the five primary hydrogeological clusters validated via Pearson affinity analysis, reference network (66 wells, records ≥ 100 months to February 2026). Points coloured by k-means assignment. Symbolism denotes Pearson affinity confidence: circles = Core members ($\Delta r > 0.05$), diamonds = Fuzzy members ($\Delta r < 0.05$; number inside diamond = secondary cluster), stars = Spy classifications (Pearson best-match $\neq$ k-means-assigned cluster; number inside star = best-match cluster). Large hollow overlays identify Multi-Cluster Affinity (MCA) wells: convergence $r > 0.90$ with three or more cluster centroids; overlay shape encodes the specific MCA combination. LiDAR: © NRW & OS (Section 10). (Source: 05_pear_01_spatial_confidence_map.png)

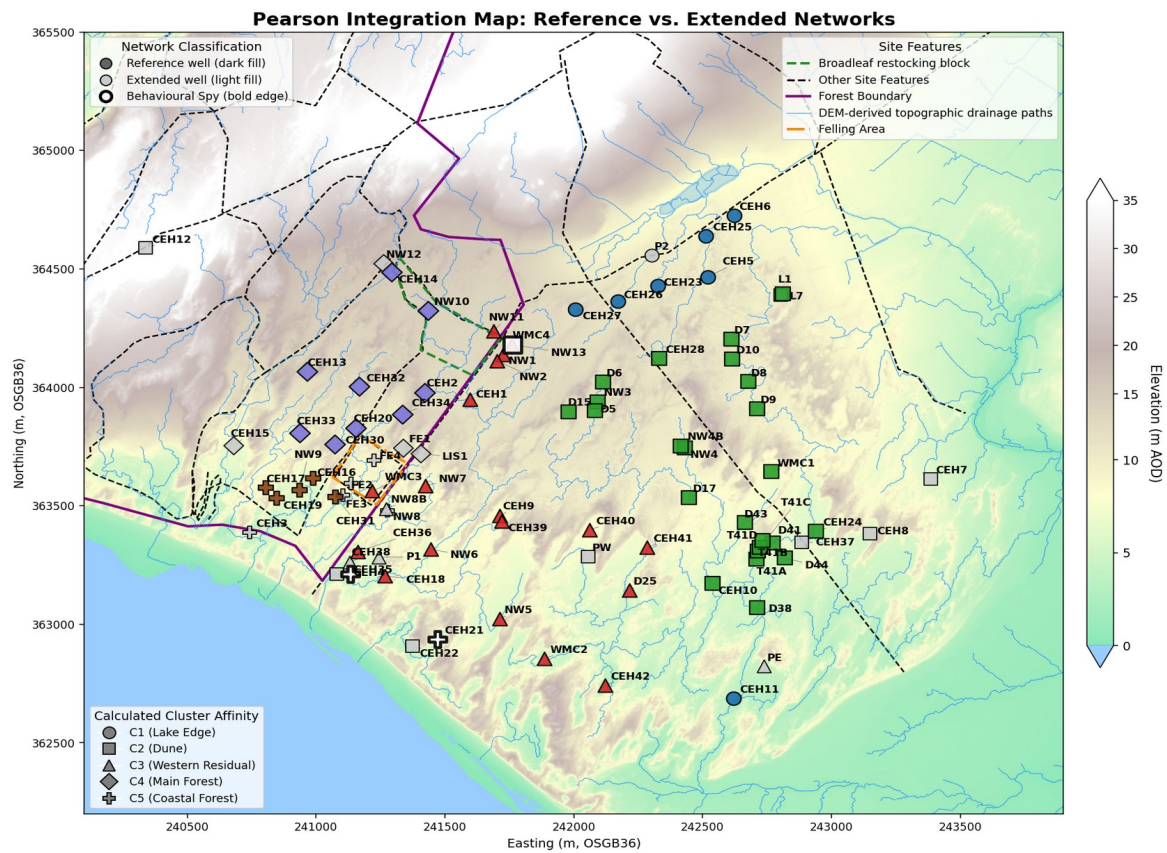


Figure 1.11b. Pearson affinity integration map, full 88 well network. Cluster identity encoded by marker shape; fill distinguishes network membership (reference = dark, extended = light grey, Spy = white with bold edge). LiDAR: © NRW & OS (Section 10). (Source: 06_pear_02_integration_map.png)

1.4 Model Benchmarking: State-Space Model Comparison with Traditional Linear Models

The SSM was compared to a TLM as outlined in Section 3.4.2. In one-step diagnostic mode — where each prediction is corrected with the observed water level from the previous month — both models track the observed record almost identically (median R^2 0.91 and 0.92 respectively; Table 5). The difference emerges under autonomous simulation. Over 100 months without observational correction, the TLM diverges rapidly at wells with strong head-dependent drainage (e.g. CEH6, C2 Dune: TLM NSE = -1.12 , SSM NSE = $+0.66$; Figure 15), while the SSM maintains physical consistency through its storage decay term. Across the full 66-well reference network, the SSM achieves a median iterative NSE of 0.75 compared with 0.18 for the TLM — a median gain of $+0.53$ NSE units — and returns a positive iterative NSE at 65 of 66 wells compared with 44 of 66 for the TLM (Table 5). The improvement varies systematically across the five clusters (Figure 16a, 16b): C1 Lake Edge shows the largest median gain ($+1.286$), where strong head-dependent drainage to Llyn Rhos-Ddu makes the storage decay term essential; C5 ($+0.726$), C3 ($+0.527$) and C2 ($+0.526$) show substantial gains; and C4 Main Forest shows the smallest ($+0.229$), reflecting its characteristically low β_3 (0.020) which produces a near-linear climate-to-water-table relationship that the TLM handles more adequately. The sole well where the SSM fails is CEH14 (C4, iterative NSE = -3.21), a ridge-flank well where the displacement formulation does not adequately capture the boundary condition. Spatial visualization (Figure 16) confirms this pattern geographically, with the largest gains concentrated along the forest–Warren boundary and the C1 lake-buffer sector.

The depth-dependent PET grid search (Section 3.7.1) provides an independent diagnostic of how completely the standard fixed- β_2 formulation captures the atmospheric draw signal. The grid search produced statistically significant NSE improvement at all five clusters, where λ (m^{-1}) is the exponential decay rate controlling how rapidly the PET signal attenuates with water table depth: C1 ($\Delta\text{NSE} = +0.183$, $\lambda = 1.90 \text{ m}^{-1}$), C4 ($+0.104$, $\lambda = 0.15$), C2 ($+0.069$, $\lambda = 0.80$), C5 ($+0.039$, $\lambda = 0.50$) and C3 ($+0.028$, $\lambda = 0.40$). AIC improvement is decisive at every cluster. The fitted λ values are physically interpretable as inverse $1/e$ decay depths: C1's strong decay ($1/e = 0.53 \text{ m}$) is consistent with direct evaporation from a frequently shallow watertable near the lake margin, while C4's weak decay ($1/e = 6.7 \text{ m}$) confirms that C4's high β_2 reflects low specific yield rather than a depth-dependent ET pathway. The mechanistic interpretation is developed in Section 5.6.1.

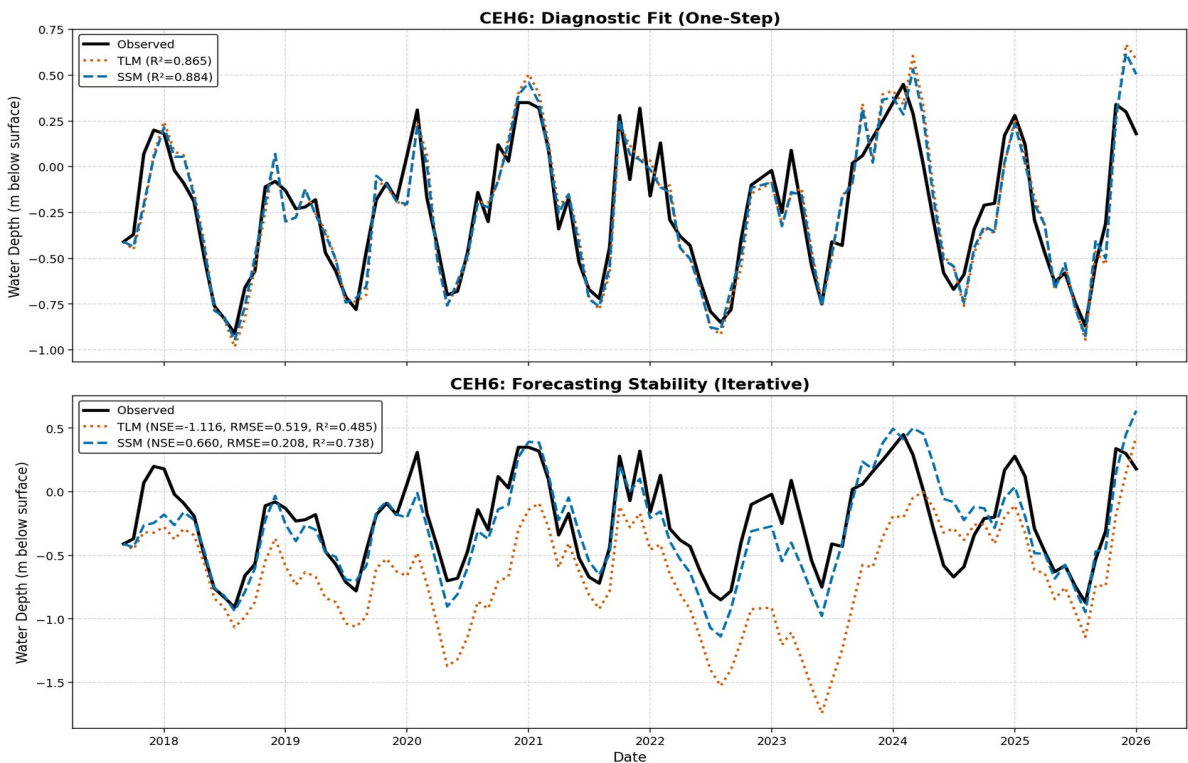


Figure 1.12: Dual-model performance comparison at dune well CEH6. Upper panel: one-step diagnostic fit — both TLM and SSM track observed water levels closely when corrected with real observations at each step. Lower panel: iterative 100-month autonomous simulation without observational correction — the TLM ($NSE = -1.12$) diverges rapidly, while the SSM ($NSE = +0.66$) maintains physical consistency through its storage decay term. CEH6 records the second-largest NSE improvement ($+1.78$) in the 66-well reference network. (Source: 08_lcsc_01_ceh6_showdown.png)

Metric	TLM	SSM	Improvement Δ
Median one-step R^2	0.91	0.92	0.01
Median iterative NSE	0.16	0.75	0.53
Wells with iterative $NSE > 0$	44 / 66	65 / 66	—
Max NSE improvement (CEH25)	-1.48	0.68	2.16
CEH6 iterative NSE	-1.12	0.66	1.78
CEH14 iterative NSE	0.43	-3.21	-3.64

Table 1.5: Model benchmarking summary comparing the Traditional Linear Model (TLM) and State-Space Model (SSM) across the 66-well reference network. Performance evaluated over the most recent 100 months at each well. One-step R^2 uses observed $h(t-1)$ at each step; iterative NSE uses simulated $h(t-1)$ over the full 100-month evaluation window without observational correction. For the median rows, Δ is the median of the per-well improvements (not the difference of the column medians). CEH25 is the well recording the largest single NSE improvement; CEH6 is the illustrative dune well shown in Figure 15; CEH14 is the sole well where the SSM underperforms the TLM. (Source: 08_lcsc_model_stats.csv).

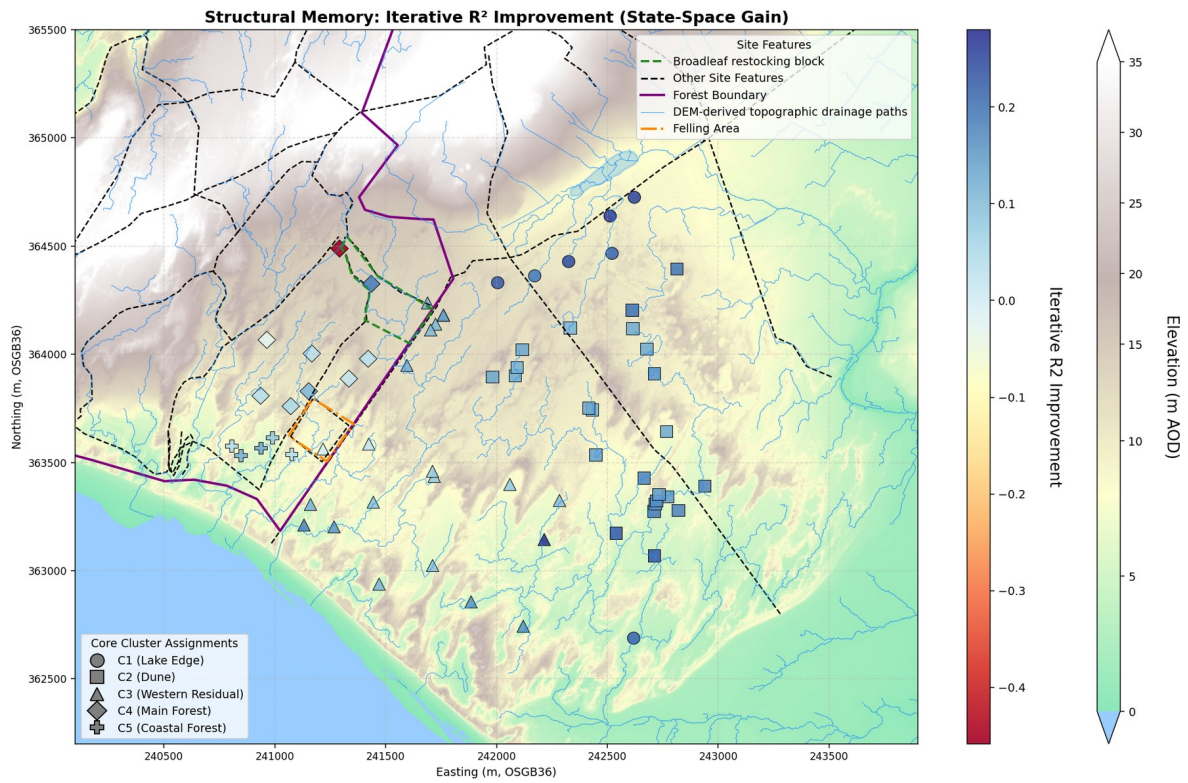


Figure 1.13a: SSM gain over TLM across all 66 reference wells. Panel (a): iterative R^2 improvement — largest gains in the Lake Edge sector and eastern dune, smallest in the Main Forest cluster where the TLM already performs adequately. (Source: 08_lcsc_02_r2_improvement_map.png) LiDAR data: © NRW & OS; see Section 10 for full attribution.

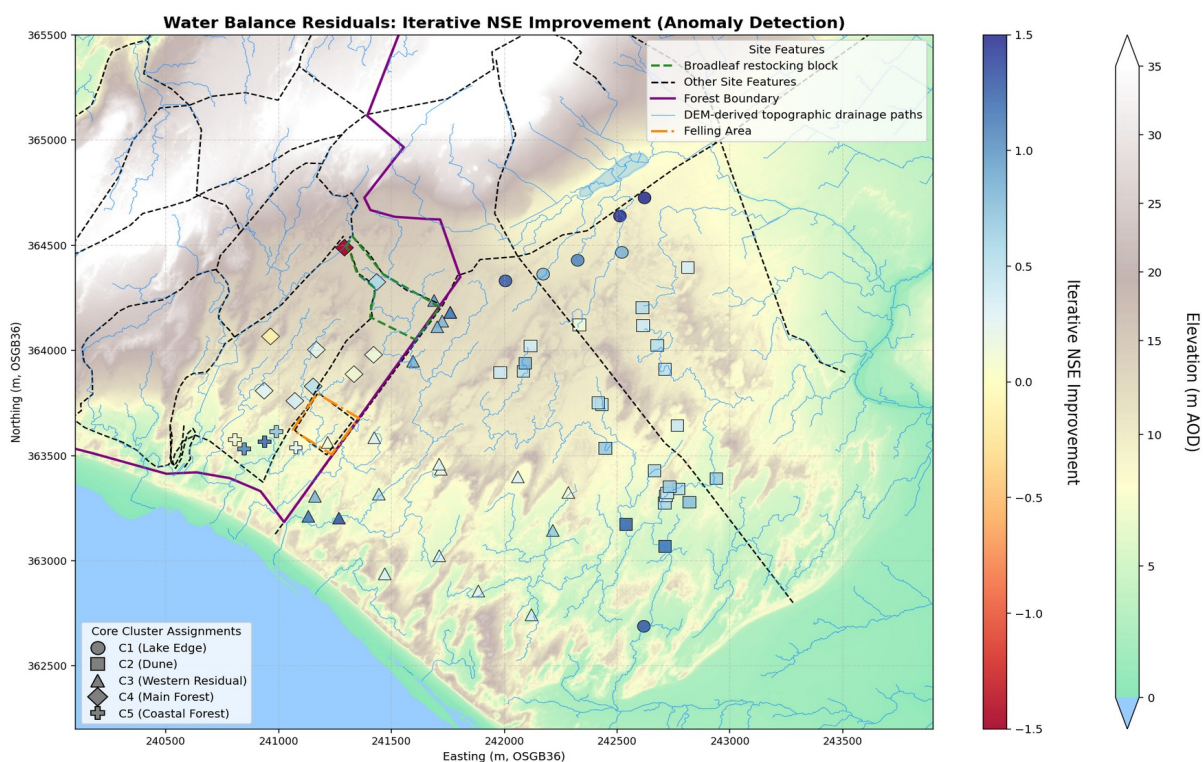


Figure 1.13b: SSM gain over TLM across all 66 reference wells. NSE improvement — corresponding spatial pattern with the largest gains concentrated in the C1 Lake Edge cluster. The spatial distribution of improvement physically delineates subsurface structural boundaries without invasive coring. (Source: 08_lcsc_03_nse_improvement_map.png) LiDAR data: © NRW & OS; see Section 10 for full attribution.

1.5 Dune Scraping Intervention Analysis

1.5.1 Background Environmental Drift

Three dipwells within the dune slack system — CEH36, CEH18 and CEH21 — were subject to ground surface lowering by mechanical scraping, with CEH36 scraped in April 2015 and CEH18 and CEH21 scraped in October 2023. The analysis uses a two-tier CUSUM-BACI framework (Section 3.5) to isolate the scraping signal from background environmental drift, with CEH4 and CEH22 serving as paired controls. Tier 1 analysis at CEH4 and CEH22 both show accelerating negative CUSUM trajectories across the monitoring period, reaching -10.55 m and -19.02 m respectively by early 2026 (Section 3.5.4; Figure 17). The convergent deterioration of two spatially separated control wells — CEH22 and CEH4 near the coast — independently of any management action confirms that progressive coastal erosion of the dune edge is driving a structural decline in water table depth across this part of the network. CEH22's trajectory is the more severe, dominated by a coastal drawdown signal that accelerated markedly from 2021 onwards.

Tier 1 - Background Environmental Drift (CUSUM Analysis)

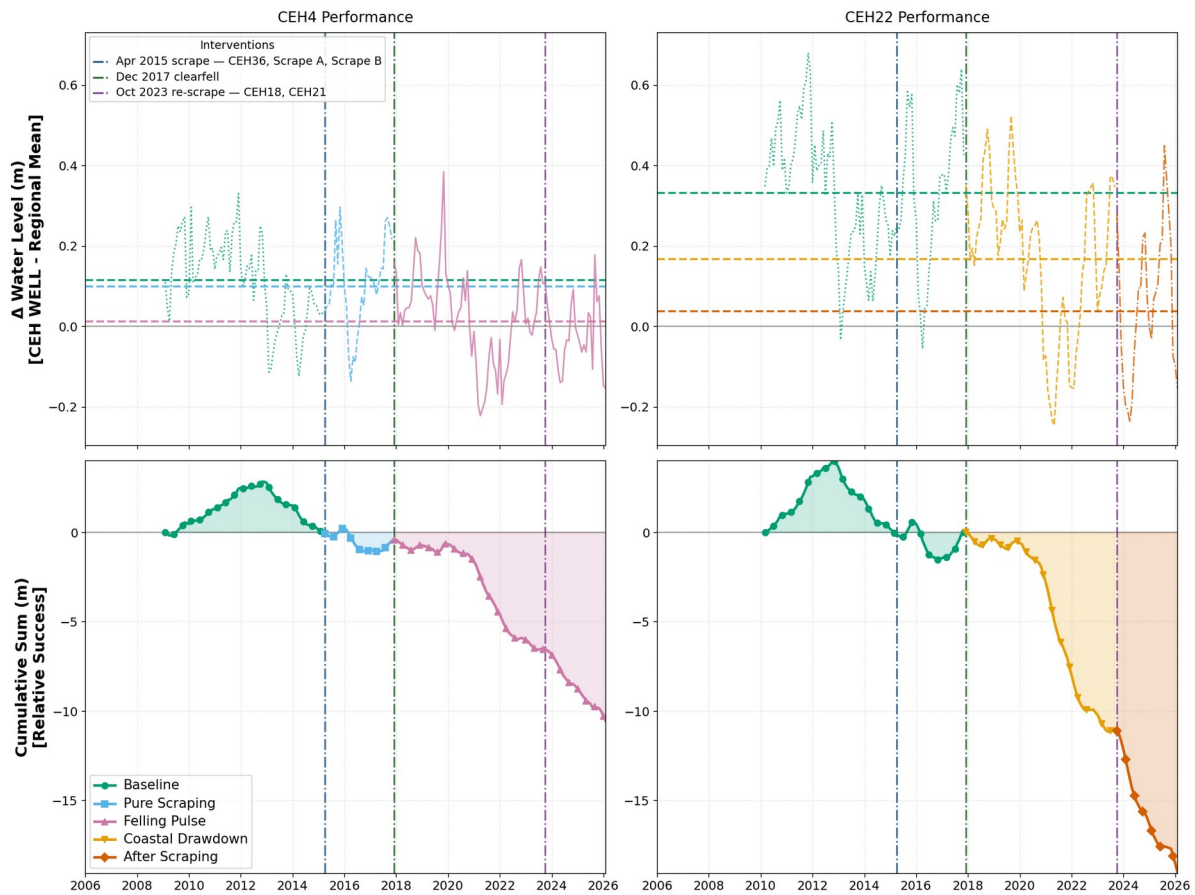


Figure 1.14: Tier 1 background environmental drift analysis. CUSUM time series for control wells CEH4 (left) and CEH22 (right) decomposed by management era relative to the regional mean. Upper panels show monthly Δ water level (treatment well minus regional mean) with era mean levels indicated by dashed horizontal lines. Lower panels show cumulative CUSUM trajectories. A rising CUSUM indicates the well is performing above the regional mean (shallower water table); a falling CUSUM indicates below-average performance (deeper water table). The accelerating negative CUSUM at both control wells confirms a system-wide deterioration trend independent of local management interventions. (Source: 09_scrape_05_tier1_background_drift.png)

Against this background of coastal erosion, CEH4 shows an additional secondary deterioration signal from December 2017 onwards consistent with the clearfell felling pulse, though this is superimposed on and inseparable from the longer-run erosion trend. These Tier 1 trajectories establish that both local controls are experiencing substantial background decline that must be accounted for before any scraping benefit can be assessed.

1.5.2 Paired BACI and Robustness

In this context, Tier 2 paired analysis shows a modest positive signal at CEH36 during the Pure Scraping era (April 2015 to December 2017), the only well and era combination to show a net benefit after Tier 1 correction. Three independent methods all return a positive scraping benefit: the two control-differenced methods agree closely — raw BACI vs CEH4 (+0.129 m) and synthetic control from 11 donor wells (+0.137 m) — while the SSM forward residual, an autonomous-forecast estimator and the most conservative of the three, is lower at +0.081 m — reflecting permanent ground surface lowering rather than a hydrodynamic deviation from the climate-driven trajectory (Figure 18). The cumulative CUSUM for CEH36 rises strongly from 2018 onwards, though this reflects the accelerating deterioration of CEH4 as much as any genuine improvement at CEH36, which is why the synthetic control method is critical to isolating the true scraping signal from CEH4's own trajectory (Figure 19). CEH18 sits consistently negative relative to CEH4 throughout the full record including the pre-scraping baseline, indicating that this well has always sat deeper than its control rather than having been driven down by the clearfell; only a modest recovery is visible after the October 2023 scraping. CEH21's rising CUSUM from 2022 onwards should be interpreted cautiously as its paired control CEH22 is itself deteriorating through coastal erosion, which inflates the apparent relative improvement at CEH21 independently of any scraping benefit; these results are noted for completeness only.

After Tier 1 correction for background drift, the net benefit at CEH36 reversed to -0.049 m during the Felling Pulse era, as the coastal erosion signal in CEH4 — compounded from 2018 by the secondary clearfell pulse — progressively eroded the scraping gain. The raw Tier 2 CUSUM tells a different story, with CEH36 rising strongly relative to CEH4 through this period — but as established above, this reflects CEH4's own deterioration rather than genuine improvement at CEH36. At CEH18 and CEH21, scraped in October 2023, no scraping-specific response survived Tier 1 correction for background drift. CEH18's corrected net benefit was -0.073 m after scraping; CEH21's paired control (CEH22) is itself on a coastal erosion trajectory, so a net benefit cannot be reliably estimated. The monitoring window for both wells is short (two post-scraping summers) and the analysis is likely underpowered to detect benefits of the magnitude observed at CEH36.

The annual summer minimum records for all three treatment wells, partitioned by management era and plotted against the Curreli et al. (2013) ecological thresholds, are presented in Figure 21. The progressive deepening of the CEH4 control well across all eras — from 1.16 m at baseline to 1.18 m during the Pure Scraping era and 1.40 m during the Felling Pulse — is consistent with the system-wide climate drying trend documented in Section 4.8.1 and confirms that management benefit must be assessed against this background trajectory rather than in absolute terms.

Tier 2 - Pure Scraping Signal (Paired CUSUM Analysis)

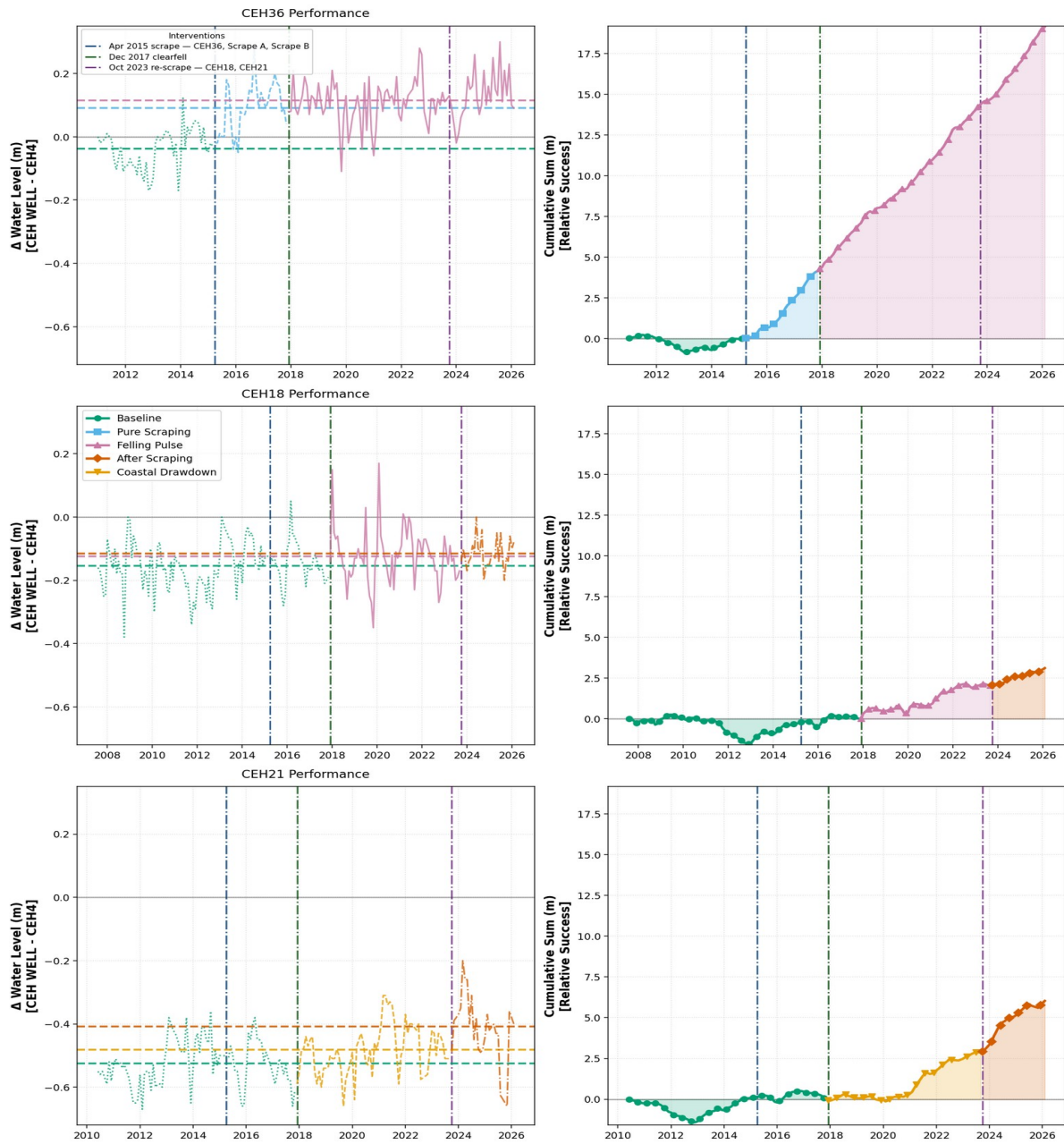


Figure 1.15: Tier 2 paired CUSUM analysis at treatment wells CEH36 (top), CEH18 (centre) and CEH21 (bottom), differenced against matched controls (CEH36 and CEH18 vs CEH4; CEH21 vs CEH22). Left panels: monthly Δ water level relative to control with era means as dashed lines. Right panels: cumulative BACI CUSUM — rising indicates the treatment well becoming shallower relative to control, falling indicates deepening. CEH36 shows a rising CUSUM from 2018 onwards, though this partly reflects the progressive deterioration of CEH4 rather than genuine improvement at CEH36. Neither CEH18 nor CEH21 shows a scraping response that survives Tier 1 correction; CEH21's apparent post-2022 rise reflects deterioration of its control (CEH22) through coastal erosion rather than a scraping benefit. (Source: 09_scrape_06_tier2_scraping_signal.png).

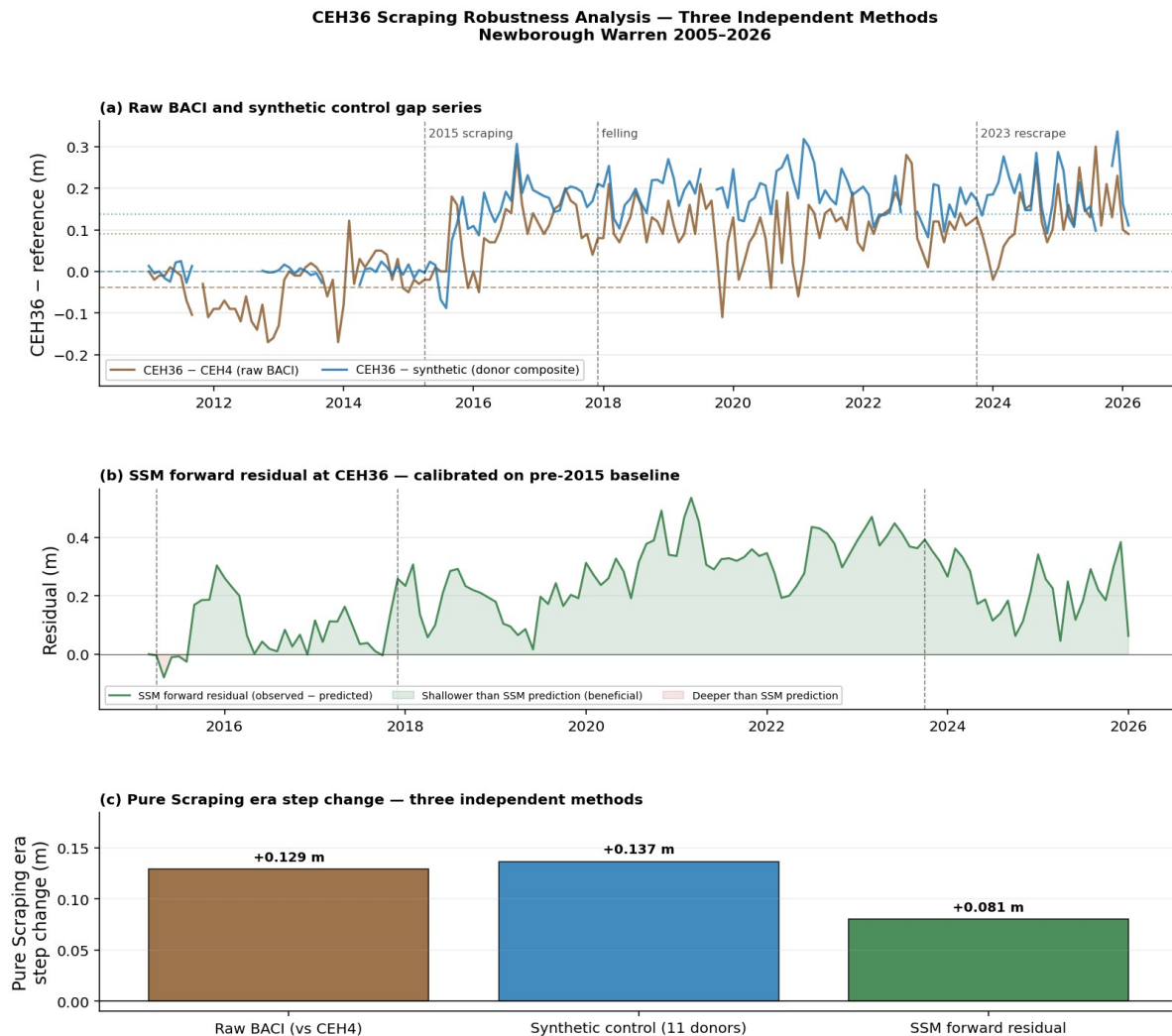


Figure 1.16: Three-method robustness assessment of the scraping signal at CEH36. (a) Raw BACI gap series (CEH36 minus CEH4, brown) and synthetic control gap series (CEH36 minus donor composite, blue), with intervention dates marked. The persistent positive offset after April 2015 confirms the scraping step change; the widening gap between the two series from 2018 onwards reflects the progressive deterioration of CEH4 through clearfell and coastal erosion effects. (b) SSM forward residual at CEH36, computed by calibrating the model on the pre-scraping period and running forward iteratively. Green shading indicates periods where CEH36 is shallower than the SSM prediction. The residual is persistently positive throughout the post-scraping record, confirming that the water table sits systematically above the level predicted by the pre-intervention model — consistent with a permanent structural benefit from scraping. The residual increases further post-clearfell (2018 onwards), suggesting the combined effect of scraping plus canopy removal exceeds the pre-intervention SSM baseline by 0.3–0.5 m in peak winters. (c) Pure Scraping era step changes from three independent methods: Raw BACI vs CEH4 (+0.129 m), synthetic control (+0.137 m), and SSM residual (+0.081 m). All three return a positive benefit, with the two control-differenced methods near 0.13 m. (Source: 09_scrape_08_ceh36_robustness.png)

1.5.3 Drainage Coefficient Response

Era-specific β_3 estimates show directional increases at both treatment and control wells (Table 6), CEH36 increased from 0.091 at baseline to 0.142 during the Pure Scraping era; CEH4, its unmanipulated control, showed a comparable increase from 0.071 to 0.164 over the same period. CEH18 and CEH21 show similar patterns against their respective controls (Figure 20). Because the β_3 increases extend to unmanipulated wells, they are more consistent with SSM fitting artefacts — arising from calibration windows straddling transitions between hydrological equilibria at wells undergoing long-run coastal erosion — than with a genuine drainage response to scraping. The per-well era estimates are reported in Table 6; they should not be interpreted as evidence of management-driven drainage change.

Well	Role	Era	β_3	95% CI	p-value	Sig
CEH36	Treatment	Baseline	0.091	[0.014, 0.167]	0.021	*
CEH36	Treatment	Pure Scraping	0.142	[0.062, 0.223]	0.001	**
CEH36	Treatment	Felling Pulse	0.124	[0.065, 0.183]	<0.001	***
CEH18	Treatment	Baseline	0.076	[0.038, 0.115]	<0.001	***
CEH18	Treatment	Felling Pulse	0.107	[0.025, 0.189]	0.011	*
CEH18	Treatment	After Scraping	0.187	[0.057, 0.318]	0.007	**
CEH21	Treatment	Baseline	0.106	[0.055, 0.156]	<0.001	***
CEH21	Treatment	Coastal Drawdown	0.060	[-0.001, 0.121]	0.055	ns
CEH21	Treatment	After Scraping	0.104	[-0.012, 0.220]	0.076	ns
CEH4	Control	Baseline	0.071	[0.028, 0.115]	0.002	**
CEH4	Control	Pure Scraping	0.164	[0.081, 0.246]	<0.001	***
CEH4	Control	Felling Pulse	0.122	[0.072, 0.173]	<0.001	***
CEH22	Control	Baseline	0.138	[0.085, 0.191]	<0.001	***
CEH22	Control	Coastal Drawdown	0.155	[0.077, 0.233]	<0.001	***
CEH22	Control	After Scraping	0.271	[0.113, 0.429]	0.002	**

Table 1.6: Era-specific β_3 storage decay coefficients at scraping treatment and control wells with 95% confidence intervals. Treatment wells (CEH36, CEH18, CEH21) shaded blue; control wells (CEH4, CEH22) shaded grey. β_3 increases at control wells reflect the system-wide hydraulic adjustment associated with the 2018 plantation clearance rather than scraping. Treatment-well era-specific β_3 estimates are plotted in Figure 20. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (Source: 09_scrape_04b_beta3_era_summary.csv).

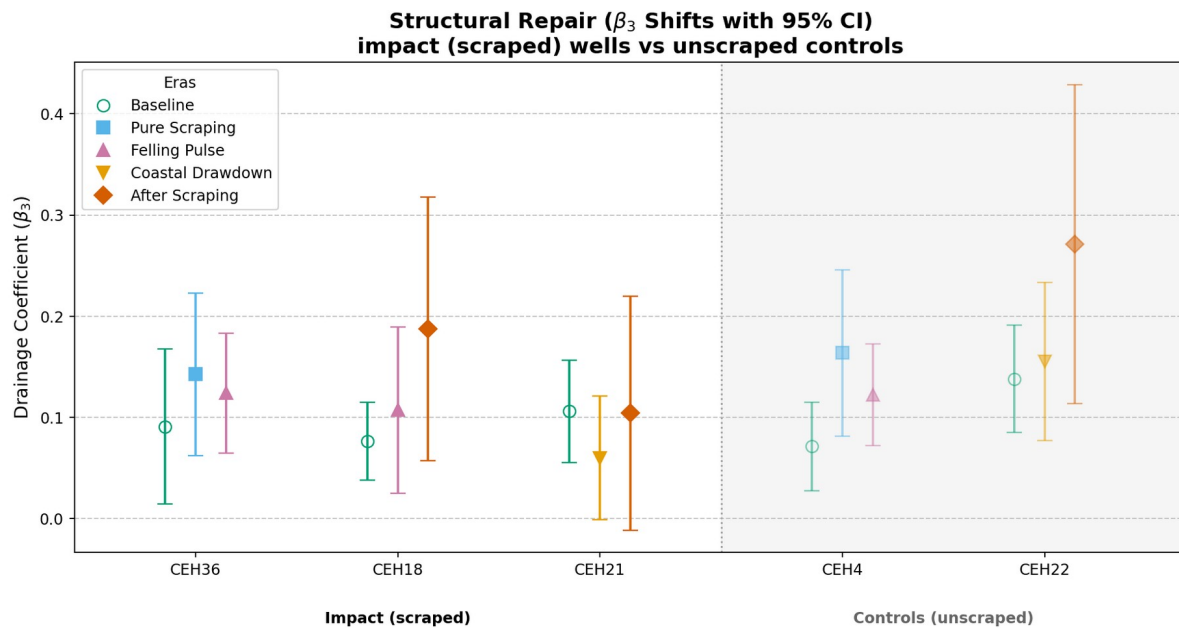


Figure 1.17: Era-specific β_3 storage decay coefficients with 95% confidence intervals at scraping treatment wells (CEH36, CEH18, CEH21) and their unscraped controls (CEH4, CEH22). Each symbol represents a management era: Baseline (hollow circle), Pure Scraping (filled square), Felling Pulse (triangle), After Scraping (diamond) and Coastal Drawdown (inverted triangle). Where confidence intervals overlap between eras, β_3 changes are not statistically distinguishable. CEH36 shows a directional increase from Baseline (0.091) through Pure Scraping (0.142) and Felling Pulse (0.124), though the confidence intervals overlap across all three eras. CEH18 shows a similar directional increase from Baseline (0.076) through Felling Pulse (0.107), with a further increase after the October 2023 scraping (0.187); the latter is significant ($p = 0.007$) though the confidence interval remains wide. CEH21's Coastal Drawdown estimate (0.060 ns) and After Scraping estimate (0.104, ns) are not significant. Control wells (CEH4, CEH22) are drawn faded, to the right of the dashed separator; their β_3 rises parallel the treatment wells, which is why the cross-era increase is not read as a scraping effect (full control values in Table 6). (Source: 09_scrape_07_beta3_confidence.png).

1.5.4 Summer Minima

The annual summer minimum depth records for the three scraping treatment wells and the CEH4 control well, partitioned by management era, are shown in Figure 21. CEH36 shows a clear reduction in summer minimum depth during the Pure Scraping era (April 2015 to December 2017), with the era mean shallowing from -1.115 m (pre-scraping, $n = 4$ summers) to -0.917 m (2015–17, $n = 3$ summers) — a raw improvement of 198 mm (Figure 21). This benefit partially erodes during the post-felling era (era mean -0.986 m), consistent with the site-wide drying trend compounding the clearfell pulse. Neither CEH18 nor CEH21 shows a clearly established summer minimum response to the October 2023 scraping — the post-rescape record comprises only two summers with high inter-annual variability, and CEH21's apparent improvement does not survive background correction against CEH22. The progressive deepening of the CEH4 control well across all eras (-1.075 m pre-scraping to -1.148 m during the Felling Pulse) confirms that management benefit must be assessed against the background coastal erosion and climate drying trajectory rather than in absolute terms.

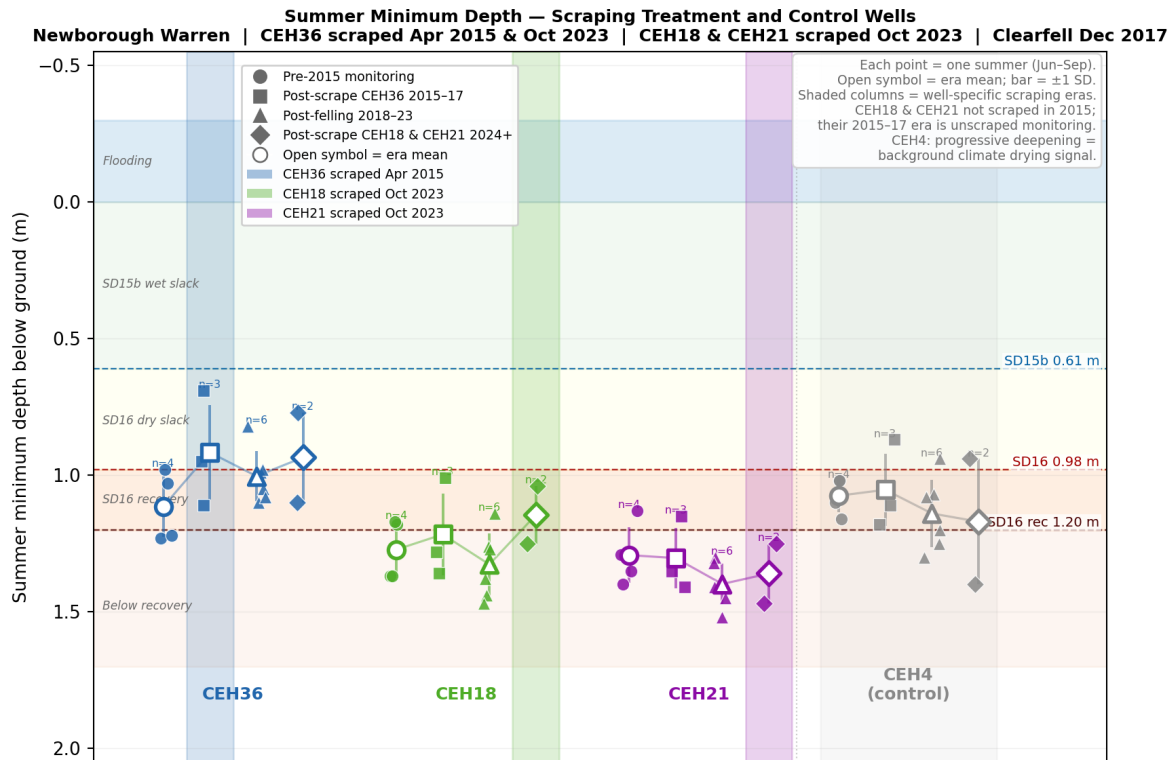


Figure 1.18: Annual summer minimum depth below ground surface at scraping treatment wells (CEH36, CEH18, CEH21) and control well (CEH4), partitioned by management era. Each point represents the mean annual summer minimum depth for one year (June–September). Open symbols show era means; vertical bars = ± 1 SD. CEH36 was scraped in April 2015 only (blue band, 2015–17 Pure Scraping era); CEH18 and CEH21 were scraped in October 2023 (green and purple bands, post-scraping era). Post-scraping era means for CEH18 and CEH21 are based on two summers (2024 and 2025) only. CEH18 and CEH21 were not scraped in 2015 — their 2015–17 records represent unscraped monitoring. CEH4 (grey) provides a climate-corrected control; its progressive deepening reflects primarily coastal erosion of the ground surface, compounded from 2018 by the clearfell pulse. Ecological zone colouring follows Curreli et al. (2013) thresholds: SD15b wet slack (0–0.61 m), SD16 dry slack (0.61–0.98 m), SD16 recovery (0.98–1.20 m). (Source: 21_forestry_03_scraping_eras.png).

The paired BACI summer minima analysis (09c) confirms the scraping benefit at CEH36 is statistically significant against the paired control CEH4 (+195 mm, $p = 0.004$) and also significant against the climate control centroid (+161 mm, $p = 0.006$). Neither CEH18 (+56 mm, $p = 0.20$) nor CEH21 (–8 mm, $p = 0.89$) showed significant paired shifts... CEH4 showed a non-significant change against the climate centroid (–34 mm, $p = 0.33$); the background drying trajectory is evidenced instead by the Tier-1 CUSUM decline and the era-mean progression (Section 4.8.1) (Figures 22, 23). The distributional view (Figure 33) confirms this pattern: Impact and Edge tier summer minima track the Forest Control trajectory without detectable divergence after felling, while the Coastal Control shows progressive deepening consistent with coastal erosion.

The same paired comparison on the five-year mean spring water level (MSL5; Section 4.8.4) corroborates the benefit on the baseline vegetation metric. MSL5 at CEH36 shallows from -0.72 m at window-end 2015 to approximately -0.50 m across window-ends 2017–2023, opening a sustained separation of roughly $+0.05$ to $+0.10$ m from the paired control CEH4 — which itself holds near -0.60 m and deepens late in the record under the coastal-erosion and drying trajectory. Because MSL5 is a rolling five-year mean, the scraping signal enters the metric gradually as post-scrape springs populate the window rather than as a discrete step, and part of the 2015–2017 shallowing is shared with CEH4 as the wet springs of that period enter both wells' windows; the sustained CEH36-minus-CEH4 separation that persists through the post-scrape record is, however, consistent in direction and approximate magnitude with the $+195$ mm paired summer-minimum benefit (Figure 23) and the $+129$ mm BACI-corrected monthly benefit (Section 4.5.2). The scraping benefit is therefore expressed across all three metrics — the monthly mean, the summer minimum, and the five-year spring baseline — in contrast to the clearfell, whose recovery is confined to the monthly mean and does not reach either ecological-threshold metric (Section 4.6.4).

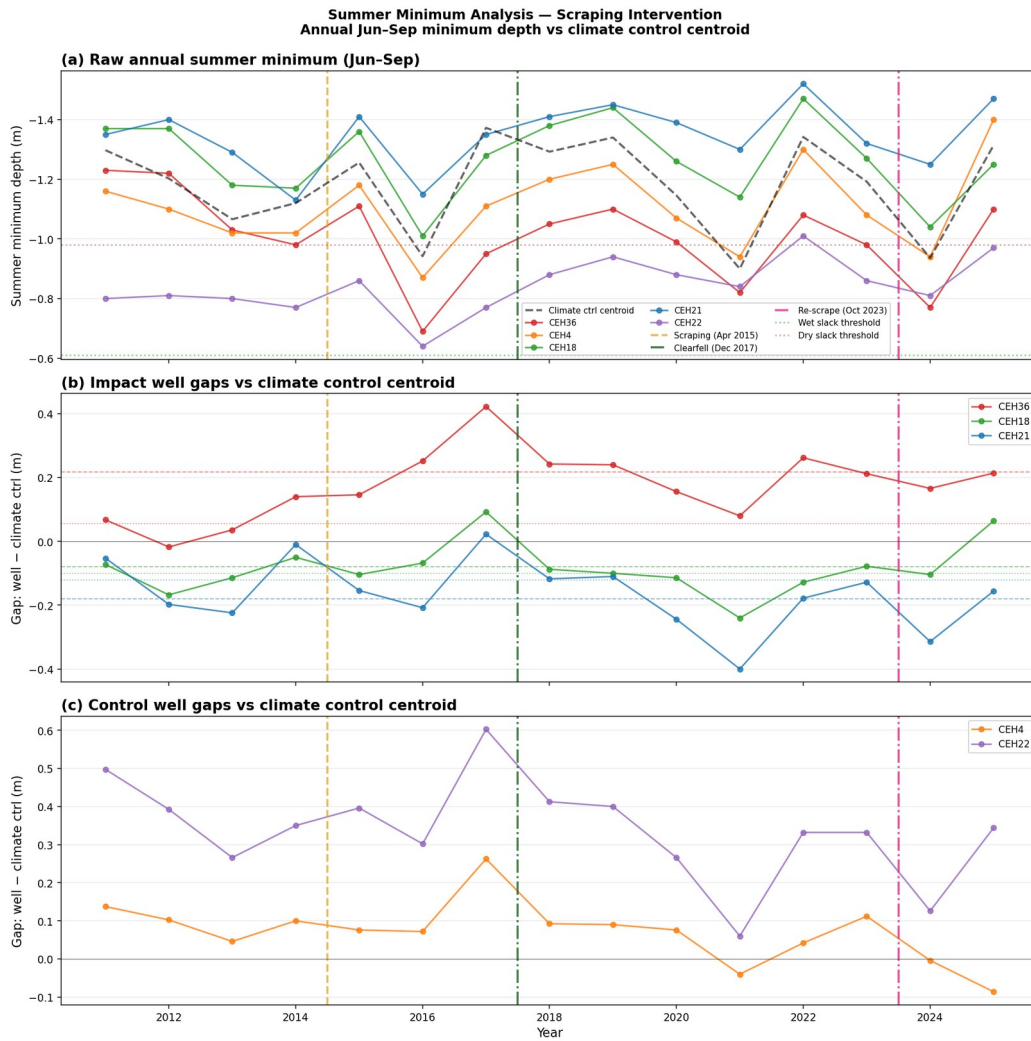


Figure 1.19: Summer minima analysis for the scraping intervention against the climate control centroid. (a) Raw annual summer minimum (June–September) for treatment and control wells. (b) Impact well gaps vs climate control centroid, pre/post scraping. (c) Control well gaps vs climate control centroid (diagnostic). (Source: 09c_03_summer_minima_climate_ctrl.png)

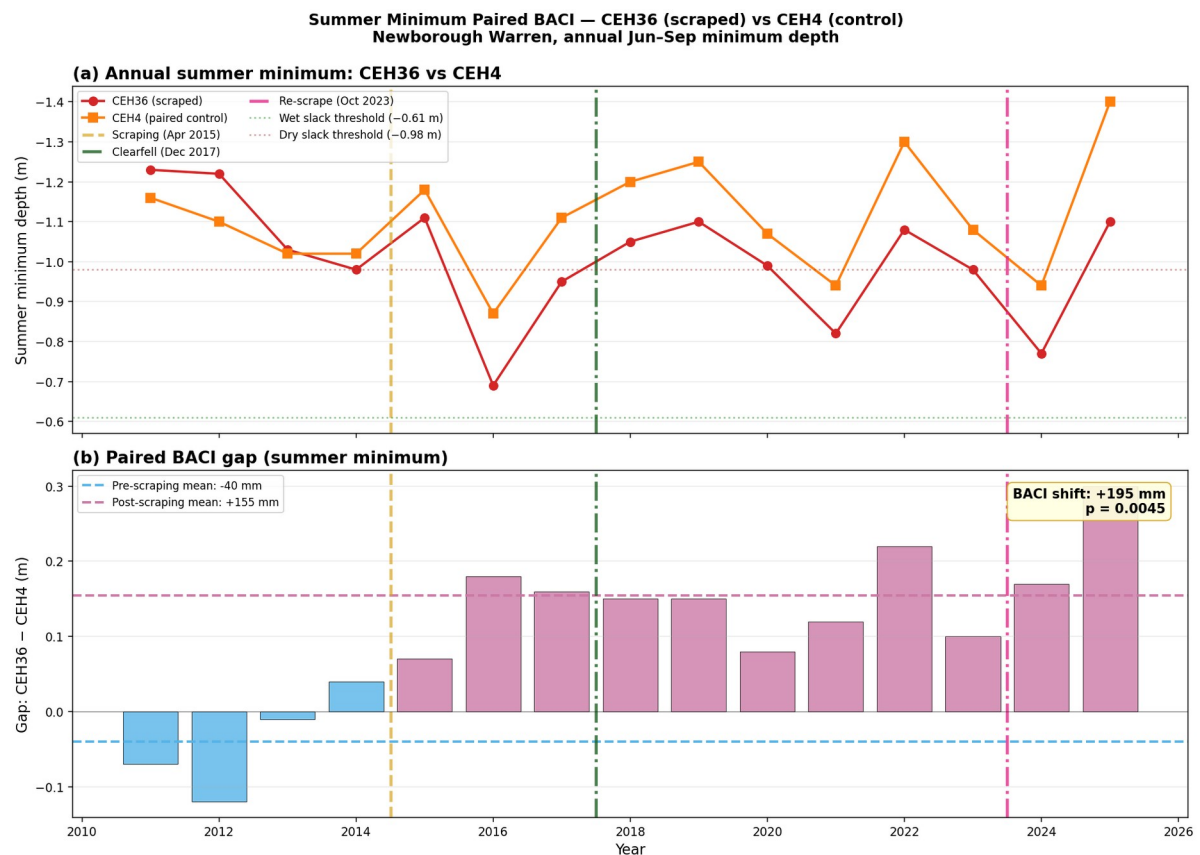


Figure 1.20: Paired BACI summer minimum analysis: CEH36 (scraped) vs CEH4 (control). (a) Annual summer minimum depth with Curreli et al. (2013) ecological thresholds. (b) BACI gap (CEH36 minus CEH4) by year, with pre- and post-scraping means. BACI shift: +195 mm, $p = 0.004$. (Source: 09c_04_summer_minima_paired.png).

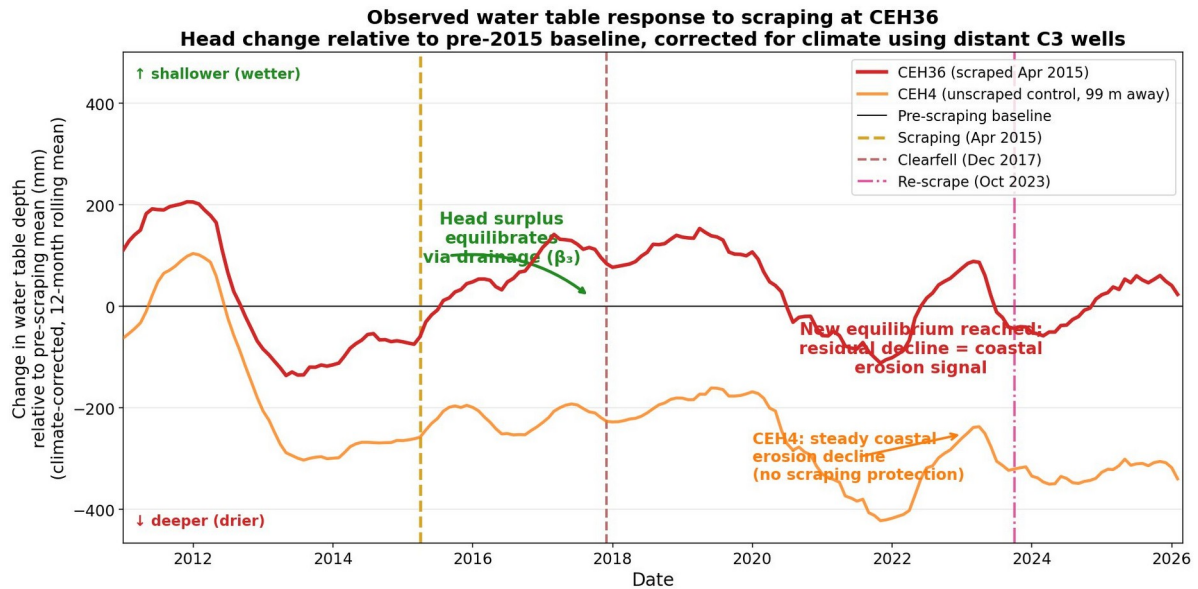


Figure 1.21: Climate-corrected anomaly trajectory at CEH36 (scraped) vs CEH4 (control), showing the divergence following April 2015 scraping and the protection of CEH36 against the coastal erosion trend visible in CEH4. (Source: 09b_03_ceh36_equilibration.jpg)

1.5.5 Scraping Propagation into the Surrounding Aquifer

A supplementary analysis tested whether the CEH36 scraping signal propagated uphill into the surrounding aquifer, comparing split-window SSM fits at ten wells between 250 m and 780 m from CEH36, with BACI correction against seven distant C3 control wells. Under a full three-coefficient refit, all eight wells within 575 m returned a positive BACI-corrected change in the drainage coefficient (mean $\Delta\beta_3 = +5.0 \times 10^{-3}$; one-sample $t = 4.98$, $p = 0.002$). Taken at face value this might suggest an uphill propagation of the scraping perturbation, but two independent lines of evidence indicate it does not represent a physical drainage response.

First, the uphill wells traverse documented pre-existing spatial gradients in the drainage coefficient that are unrelated to the scraping event. The wells nearer CEH36 lie within the C3 Western Residual cluster, which carries a significant internal β_3 gradient along the south-west to north-east axis (Section 4.9.2; β_3 increasing inland, $r = +0.58$, $p = 0.006$); the higher uphill wells lie within the C4 and C5 forest zone, where β_3 shows a significant elevation dependence (Section 4.9.5; $r = -0.77$, $p = 0.001$). A split-window fit comparing CEH36 against wells arranged along these gradients cannot separate a genuine propagating signal from the static spatial structure already present in the aquifer. Consistent with a spatial-structure origin, the BACI-corrected $\Delta\beta_3$ shows no decay with distance from CEH36 — the signature of a co-located gradient rather than a perturbation radiating from a point source.

Second, the aquifer diffusivity timescale places a physical bound on how far a genuine drainage perturbation could propagate within the observation window. With $L^2\text{Sy/K}$

exceeding seven years at a separation of 570 m (Fetter, 2001), a real drainage front could not traverse the uphill well array within the 31-month post-scraping record. The scraping benefit at CEH36 is therefore best understood as a direct geometric perturbation confined to the scraped well itself (Figure 25): the inferred ground surface lowering of approximately 0.42 m (back-calculated as $D = H_0/Sy$ from the measured BACI step $H_0 \approx 0.13$ m and open-dune $Sy \approx 0.31$) produces an immediate water-table shallowing that partially re-equilibrates over approximately two years, retaining +129 mm as the BACI-corrected benefit during the pure scraping era (Section 4.5.2). The SSM forward residual analysis (Figure 19b) confirms near-zero deviation during this period, indicating a permanent surface-lowering effect rather than a change in the aquifer's dynamic response properties.

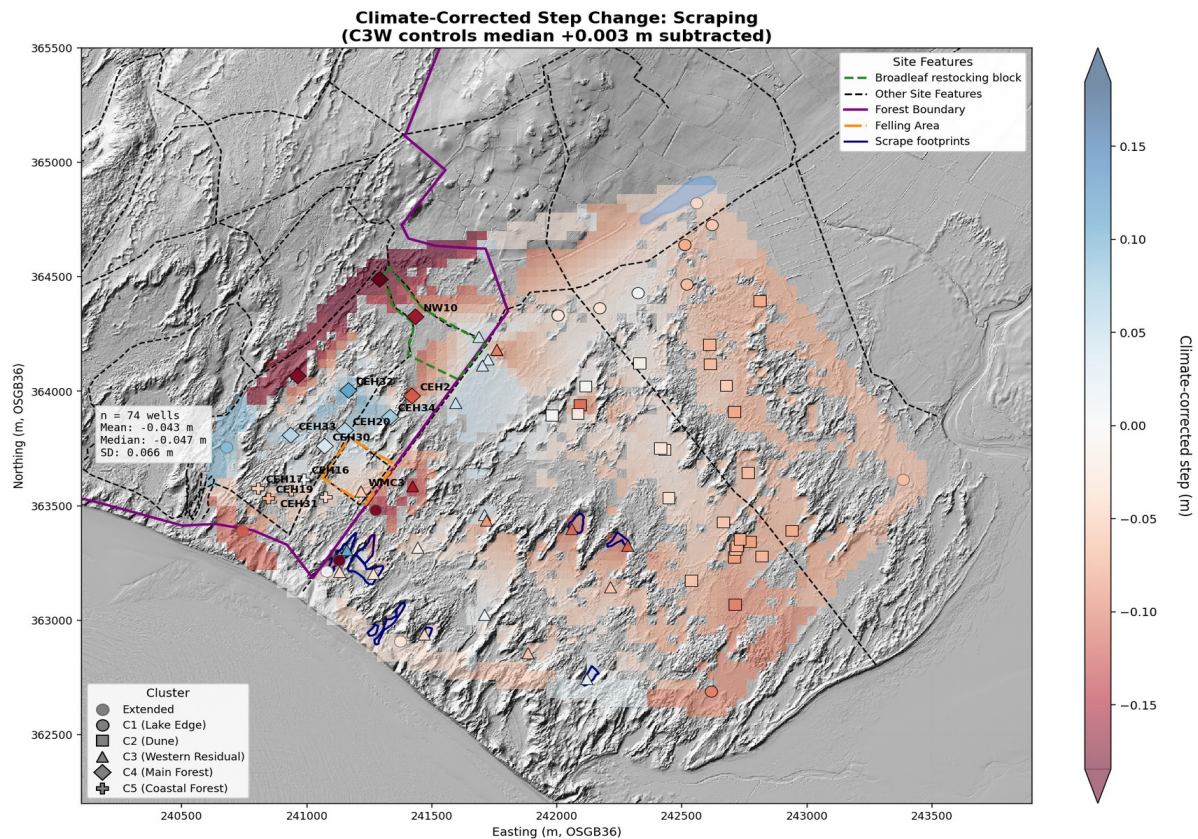


Figure 1.22: Climate-corrected spatial step-change map for the scraping era (April 2015 – November 2017). Each well's step relative to its pre-scraping mean, corrected by subtracting the C3 control centroid step. The localized response at CEH36 and absence of a spatial gradient uphill at well distances of 850–1100 m within the 31-month observation window are consistent with the scraping effect being either spatially confined or propagating on timescales longer than the available record (Section 5.4.3). (Source: 10b_spatial_scrape_corrected.png)

1.5.6 Scenario Contextualization

To contextualize the observed scraping benefit at CEH36, five alternative interventions were compared at the same well under two climate forcings (09d). Under annual-mean forcing (Figure 26), the observed scraping benefit (+45.2 mm w.e./month) substantially exceeds any alternative: clearfell +15.4, thinning +7.7, broadleaf +4.5, and climate wet +7.7 mm/month, against -13.8 mm/month under climate dry. Evaluated under summer forcing (Figure 27) the ranking is unchanged and the forestry gains are marginally smaller (clearfell +13.5, thinning +6.7, broadleaf +2.9), while the climate-dry penalty is marginally larger (-14.4); both figures report the same equilibrium volumetric quantity and are directly comparable. Both figures also show the modelled off-site neighbour drawdown the scrape imposes on the surrounding water table (-30.3 mm w.e./month at 100 m, decaying to -15.6 at 250 m): scraping raises the slack locally but draws its neighbours down, a dipole absent from canopy management. The existence and direction of this off-site drawdown — though not this specific modelled magnitude or currency — are corroborated by the measured, reproducible level fall at the nearest uphill well, WMC3 (Section 4.6.3). Even at a single well, scraping delivers an order-of-magnitude larger on-site response than canopy management, confirming that it operates through a fundamentally different mechanism — creating a topographic drain rather than modifying the surface energy balance. (For reference, the observed paired-BACI summer-minimum shift at CEH36 was +165 mm; the equilibrium framework does not resolve summer minima, so it is not plotted.)

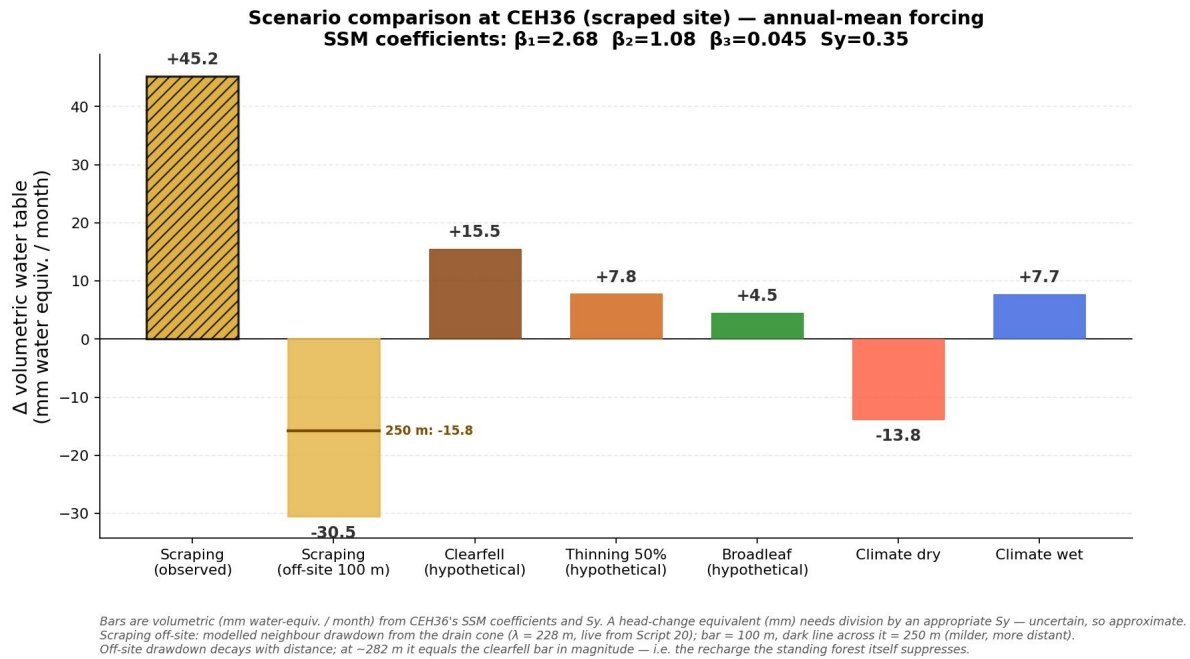


Figure 1.23: Scenario comparison at CEH36 (scraped site) — annual-mean forcing. Equilibrium volumetric water table change (mm water-equivalent per month) under each scenario. Observed scraping benefit (+45.2) compared with hypothetical forestry scenarios (clearfell +15.4, 50% thinning +7.7, broadleaf +4.5 — ‘what if CEH36 had pine canopy?’) and UKCP18 RCP8.5 climate projections (wet +7.7, dry -13.8). A second scraping bar shows the modelled off-site neighbour drawdown at 100 m (-30.3), with a reference line marking the milder drawdown at 250 m (-15.6), from the drain-cone model of Section 4.9.3. Broadleaf applies a seasonally varying β_2 adjustment (winter $\times 0.882$, summer $\times 1.075$). Converting a volumetric bar to a head change requires dividing by an appropriate, uncertain specific yield and is only approximate. All computed using CEH36's own SSM coefficients ($\beta_1 = 2.68$, $\beta_2 = 1.08$, $\beta_3 = 0.046$) and $S_y = 0.35$. (Source: 09d_01_scenario_comparison.jpg)

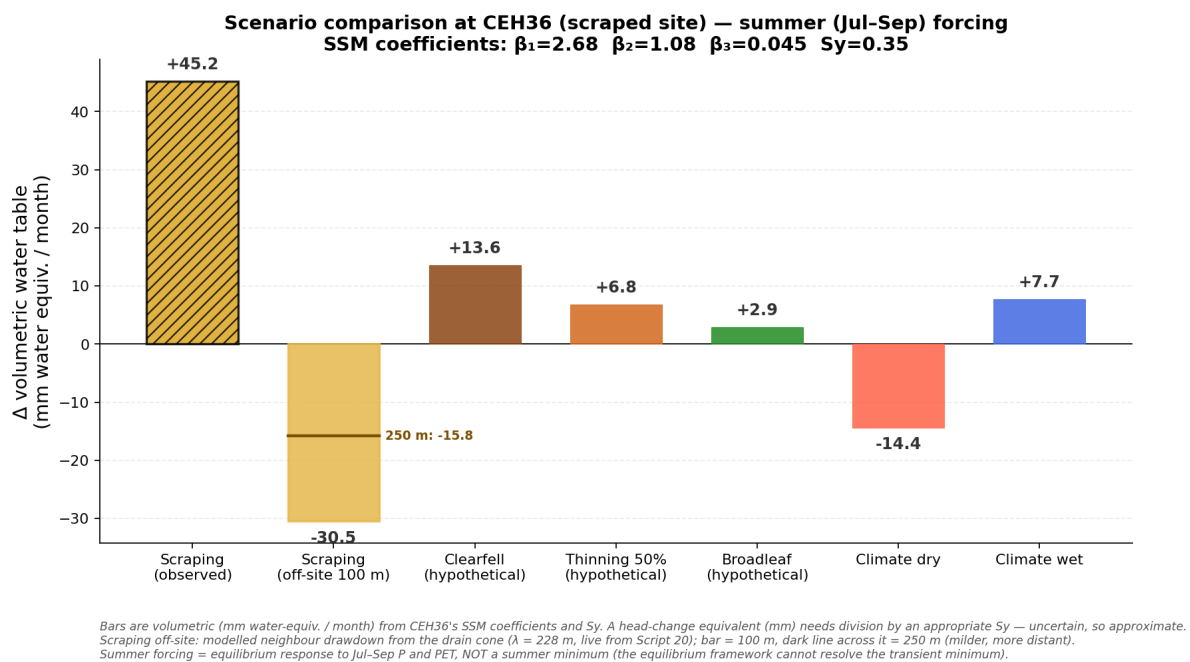


Figure 1.24: Scenario comparison at CEH36 (scraped site) — summer (July–September) forcing. As Figure 26 but with the SSM equilibrium evaluated under summer P and PET; bars are the same volumetric quantity (mm water-equivalent per month) and directly comparable to Figure 26. Observed scraping +45.2 (on-site) and -30.3 off-site at 100 m (reference line at 250 m, -15.6); clearfell +13.5, thinning +6.7, broadleaf +2.9, climate dry -14.4, wet +7.8. This is the equilibrium response to summer forcing, not a summer minimum, which the equilibrium framework cannot resolve; for reference, the observed paired-BACI summer-minimum shift at CEH36 was +165 mm. Broadleaf applies the seasonal β_2 adjustment of Section 4.10.2. As Figure 26, converting to a head change requires an appropriate specific yield and is approximate. (Source: 09d_02_summer_scenario_comparison.png)

1.6 The December 2017 Experimental Clearfell: Monitoring and Retrospective Analysis

1.6.1 Interpretive Context

Three features of the monitoring record bear on interpretation of the clearfell results. First, wells within the clearfell footprint (FE1–FE4, LIS1) and the scraping treatment well CEH36 were excluded from the BACI design on the grounds of insufficient or confounded pre-felling baselines (Section 3.5.4). FE1 and FE2, which have approximately 28 months of pre-felling record but no pre-scraping baseline, were excluded from the primary BACI design because their late start dates create an unstable impact centroid when pooled with WMC3 — the changing well composition introduces spurious level shifts that the ANCOVA conflates with the intervention steps. These wells were instead included as a robustness check via donor-regression synthetic extension, which provides a stable-composition centroid across all three eras (Section 4.6.7). Second, all clearfell analyses use the period from 2010 onwards as the effective baseline, because only three of nine C4 wells were operational before mid-2010 and these are unrepresentative of the cluster (Section 4.2.1). Third, the April 2015 scraping event at CEH36 produced a significant level shift 32 months before the clearfell; the ANCOVA's distance-weighted scraping covariate absorbs this, but the short pre-clearfell window encompasses only three summer minima, limiting baseline power.

1.6.2 Climate Sensitivity

The $CWB \times \text{clearfell}$ interaction was non-significant in the Forest control models (Impact $p = 0.40$, Edge $p = 0.16$), indicating that the rate at which the BACI gap responds to cumulative water balance deficit did not detectably change after felling when that response is modelled as a single linear slope. The relationship is shown in Figure 28: the fitted pre-felling and post-felling climate-response lines are effectively parallel, consistent with canopy removal having raised the mean water table level (Section 4.6.3) without altering the linear sensitivity of the system to climate forcing.

A linear slope is, however, a blunt instrument for the buffering hypothesis as it would most plausibly operate. If the canopy moderated the water table specifically against climate *extremes*, the effect would be concentrated at the wet or dry tails of the cumulative water balance range and could average close to zero across the full range, leaving a linear interaction undetectably small. To test this directly, a non-linear extension of the headline model was fitted, adding a centred CWB^2 term and a $CWB^2 \times \text{clearfell}$ interaction to the full ANCOVA specification. The $CWB^2 \times \text{clearfell}$ interaction is significant at both zones (Impact -27.5 mm per $(100 \text{ mm } CWB)^2$, $p = 0.016$; Edge -22.2 mm per $(100 \text{ mm } CWB)^2$, $p = 0.008$), the curvature model is preferred over the linear model by ΔAIC of -4.2 at Impact and -10.7 at Edge, and the joint F-test of the two added curvature terms is significant at both zones. The CWB^2 main effect is non-significant — the pre-felling climate response is linear; it is specifically the post-felling response that is curved. The negative interaction sign means

the post-felling response is concave in cumulative water balance: the felling-attributable rise in the water table is larger in dry, low-water-balance conditions than in wet ones.

This is reported as a preliminary finding rather than a confirmed buffering effect. It rests on a single interacted quadratic term and one felling event at one site, and a climate-state-dependent felling response has alternative explanations — post-felling non-stationary drift, or the coastal-retreat gradient — that this analysis does not fully exclude. It is also a sensitivity variant only: the headline clearfell step and the linear ANCOVA model of Section 4.6.3 are unchanged. With those caveats, the result indicates that the clearfell's effect on the water table was not uniform across climate conditions but concentrated in dry periods, which is the pattern a dry-season canopy-buffering mechanism would produce. The interpretation is developed in Section 5.

The Climate-control models — which contrast the felled wells against the unforested Western Block — return non-significant $CWB \times \text{clearfell}$ interactions (Impact $p = 0.55$, Edge $p = 0.28$). The Forest-control comparison is the position-matched test and its results, linear and non-linear, are the headline findings; the Climate-control interactions are not interpreted as a clearfell-specific change in canopy water-balance physics.

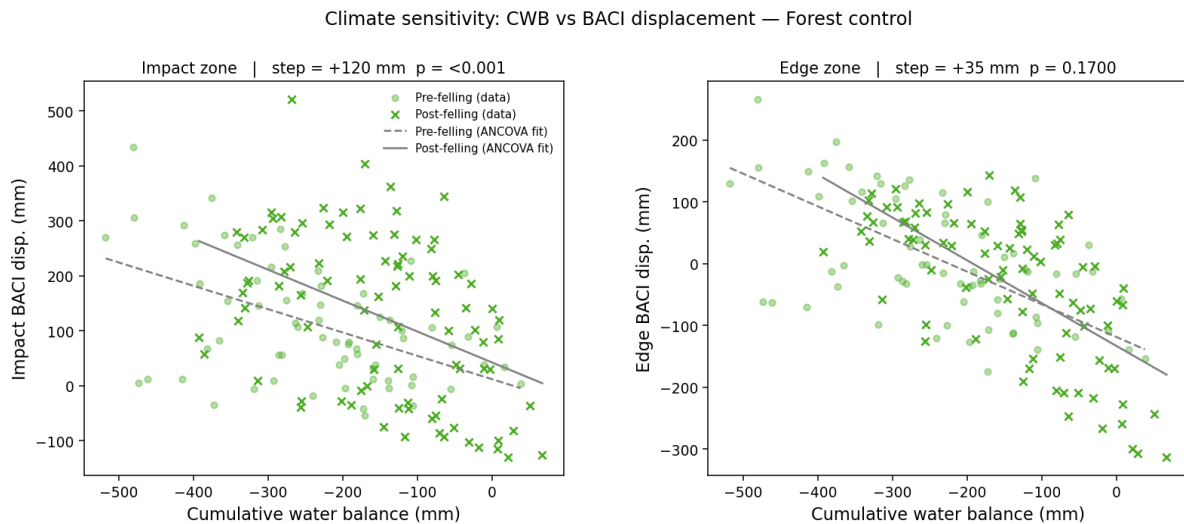


Figure 1.25: Climate sensitivity: cumulative water balance (CWB) vs BACI displacement against the Forest control, pre-felling (circles) and post-felling (crosses), for Impact (left) and Edge (right) zones. Solid and dashed lines are the ANCOVA-fitted climate-response lines for the post- and pre-felling eras respectively; their near-parallel form reflects the non-significant linear $CWB \times \text{clearfell}$ interaction. (Source: 10a_06_climate_sensitivity.png).

1.6.3 ANCOVA-BACI Results

The ANCOVA-BACI model (Section 3.5.4) was run three times with different control centroids (Table 7).

Control	Zone	Clearfell step (m)	95% CI (m)	p	Sig	Scraping step (m)	Scraping p	R ²	n
Forest	Impact	+0.120	[+0.050, +0.189]	<0.001	***	+0.344	0.002	0.274	162
Forest	Edge	+0.033	[−0.016, +0.082]	0.193	ns	+0.583	<0.001	0.484	158
Climate	Impact	−0.016	[−0.065, +0.034]	0.537	ns	+0.007	0.956	0.191	146
Climate	Edge	−0.112	[−0.174, −0.050]	<0.001	***	+0.191	0.633	0.219	143
Combined	Impact	+0.072	[+0.037, +0.108]	<0.001	***	+0.080	0.291	0.268	144
Combined	Edge	−0.022	[−0.049, +0.005]	0.117	ns	+0.287	0.021	0.451	141

Table 1.7: Three-counterfactual ANCOVA-BACI results for the December 2017 clearfell. Each row reports the clearfell step estimate from a separate ANCOVA model fitted to the monthly BACI gap (target tier centroid minus control centroid). Covariates: centred cumulative water balance (CWB), distance-weighted scraping dummy ($\lambda = 300$ m), CWB $\times$ clearfell interaction, and easting $\times$ time gradient. The Forest control is the most direct test of the clearfell hypothesis: it compares felled wells against unfelled forest sharing the same canopy and substrate, isolating the felling response from the coastal erosion gradient. All comparisons use a record-length-balanced window beginning January 2011 — the date from which every well in the five-tier comparison network has continuous record. The control centroid is computed on a fixed-membership basis: a month enters the BACI series only when every control-roster well has a reading, so the centroid is not silently re-weighted by control wells going offline mid-record (clearfell_common v1.7.0). Sample sizes therefore differ between comparisons. (Source: 10a_01_ancova_comparison_table.csv)

Against the Forest control, the clearfell step was +120 mm for Impact ($p < 0.001$) and +33 mm for Edge ($p = 0.19$) (Figure 30, Figure 31). The Impact step is statistically significant; the Edge step is not. Against the Climate control, the clearfell step was −16 mm for Impact ($p = 0.54$) and −112 mm for Edge ($p < 0.001$). Against the Combined control, the clearfell step was +72 mm for Impact ($p < 0.001$) and −22 mm for Edge ($p = 0.12$). The interpretation of the contrasting Forest, Climate and Combined results is taken up in Section 5.

The scraping covariate was significant in both Forest-control models: +344 mm at Impact ($p = 0.002$) and +590 mm at Edge ($p < 0.001$). Sensitivity analysis across three decay length scales ($\lambda = 200, 300, 500$ m) produced consistent clearfell coefficients, confirming that the scraping and clearfell terms are temporally well-separated. These covariate coefficients index the temporal separation of the two interventions in the model; they are not a physical estimate of the scraping effect on the impact well. In the raw, uncorrected BACI gap the two scraping events each *lower* the WMC3-minus-forest-control level — consistent with drainage of the neighbouring aquifer toward the lowered scrape surface — whereas the climate-corrected covariate carries the opposite sign because it is estimated jointly with the cumulative-water-balance and easting $\times$ time terms. Figure 29 shows the raw gap and its per-era steps directly.

The raw before-and-after evidence for all three interventions is shown in Figure 29. In the uncorrected WMC3-minus-forest-control gap, the December 2017 clearfell registers a +41

mm difference-in-differences step — the raw counterpart to the +120 mm climate-corrected ANCOVA step above, the two differing because the raw step retains the drier-than-average pre-felling climate the ANCOVA removes. The gap also steps downward at each scraping event, by a near-identical amount (–55 mm in 2015, –54 mm in 2023). The direction and the consistency across two independent events are what the drainage-drawdown mechanism would produce: the lowered scrape surface acts as a local drainage focus and draws down heads in the surrounding aquifer, including at WMC3 262 m from CEH36. This is a head-level effect, distinct from the drainage-coefficient ($\Delta\beta_3$) propagation tested in Section 4.5.5 — where WMC3 itself returned an essentially zero BACI-corrected $\Delta\beta_3$ — so the two observations are consistent: the well is drawn down in level without its intrinsic drainage response changing. The drainage response in the CEH36 area is rapid relative to the record — the fitted drainage half-life ($\ln 2 / \beta_3$) is 13–21 months at the wells within ~400 m of the scrape — so the local aquifer turns over roughly twice within the 31-month post-scraping window, fast enough for a scraping-induced drawdown to register at WMC3 within the record. The 2015 drawdown is the same scraping signal that confounds the pre-felling baseline of the surrounding within-footprint wells; WMC3 alone has a pre-2015 record that lets the scraping step be separated, which is why it is retained as the sole impact well while CEH36 and the FE wells are excluded (Section 4.6.1). The steps remain uncorrected for differential climate sensitivity between WMC3 and the forest control, so they are presented as consistent with scraping drawdown rather than as a confirmed magnitude.

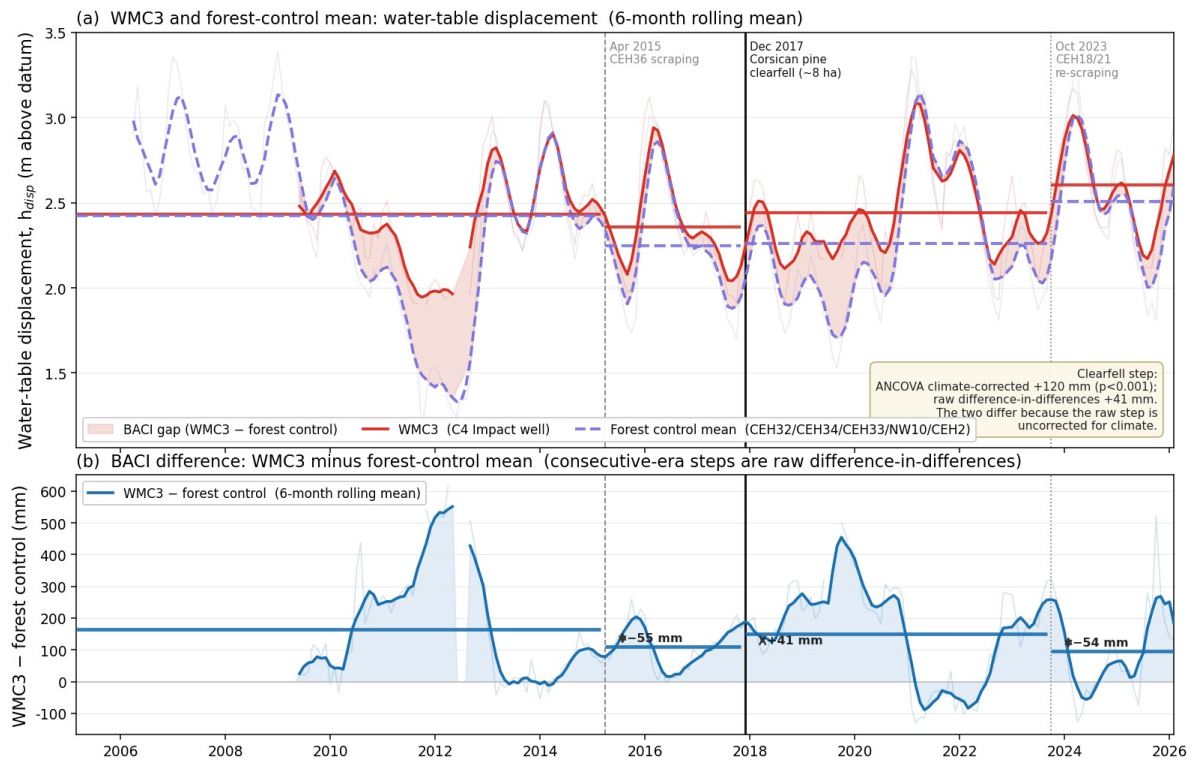


Figure 1.26: WMC3 (Impact well) and the forest-control mean across the three interventions, in water-table displacement ($h_{disp} = \text{datum} + \text{depth}$). (a) Six-month rolling means (bold) over faint monthly values, with the WMC3-minus-forest-control gap shaded and per-era mean bars; vertical lines mark April 2015 scraping (grey dashed), December 2017 clearfell (black solid) and October 2023 re-scraping (grey dotted). The boxed note reconciles the raw and climate-corrected clearfell steps. (b) The BACI gap (WMC3 minus forest control) as a six-month rolling mean over faint monthly values, with per-era mean bars; annotations give the consecutive-era difference-in-differences steps. The two scraping events each lower the gap (-55 mm in 2015, -54 mm in 2023), consistent with drainage toward the lowered scrape surface; the clearfell raises it (+41 mm). These raw difference-in-differences steps are uncorrected for climate and are not a substitute for the climate-corrected ANCOVA step of Section 4.6.3. (Source: 10m_02_wmc3_baci_dual.png)

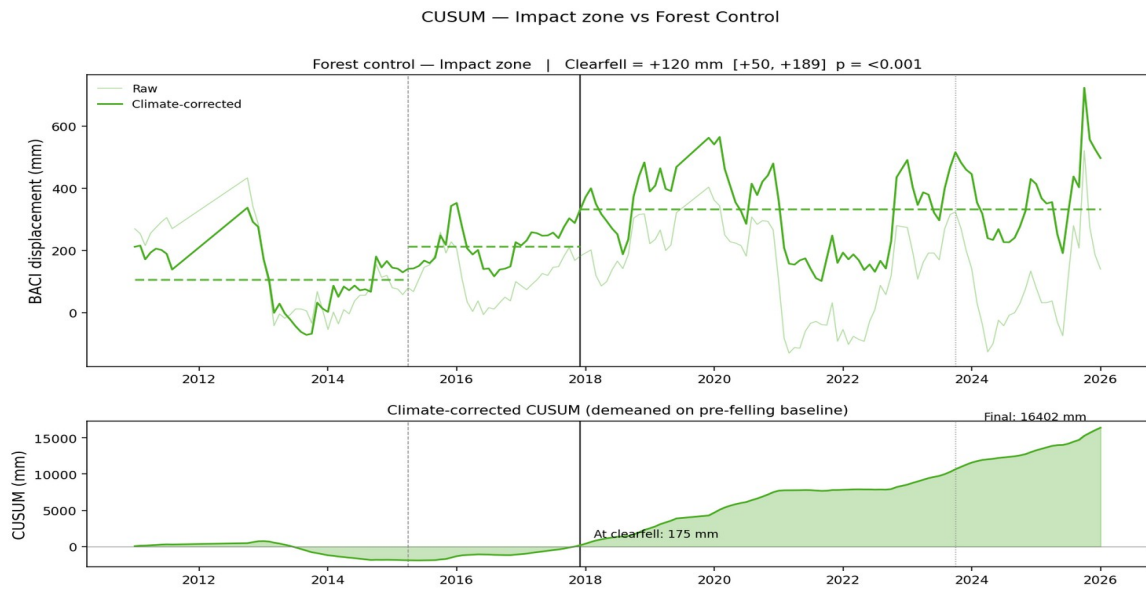


Figure 1.27: Forest control BACI timeseries and CUSUM for the Impact tier (WMC3). Upper panel: monthly BACI gap (WMC3 minus Forest control centroid) with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; sustained positive accumulation after December 2017 confirms the clearfell step. Vertical lines: April 2015 scraping (grey dashed), December 2017 clearfell (black solid), October 2023 re-scraping (grey dotted). (Source: 10a_07_cusum_impact.png).

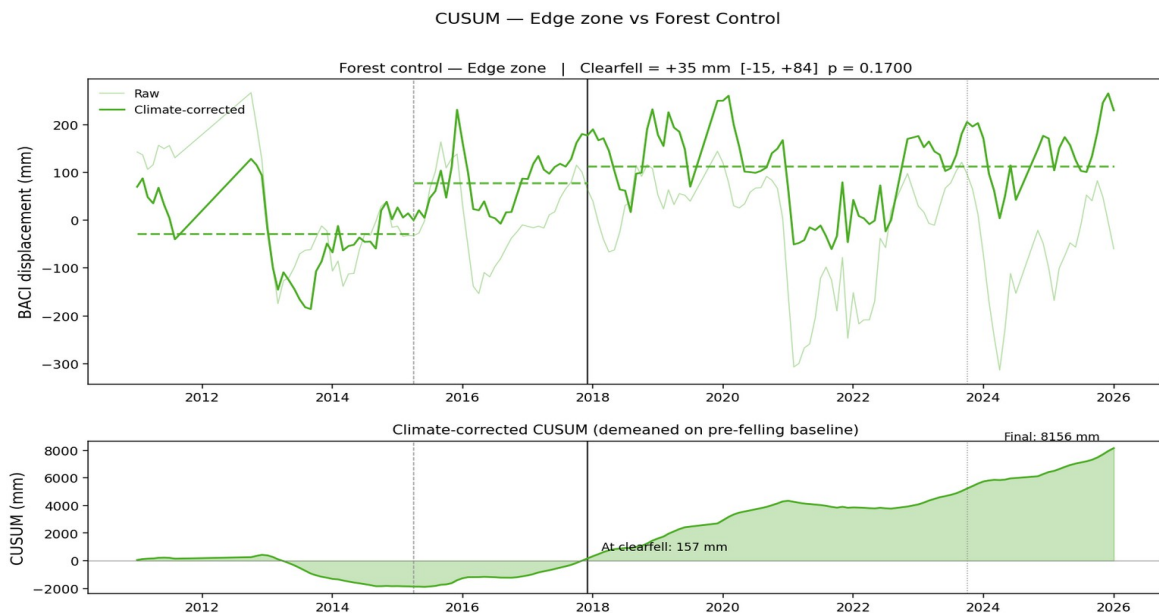


Figure 1.28: Forest control BACI timeseries and CUSUM for the Edge tier (CEH31, CEH20, CEH30, CEH16). Upper panel: monthly BACI gap (Edge centroid minus Forest control centroid) with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; sustained positive accumulation after December 2017 confirms the clearfell step. Vertical lines: April 2015 scraping (grey dashed), December 2017 clearfell (black solid), October 2023 re-scraping (grey dotted). (Source: 10a_08_cusum_edge.png)

The easting $\times$ time covariate in Table 7's ANCOVA design absorbs the differential coastal-retreat drift between impact and control wells across the post-felling era. The validity of this correction can be independently checked against the network-scale coastal-retreat gradient model fitted in Section 4.8.1. For each impact zone $\times$ control tier pair, the gradient model predicts a coastal-retreat-attributable differential deepening between the impact and control mean coast-distances, against which the BACI's fitted easting $\times$ time coefficient implies an absorbed differential (Table 8). For the headline Forest Impact comparison the two are consistent within sampling error: the BACI absorbs $-16.8 \pm 5.0 \text{ mm yr}^{-1}$ against an independent model prediction of -11.1 mm yr^{-1} ($z = -1.16$; consistent at the $|z| < 2$ threshold), supporting the reading that the +120 mm Impact-tier clearfell step is not a residual of incomplete gradient correction. The Forest Edge comparison is also consistent (BACI absorbs $-11.2 \pm 4.5 \text{ mm yr}^{-1}$, model predicts -10.0 mm yr^{-1} , $z = -0.28$). The two Climate-tier comparisons are less cleanly resolved — the Climate Impact pair is not consistent ($z = +2.17$) and the Climate Edge pair is marginal ($z = +1.96$) — reflecting the much larger easting separations of the Climate-control wells, across which the easting $\times$ time term absorbs spatial drift in components other than coastal retreat; the implications are discussed in Section 5.5.

Control tier	Impact zone	ΔE (m)	BACI absorbs (mm yr ⁻¹)	Gradient model predicts (mm yr ⁻¹)	z	Consistent?
Forest	Impact	-41	-16.8 ± 5.0	-11.1	-1.16	yes
Forest	Edge	-187	-11.2 ± 4.5	-10.0	-0.28	yes
Climate	Impact	-418	$+10.0 \pm 5.0$	-0.9	+2.18	no
Climate	Edge	-563	$+13.5 \pm 6.8$	+0.2	+1.96	yes

Table 1.8: Independent corroboration of the BACI ANCOVA's easting $\times$ time covariate against the network-scale coastal-retreat gradient model of Section 4.8.1. ΔE is the easting separation between impact and control centroids (m). "BACI absorbs" is the differential deepening absorbed by the easting $\times$ time coefficient, computed as coefficient $\times \Delta E \times 12 \text{ months} \times 1000 \text{ (m} \rightarrow \text{mm)}$, with ± 1 SE propagated from the coefficient SE. "Gradient model predicts" is the difference in fitted gradient (Section 4.8.1; $\delta_0 = -29.0 \text{ mm yr}^{-1}$, $L = 894 \text{ m}$) evaluated at the impact and control mean perpendicular distances to the western foreshore. The Forest Impact comparison agrees within sampling error; the Edge and Climate-tier comparisons absorb additional non-coastal-retreat drift discussed in Section 5.5. (Source: 25_04_baci_corroboration.csv)

Four-zone pooled-panel cross-check. The three-counterfactual ANCOVA above fits the felling step six times — once per (control, zone) pair — and each fit has its own dependent variable, because the BACI gap is the target centroid minus a control centroid and the control centroid differs between the Forest, Climate and Combined configurations. The six steps therefore do not share a covariance matrix and are not subtractable. As a cross-check, a four-zone pooled-panel model (Script 10k) was fitted: the Forest control, a second western-dune control (C3/Warren), the Edge tier and the Impact well are stacked into one long-form panel, and every zone contrast is read off a single internally-consistent joint fit. This

supplements, and does not replace, the three-counterfactual ANCOVA, which remains the principal clearfell analysis.

The joint fit returns three primary zone-versus-Forest monthly contrasts: Impact–Forest $+0.032$ m (95% CI $[+0.010, +0.053]$, $p = 0.004$), Edge–Forest -0.030 m ($p = 0.13$, ns), and C3/Warren–Forest $+0.005$ m ($p = 0.65$, ns); joint model $R^2 = 0.856$, $N = 2475$ well-months. The four-zone result is consistent with the principal analysis. The joint-fit Impact–Forest step ($+0.032$ m) is smaller in magnitude than the three-counterfactual Forest $\times$ Impact step ($+0.120$ m); the two estimators partition climate variance differently — the joint panel estimates climate sensitivity from the full cross-zone monthly record, so a portion of the monthly variance the separately-fitted ANCOVA attributes to the felling step is attributed to climate in the joint fit. The felling response is positive and statistically significant under both estimators, and only the magnitude is sensitive to the partitioning choice. Dropping the easting $\times$ time covariate moves the joint-fit Impact step only from $+0.032$ m to $+0.033$ m, significance unchanged.

The Impact–Edge difference of $+0.061$ m is an exact arithmetic combination of two primary contrasts; its standard error is covariance-dependent and not comparable to a primary contrast, so it is not cited with a p-value here. It is instead corroborated independently by the two-zone Impact-versus-Edge estimator (Script 10j), which returns a monthly Impact–Edge step of $+0.063$ m ($p = 0.0002$) — the close agreement between the two estimators is the warrant for the contrast.

1.6.4 Summer Minima

The annual summer minimum depth (June–September) — the ecologically critical metric determining dune slack habitat viability (Curreli et al. 2013) — was analysed for each well in the 17-well network. Four complementary analyses are reported: per-well Welch t-tests on the annual summer-minimum gap (Table 9); a mixed-effects model pooling within-tier (Table 10); a directly-fitted Jun–Sep ANCOVA on monthly observations, using the same three-counterfactual specification as the annual analysis (Section 3.5.4) restricted to the June–September window; and a four-zone pooled-panel cross-check fitted on the annual summer-minimum series (Section 3.5.4). The two ANCOVA-based analyses differ in estimator and in unit of observation: the Jun–Sep ANCOVA estimates the felling step from monthly observations as a differenced centroid, while the four-zone model estimates every zone contrast from a single joint fit on annual summer-minimum well-years — the annual deepest point rather than the summer-window mean.

Per-well summer-minimum shifts. Pre- vs post-felling Welch t-tests on the annual summer-minimum gap (well minus Forest control centroid) found no significant clearfell effect at any well in the Impact or Edge tier (Table 9). The Impact well WMC3 showed a shift of -7 mm ($p = 0.91$). Individual Edge well shifts ranged from -9 mm at CEH30 to -97 mm at CEH16, all non-significant.

Well	Tier	Control	n pre	n post	Pre gap (m)	Post gap (m)	Shift (mm)	p	Sig
WMC3	Impact	Forest	6	7	+0.127	+0.120	−7	0.905	ns
CEH31	Edge	Forest	7	8	−0.075	−0.132	−56	0.581	ns
CEH20	Edge	Forest	7	7	+0.122	+0.111	−10	0.739	ns
CEH30	Edge	Forest	7	8	−0.311	−0.320	−9	0.859	ns
CEH16	Edge	Forest	7	8	+0.603	+0.506	−97	0.297	ns

Table 1.9: Per-well pre- vs post-felling shifts in annual summer minimum depth gap (well minus Forest control centroid), with Welch t-test. (Source: 10d_02_summer_minima_shifts.csv)

Pooled mixed-effects model. A mixed-effects model with random intercept per well and fixed effects for post-felling and post-scraping indicators, fitted to the same Forest-control centroid gap series (Table 10), produced non-significant clearfell-step estimates across all five tiers. The Forest × Edge pooled estimate was −64 mm ($p = 0.12$); the Forest × Coastal Control estimate of −176 mm ($p = 0.06$) sits just outside conventional significance.

Control	Tier	Model	Clearfell coef (m)	SE	p	n obs
Forest	Impact	OLS (single well)	−0.001	0.078	0.991	13
Forest	Edge	Mixed-effects	−0.064	0.042	0.123	59
Forest	Forest Ctrl	Mixed-effects	−0.008	0.016	0.618	72
Forest	Coastal Ctrl	Mixed-effects	−0.176	0.093	0.059	29
Forest	Climate Ctrl	Mixed-effects	−0.046	0.057	0.425	69

Table 1.10: Mixed-effects model results: pooled clearfell step by tier against Forest control, with random intercept per well, except the single-well Impact tier which is fitted by OLS. (Source: 10d_03_mixed_model_results.csv).

Directly-fitted Jun–Sep ANCOVA. A directly-fitted summer-window ANCOVA was run on the same five-tier specification as the annual analysis (Section 3.5), restricted to June–September observations ($n = 52$ months). Against the Forest control, the Jun–Sep clearfell step at the Impact tier was +46 mm (95% CI [−69, +162], $p = 0.44$), with $R^2 = 0.34$. A sensitivity variant dropping the CWB covariate and the CWB × clearfell interaction from the design matrix produced an estimate of +123 mm (95% CI [−1, +247], $p = 0.06$), with $R^2 = 0.10$. The contrast between the full-specification and CWB-dropped estimates is discussed in Section 5.

Distributional comparison. Figure 33 shows the distribution of summer minima across the five BACI tiers across three phases: pre-scraping baseline (2007 to April 2015), the 2015–17

era, and post-felling 2018 onwards. The Forest Control group shows stable summer minima throughout. The Impact and Edge tiers track the Forest Control trajectory across the three phases. The Coastal Control tier (CEH19, CEH17) shows progressive deepening across all three phases.

Four-zone pooled-panel cross-check (summer minimum). Whereas the directly-fitted Jun–Sep ANCOVA above estimates the felling step from monthly observations in the Jun–Sep window as a differenced centroid, the four-zone pooled-panel model (Section 3.5.4) estimates every zone contrast from a single joint fit on the annual summer-minimum series (Script 10l) — the annual deepest point rather than the summer-window mean. The three primary zone-versus-Forest contrasts were Impact–Forest -7 mm (95% CI $[-23, +8]$, $p = 0.36$), Edge–Forest -29 mm ($p = 0.16$) and C3/Warren–Forest $+12$ mm ($p = 0.26$); joint model $R^2 = 0.724$, $N = 181$ well-years. No zone contrast resolves a significant differential felling response at the summer minimum.

The four summer-window analyses converge: the per-well Welch tests, the pooled mixed-effects model, the directly-fitted Jun–Sep ANCOVA and the four-zone pooled-panel cross-check all return non-significant clearfell steps at the Impact and Edge tiers. The felling produced a measurable rise in the felled compartment's monthly-mean water table (Section 4.6.3) but no detectable change at the summer minimum — the driest and most ecologically critical point of the year, and the level against which the Curreli SD15b and SD16 slack thresholds are defined.

Five-year mean spring water level. The same absence of a discrete felling response is seen in the five-year mean spring water level (MSL5; Section 4.8.4), the baseline metric most directly tied to dune-slack vegetation (van Willegen et al. 2025). At the Impact well WMC3, MSL5 carried no step at the December 2017 felling: the window-end values deepen through the post-felling years (-1.00 m at window-end 2018, -1.12 m at 2019, holding near -1.05 m through 2023) before recovering to -0.90 m at window-end 2024. The four Edge wells (CEH16, CEH20, CEH30, CEH31) show the same pattern — flat to slightly deepening from 2018 to 2023, then a recovery first appearing at window-end 2024. The 2024 recovery is not clearfell-attributable: it appears synchronously across the felled wells, the unfelled Forest control, and every other cluster in the network (Section 4.8.4, Figure 43), six years after the intervention, and reflects the 2023–24 hydrological year entering the rolling five-year window rather than a delayed canopy-removal response. On the baseline metric, as on the summer minimum, the clearfell produced no detectable step against the background climate signal (Figure 32 shows the summer-minimum result by tier). The mean-monthly recovery established by the ANCOVA (Section 4.6.3) is thus specific to the monthly-mean level; it is not expressed in either of the two ecologically referenced metrics — the summer minimum or the five-year spring baseline — against which dune-slack viability is assessed.

Summer minima analysis — Forest control centroid

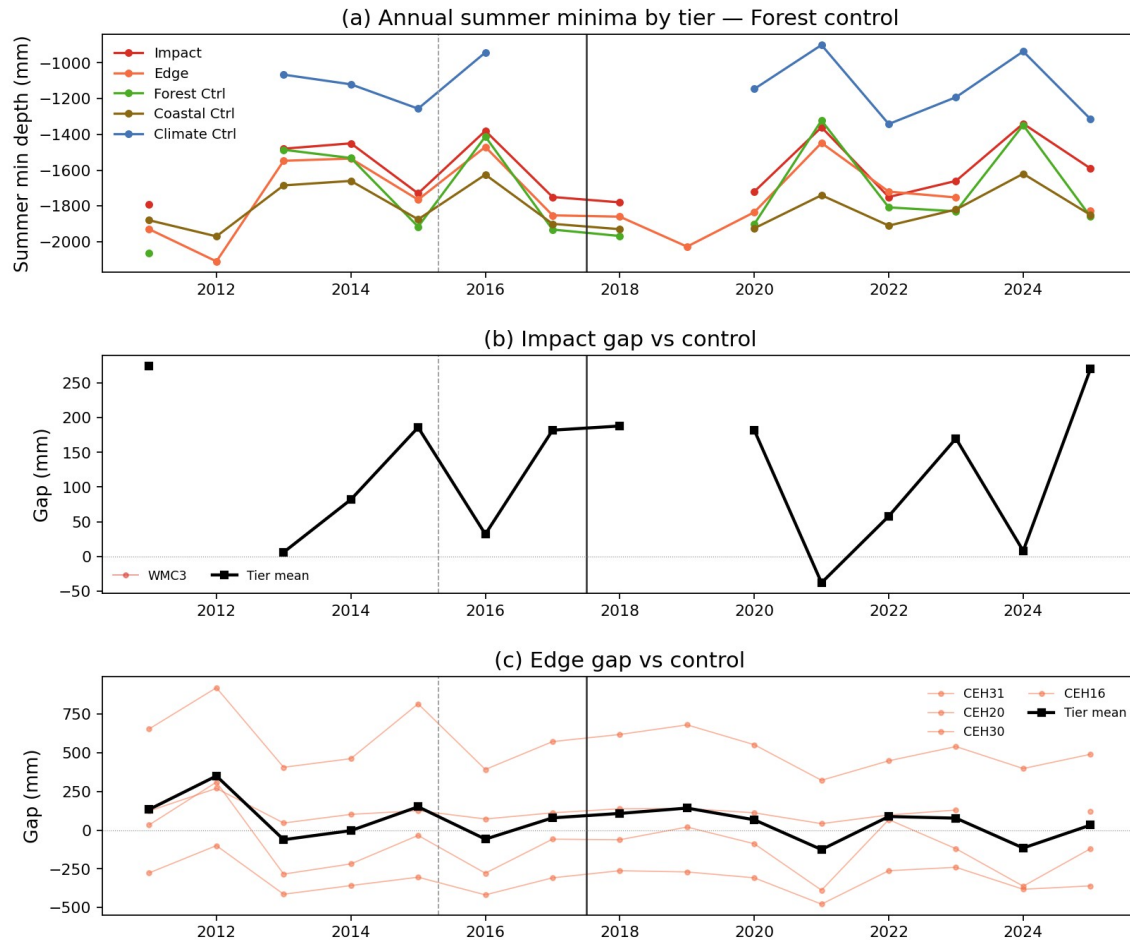


Figure 1.29: Summer minima analysis against the Forest control. (a) Raw annual summer minimum depth (June–September) by tier. (b) Impact well gap (WMC3 minus Forest control centroid), pre- vs post-felling. (c) Edge well gaps vs Forest control centroid. No significant clearfell effect on summer minima was detected at the Impact or Edge tier; the mixed-effects Edge estimate (-64 mm, $p = 0.12$) is non-significant. (Source: 10d_04_summer_minima_forest_ctrl.png).

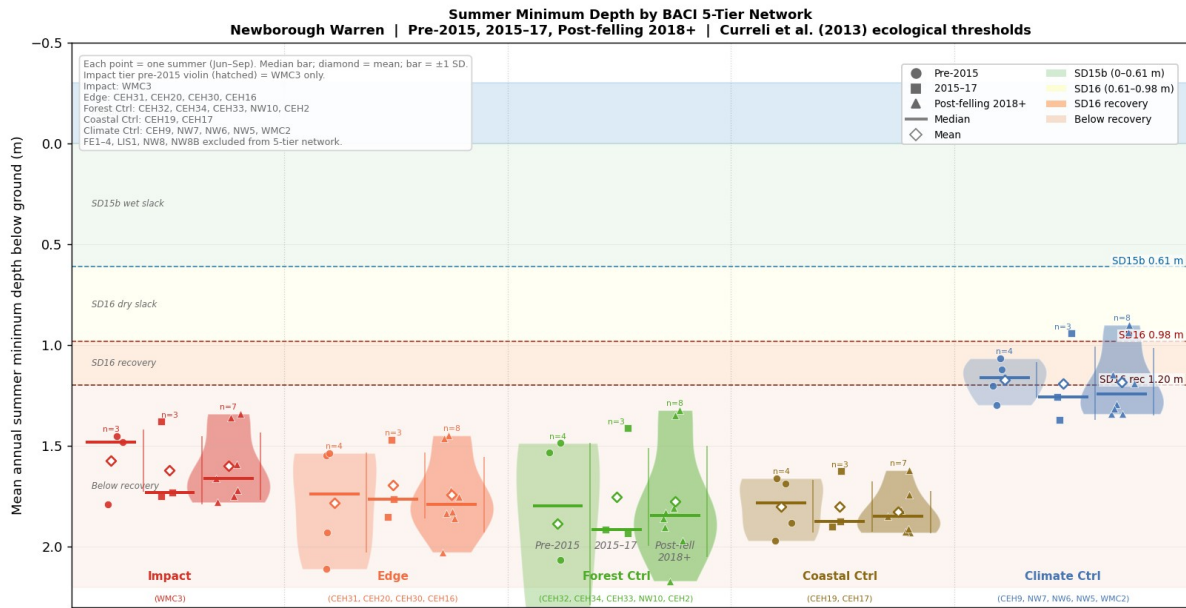


Figure 1.30: Summer minimum depth distributions by BACI tier, Newborough Warren 2007–2026. Three phases: pre-scraping baseline, 2015–17 era, post-felling 2018+. Violin width indicates frequency; diamond = mean; bar = median. The Forest Control group shows stable summer minima throughout. Both Impact and Edge tiers track the Forest Control trajectory, with no detectable divergence in summer extremes after felling despite the significant positive shift in the monthly ANCOVA (Section 4.6.3). The Coastal Control tier shows progressive deepening consistent with coastal erosion. Ecological thresholds from Curreli et al. (2013). (Source: 21_forestry_04_baci_zone_violin.png).

1.6.5 Spatial Step-Change Maps

Climate-corrected step-change maps for the clearfell era are shown in Figure 34. Each well's step is computed relative to its own pre-intervention mean, corrected by subtracting the C3 open-dune control centroid step to remove the background climate signal. This correction removes shared climate forcing but does not account for the coastal erosion gradient, which produces a systematic east–west trend in the corrected steps. The map shows spatially uniform changes with no clear distance-decay from the felling compartment (Figure 34). The transect analysis confirms the absence of a distance gradient (Figure 36). The ANCOVA (Section 4.6.3), which explicitly models the easting $\times$ time gradient, provides a cleaner estimate of the clearfell-specific effect.

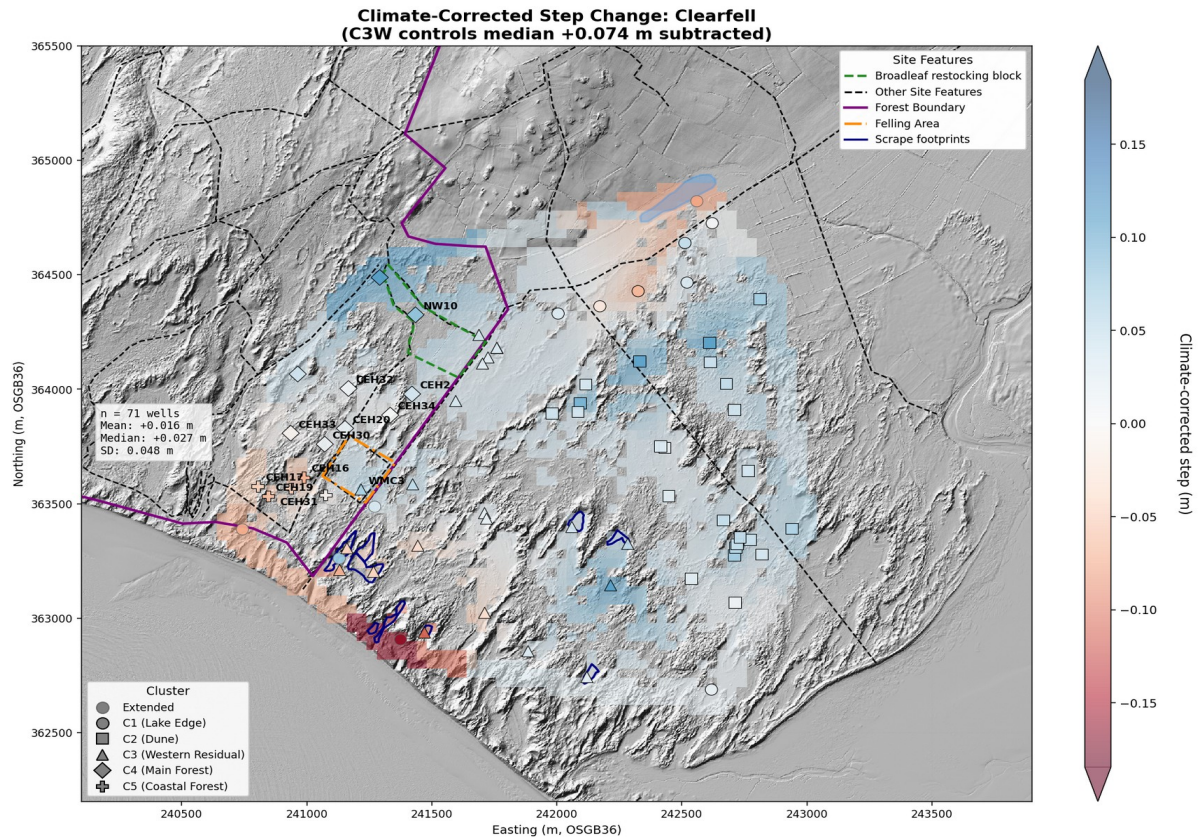


Figure 1.31: Climate-corrected spatial step-change map for the clearfell era (December 2017 – end of record). Each well's step is computed relative to its pre-felling mean, corrected by subtracting the C3 open-dune control centroid step to remove the background climate signal. Step changes are spatially uniform across the network with no gradient consistent with a localized felling response. The absence of a distance-decay from the felling compartment centroid is confirmed by the transect analysis (Figure 36). Source: 10b_spatial_fell_corrected.png.

1.6.6 Coefficient Decomposition

The SSM was fitted independently to Before (record-length-balanced pre-felling window) and After (felling to end of record) windows for each well in the 17-well network, yielding per-well coefficient shifts $\Delta\beta_1$, $\Delta\beta_2$ and $\Delta\beta_3$ (Script 10e). This decomposition is a mechanistic diagnostic: it characterizes *which* SSM pathway — recharge, atmospheric draw or drainage — shifted between eras and in *what direction*. It is not an estimate of the clearfell magnitude; the clearfell magnitude is the BACI step reported in Section 4.6.3. The per-era fits use the canonical no-intercept SSM (Section 3.5), consistent with Script 03 and the master coefficient table.

β_1 (recharge) declined at most wells across all five tiers: mean $\Delta\beta_1 = -0.10$ at Impact, -0.27 at Edge, -0.05 at Forest Control, -0.32 at Coastal Control and -0.13 at Climate Control. The network-mean shift across the seventeen-well BACI network is $\Delta\beta_1 = -0.16$, a 5.8% reduction against a pre-felling network mean of 2.81. The decline is present at the unfelled

Climate Control and Forest Control tiers as well as at the treatment tiers, indicating a site-wide phenomenon rather than a clearfell-specific response — its candidacy as a mechanism for the network-wide summer-minimum decline is taken up in Section 4.8 and Section 5.5.

β_2 (atmospheric draw) increased at the unfelled Forest-Control wells (mean $\Delta\beta_2 = +0.15$) with no comparable rise at the non-forest tiers — the direction and tier-specificity expected if progressive canopy maturation raises the canopy's share of the PET budget at mature unfelled pine. The shift is not a climate-sampling effect: refitting β_2 on a common PET support across the two eras leaves the Forest-Control increase positive while the Climate-Control tier remains negative, so a change in climate forcing — common to all tiers — cannot account for the forest-specific direction. The Impact-tier β_2 shift ($\Delta\beta_2 = -0.28$) is a single-well estimate lying within one standard error of zero and is not interpreted; Edge-tier shifts were near zero in the mean (-0.03) and incoherent in sign across wells.

β_3 (drainage) shifts were small and spatially incoherent across all tiers (mean $\Delta\beta_3$ between -0.01 and 0.00 at every tier), consistent with no systematic change in the drainage pathway.

Individual wells departed from these tier means. Within the Forest Control, NW10 and CEH2 showed β_1 increases ($\Delta\beta_1 = +0.28$ and $+0.12$) against a network of general decline. These departures are retained: the per-well era coefficients carry wide confidence intervals (Figure 35) and exclusion would not be justified on physical grounds.

The decomposition therefore identifies a site-wide recharge decline and a forest-maturation signal in atmospheric draw, but does not by itself isolate a clearfell-specific coefficient signature; the clearfell effect is quantified by the BACI analysis of Section 4.6.3.

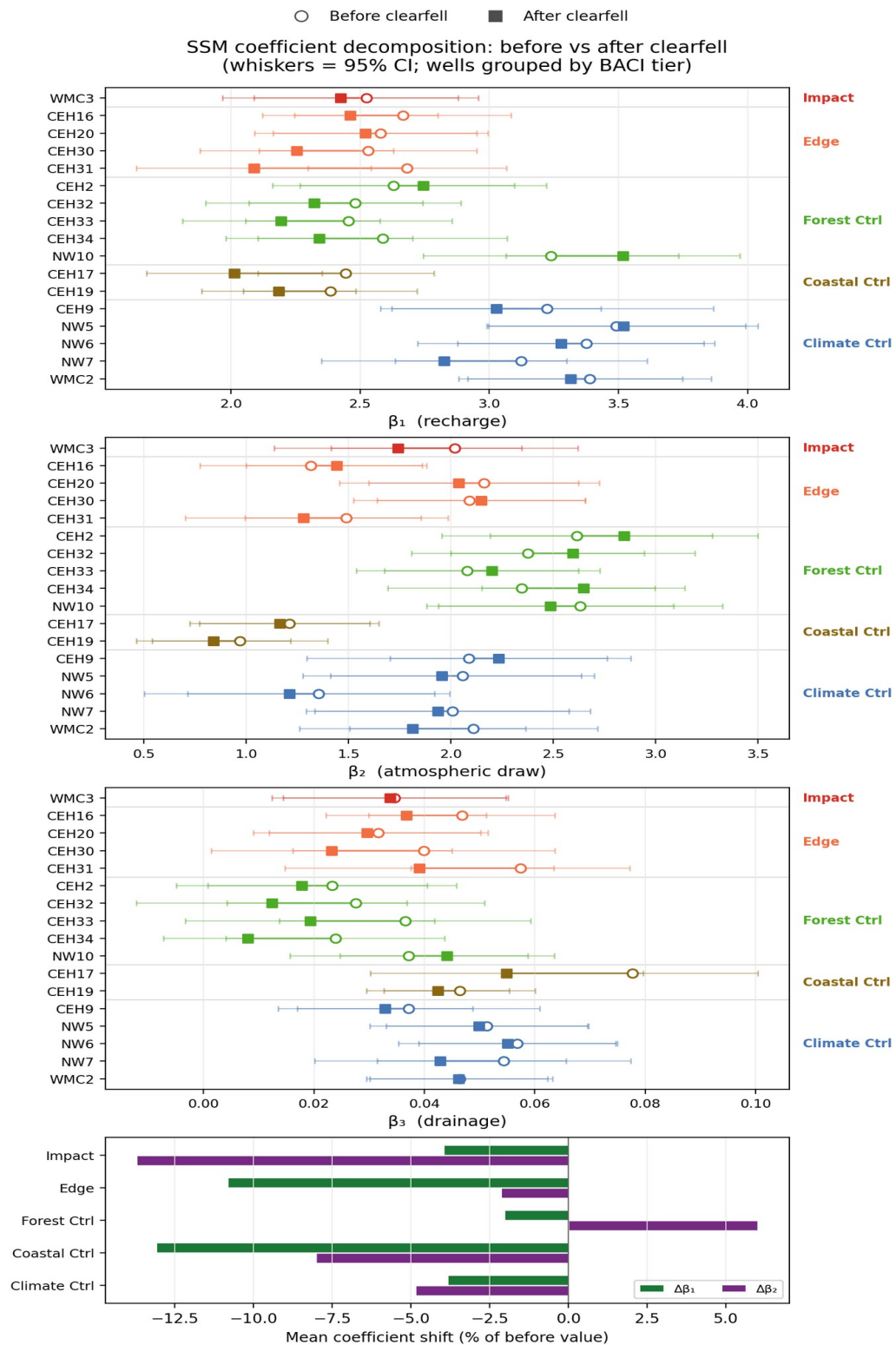


Figure 1.32: Before- and after-clearfell SSM coefficient estimates with 95% CI for all 17 wells in the BACI network, grouped by tier. (a) β_1 recharge: declined at most wells across all tiers — a site-wide pattern. (b) β_2 atmospheric draw: increased at Forest Control wells, consistent with canopy maturation. (c) β_3 drainage: small, spatially incoherent shifts. (d) Per-tier mean coefficient shift $\Delta\beta_1$, $\Delta\beta_2$, $\Delta\beta_3$. (Source: 10e_03_coefficient_shifts.png).

Well	Tier	β_1 before	β_1 after	$\Delta\beta_1$	β_2 before	β_2 after	$\Delta\beta_2$	β_3 before	β_3 after	$\Delta\beta_3$
WMC3	Impact	2.524	2.424	-0.100	2.017	1.741	-0.277	0.035	0.034	-0.001
CEH31	Edge	2.683	2.089	-0.594	1.490	1.278	-0.211	0.057	0.039	-0.018
CEH20	Edge	2.580	2.522	-0.058	2.161	2.039	-0.123	0.032	0.030	-0.002
CEH30	Edge	2.531	2.255	-0.276	2.089	2.148	+0.059	0.040	0.023	-0.017
CEH16	Edge	2.666	2.462	-0.204	1.315	1.440	+0.125	0.047	0.037	-0.010
CEH32	Forest Ctrl	2.480	2.322	-0.158	2.375	2.596	+0.221	0.028	0.012	-0.015
CEH34	Forest Ctrl	2.588	2.342	-0.245	2.344	2.646	+0.302	0.024	0.008	-0.016
CEH33	Forest Ctrl	2.456	2.195	-0.261	2.080	2.201	+0.121	0.037	0.019	-0.017
NW10	Forest Ctrl	3.239	3.517	+0.278	2.632	2.484	-0.148	0.037	0.044	+0.007
CEH2	Forest Ctrl	2.630	2.745	+0.115	2.616	2.846	+0.230	0.023	0.018	-0.006
CEH19	Coastal Ctrl	2.385	2.185	-0.200	0.969	0.840	-0.129	0.046	0.043	-0.004
CEH17	Coastal Ctrl	2.445	2.013	-0.432	1.210	1.164	-0.046	0.078	0.055	-0.023
CEH9	Climate Ctrl	3.224	3.028	-0.197	2.088	2.233	+0.145	0.037	0.033	-0.004
NW7	Climate Ctrl	3.124	2.825	-0.299	2.008	1.935	-0.073	0.054	0.043	-0.012
NW6	Climate Ctrl	3.376	3.277	-0.099	1.352	1.211	-0.142	0.057	0.055	-0.002
NW5	Climate Ctrl	3.491	3.525	+0.034	2.056	1.956	-0.100	0.051	0.051	0.000
WMC2	Climate Ctrl	3.388	3.315	-0.073	2.110	1.812	-0.298	0.046	0.046	0.000

Table 1.11: Before- and after-clearfell SSM coefficient estimates for all 17 wells in the BACI network, with per-well shifts $\Delta\beta$. Before = record-length-balanced pre-felling window; After = felling to end of record. Coefficients are from the canonical no-intercept SSM (Section 3.5). (Source: 10e_01_coefficient_shifts.csv)

1.6.7 Robustness

Three independent methods were applied to test whether a clearfell-specific signal could be detected once the ANCOVA assumptions were relaxed.

SSM residual normalization. The SSM was calibrated on the pre-felling record for each well and run forward iteratively into the post-felling era. The Impact well WMC3 showed a +56 mm normalized step — its post-felling water table 56 mm above the pre-felling SSM prediction. The Forest Control wells drifted upward by a larger mean of +82 mm over the same period, so against the Forest-control drift the Impact-specific component is approximately -26 mm: the Impact well's upward residual is slightly smaller than the unfelled forest controls', and this method does not isolate a positive clearfell-specific step.

Synthetic control. A synthetic counterfactual for WMC3 constructed from six optimally weighted donor wells showed an Impact step of +99 mm ($p = 0.001$); the corresponding Edge synthetic-control step was +41 mm ($p = 0.22$), non-significant. The synthetic-control and SSM-residual methods agree that WMC3's gross post-felling step is of order +50 to

+100 mm; they differ on how much survives normalization against the unfelled forest controls, and neither, after that normalization, resolves a clearfell-specific step distinguishable from zero.

Clearfell transect. Step changes across a transect from the plantation interior to the felling core showed no significant spatial gradient (-0.4 mm per 100 m, $p = 0.19$). A distance-decay response would be strongest at WMC3 and diminish outward; the observed pattern is statistically uniform, consistent with a common climate signal (Figure 36). The post-felling spread in the transect hydrographs is nonetheless visibly wider than pre-felling (Figure 36, panel a), with WMC3 consistently above the pack — a distance-dependent response that the seven-well transect regression lacks the power to resolve.

Synthetic FE-well extension. FE1 and FE2 — which sit inside or at the immediate edge of the clearfell compartment but lack pre-scraping baselines — were extended backwards using OLS donor regression calibrated on Forest Control wells (calibration $R^2 > 0.99$, RMSE 18–25 mm; Section 4.6.1). Against the Forest control, the centroid of WMC3 + FE2 produced a clearfell step of +103 mm (95% CI [+45, +162], $p < 0.001$, $R^2 = 0.34$), consistent with the WMC3-only ANCOVA estimate of +120 mm (Section 4.6.3). FE2's post-felling divergence from its own donor counterfactual was +28 mm ($p < 0.001$), indicating a significant within-compartment response. Extending the centroid to WMC3 + FE1 + FE2 returned a Forest-control clearfell step of +85 mm (95% CI [+39, +131], $p < 0.001$). FE1 sits approximately 20 m outside the felling boundary in standing forest, so including it attenuates the step magnitude — from +120 mm (WMC3 alone) through +103 mm (WMC3 + FE2) to +85 mm — but the within-compartment signal remains highly significant throughout.

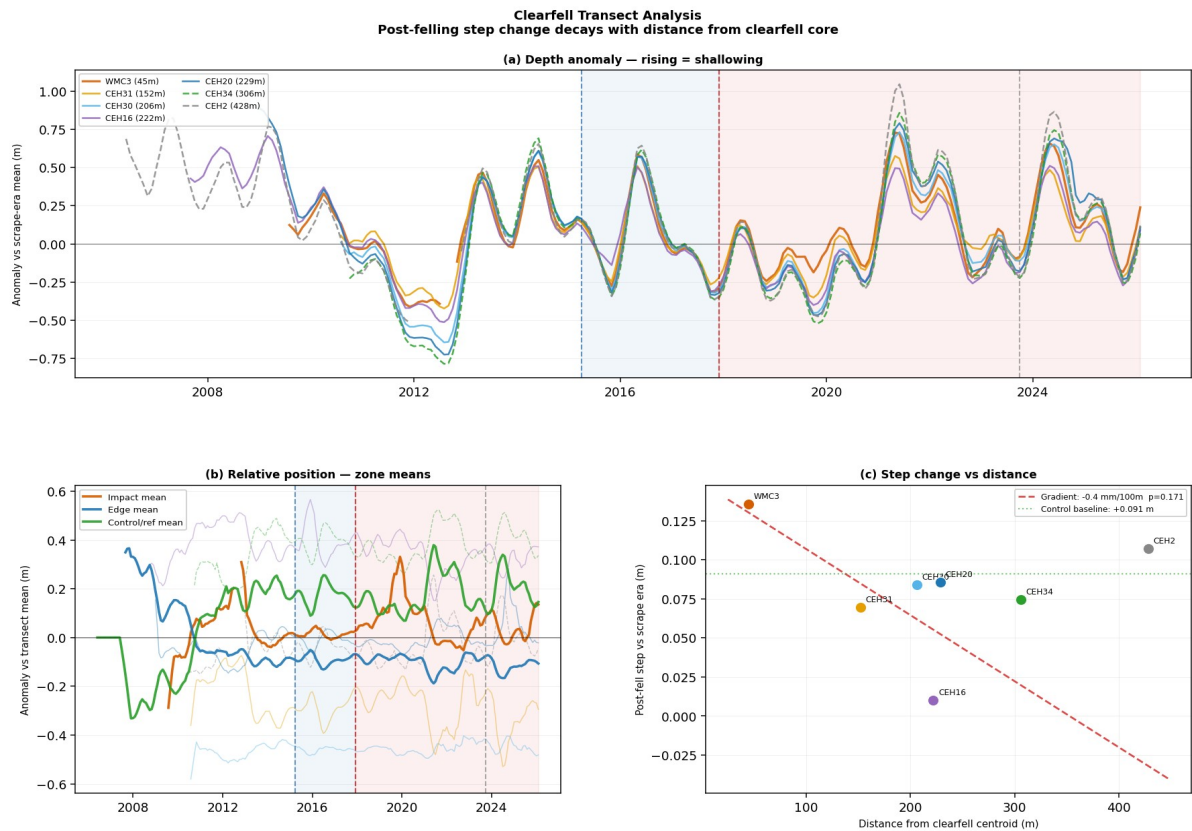


Figure 1.33: Clearfell transect: post-felling step change vs distance from felling compartment centroid. Gradient = $-0.4 \text{ mm}/100 \text{ m}$ ($p = 0.19$, ns). No spatial decay consistent with a localized clearfell response; the uniform pattern indicates a common climate signal. (Source: 10g_02_clearfell_transect.png)

1.7 Critical Rainfall Threshold Forecasting - Predicting Winter flooding and Summer droughts

1.7.1 Mechanistic State-Space Equations

The cluster-level state-space equations (Section 3.6.1) fitted without intercept produced statistically significant coefficients at all five clusters (all $p < 0.001$ except the C4 drainage coefficient β_3 , where $p = 0.002$), confirming that the physical mass-balance formulation is well-supported by the data. Model fit ranged from $R^2 = 0.813$ at Western Residual (C3) to $R^2 = 0.685$ at Coastal Forest (C5), with C5 the least well fit and C4 Main Forest mid-range ($R^2 = 0.725$) — consistent with the greater storage buffering of the forest clusters, whose higher LCSC values (Table 3) attenuate the direct rainfall–water-table coupling the SSM captures. The full coefficient values are reported in Section 4.2.2; note that the β coefficients used in the forecasting tool are expressed in m/mm rainfall, one thousand times smaller than the m/m values in Table 3.

1.7.2 Seasonal Prediction Equations

The prediction equations were fitted at hydrological-year resolution, one equation per cluster under the $k = 5$ partition (Table 12). The two predictor terms — winter rainfall total (P_{winter}) and antecedent summer minimum (h_{min}) — show a clear spatial partitioning of dominance across the site.

In the Eastern Block, winter rainfall is the dominant predictor. P_{winter} is significant at Lake Edge ($p = 0.002$) and Dune ($p = 0.004$), while h_{min} is non-significant in both ($p = 0.545$ and 0.475 respectively). This is consistent with the faster-draining, lower-memory character of the eastern aquifer, where same-season rainfall dominates the winter flooding response.

In the Western Residual and forest clusters, the pattern reverses. The antecedent summer minimum is the dominant predictor: Western Residual ($p = 0.017$, $R^2 = 0.638$), Forest ($p < 0.001$, $R^2 = 0.865$) and Coastal Forest ($p < 0.001$, $R^2 = 0.882$). Winter rainfall is non-significant at all three ($p = 0.078$, 0.919 and 0.379 respectively). The depth of the preceding summer drought is a stronger predictor of winter flooding potential than the actual winter rainfall amount in these more buffered domains. This reflects the aquifer's multi-month memory — a deep summer minimum requires progressively more winter rainfall simply to restore the water table to its pre-drought position before any flooding can occur. The significant intercepts at Forest ($p = 0.035$) and Coastal Forest ($p = 0.003$) are consistent with the strong structural offset imposed by the plantation canopy on water table depth.

Two individual coefficients require comment. The negative h_{min} coefficient at Lake Edge (-0.134) is not mechanistically meaningful given its non-significance; it is an artefact of the small sample ($n = 19$) and the dominance of P_{winter} in that cluster. The negative P_{winter} coefficient in the Forest cluster (-0.00006) reflects multicollinearity with h_{min} and should not be interpreted mechanistically.

Block	Equation	R^2	n	$p(P_{\text{winter}})$	$p(h_{\text{min}})$
Lake Edge (C1)	$h_{\text{peak}} = 0.00108 \cdot P_{\text{winter}} - 0.134 \cdot h_{\text{min}} - 0.687$	0.47	19	0.002	0.545
Dune (C2)	$h_{\text{peak}} = 0.00160 \cdot P_{\text{winter}} + 0.210 \cdot h_{\text{min}} - 0.786$	0.53	20	0.004	0.475
Western Residual (C3)	$h_{\text{peak}} = 0.00119 \cdot P_{\text{winter}} + 0.827 \cdot h_{\text{min}} - 0.053$	0.64	20	0.078	0.017
Main Forest (C4)	$h_{\text{peak}} = -0.00006 \cdot P_{\text{winter}} + 1.270 \cdot h_{\text{min}} + 1.259$	0.86	19	0.919	<0.001
Coastal Forest (C5)	$h_{\text{peak}} = 0.00040 \cdot P_{\text{winter}} + 1.631 \cdot h_{\text{min}} + 1.415$	0.88	19	0.379	<0.001

Table 1.12: Winter peak prediction equations fitted at hydrological-year resolution (Section 3.6.2), one equation per cluster under the $k = 5$ partition. P_{winter} = total winter rainfall (mm) accumulated over the cluster-specific recharge horizon (October–January for C1 and C2; October–February for C3, C4 and C5). h_{min} = preceding summer water table minimum (m below surface). (Source: 11_forecast_winter_transfer_functions.csv)

1.7.3 Summer Drought Prediction Equations

The summer drought prediction equations performed substantially better than the winter peak functions (Section 3.6.2). Both predictors — antecedent winter peak ($h_{\text{max_winter}}$) and cumulative summer precipitation (P_{summer}) — were significant in all five blocks (all $p < 0.05$). Model fit was highest for the Forest ($R^2 = 0.916$) and Coastal Forest ($R^2 = 0.905$) blocks, high for the Western Residual ($R^2 = 0.707$), moderate-to-high for the Dune ($R^2 = 0.654$), and moderate for the Lake Edge cluster ($R^2 = 0.406$). These functions provide a reliable basis for forecasting summer drought severity from the preceding winter state and are the more operationally useful of the two prediction equation pairs (Table 13).

Block	Equation	R^2	$p(P_{\text{summer}})$	$p(h_{\text{max}})$
Lake Edge (C1)	$h_{\text{min}} = 0.00139 \cdot P_{\text{summer}} + 0.419 \cdot h_{\text{max_winter}} - 1.430$	0.406	0.005	0.047
Dune (C2)	$h_{\text{min}} = 0.00184 \cdot P_{\text{summer}} + 0.540 \cdot h_{\text{max_winter}} - 1.653$	0.654	<0.001	<0.001
Western Residual (C3)	$h_{\text{min}} = 0.00121 \cdot P_{\text{summer}} + 0.547 \cdot h_{\text{max_winter}} - 1.445$	0.707	0.010	<0.001
Forest (C4)	$h_{\text{min}} = 0.00122 \cdot P_{\text{summer}} + 0.747 \cdot h_{\text{max_winter}} - 1.470$	0.916	0.007	<0.001
Coastal Forest (C5)	$h_{\text{min}} = 0.00067 \cdot P_{\text{summer}} + 0.522 \cdot h_{\text{max_winter}} - 1.331$	0.905	0.043	<0.001

Table 1.13: Summer drought prediction equations (Section 3.6.2). P_{summer} = total April–September rainfall (mm); $h_{\text{max_winter}}$ = preceding winter water table peak (m below surface). All coefficients significant at $p < 0.05$. (Source: 11_forecast_summer_transfer_functions.csv).

1.7.4 Spatial Distribution of P_{flood} and Flooding Status Across the Well Network

Application of the cluster-level P_{flood} equations derived in Section 3.6.3 to each well in the 88 well classified network (66 reference wells with per-well SSM coefficients; 22 extended wells assigned cluster-level coefficients via the Pearson affinity audit of Section 3.3), using that well's observed mean summer minimum as h_0 , produces a per-well critical rainfall threshold expressed as either millimetres of cumulative winter rainfall or as a dimensionless multiplier $\lambda = P_{\text{flood}} / \Sigma P_{\text{clim}}$, where ΣP_{clim} is the long-term mean rainfall over the cluster-specific recharge horizon (401 mm for C1/C2 over Oct–Jan; 464 mm for C3/C4/C5 over Oct–Feb). A well with $\lambda = 1.0$ would flood under an average winter; $\lambda > 1.0$ indicates that above-average rainfall is required. The network-wide picture is summarized in Figures 36–39.

The summer minimum depth map (Figure 37; Section 3.6.5) classifies each well against the five eco-hydrological bands derived from Curreli et al. (2013) and introduced in Section 2: SD15b wet slack viability (<0.61 m), SD15b-recoverable (0.61–0.75 m), SD16 dry slack (0.75–0.98 m), SD16-recoverable (0.98–1.20 m), and beyond-recoverable (>1.20 m). The two forest clusters (C4 and C5) show the deepest mean summer minima across the network, consistently below 0.98 m. Most C4 and C5 wells fall within the SD16 or beyond-recoverable zone, reflecting the suppression of net recharge by canopy interception. By

contrast, the majority of C1 Lake Edge wells maintain mean summer minima within or close to the SD15b wet slack zone, reflecting the high-frequency recharge dynamics of the shallow till substrate. Of the 88 wells in the classified network, 41 have mean summer minima deeper than 0.98 m — the SD16 dry slack threshold — of which 24 are below the SD16-recovery threshold of 1.20 m.

The P_flood map (Figure 38) quantifies the rainfall demand implied by these summer minima and reveals a progressive, cluster-ordered closure of the operational window (Table 14). Almost every well in the classified network requires winter rainfall above its climatological mean to reach the slack floor: 87 of the 88 wells have $\lambda > 1.0$, the single exception ($\lambda = 0.98$) flooding marginally below an average winter. Under mildly wet winters ($1.0 \leq \lambda < 1.5$), 59 wells (67%) become reachable; a further 24 (27%) require wet winters of varying rarity ($1.5 \leq \lambda < 2.5$). The remaining 4 wells (5%) have $\lambda \geq 2.5$ — they require at least 2.5 times the climatological mean winter rainfall, a total well beyond the wettest winter in the ~95-year RAF Valley record. For these wells the slack floor is not reachable under any winter rainfall yet observed: the wettest winter on record (745 mm in 2001, $1.51\times$ the climatological mean) delivers only 60% of the $\lambda = 2.5$ demand.

Cluster	n	P_flood range (mm)	Median (mm)	λ
C1 Lake Edge	8	418–485	431	1.08
C2 Dune	32	420–734	469	1.17
C3 Western Residual	25	545–940	589	1.27
C4 Main Forest	13	692–1,200	818	1.76
C5 Coastal Forest	10	711–1,536	1,044	2.25

Table 1.14: Per-cluster P_flood summary across the 88 well classified network. $\lambda = P_{\text{flood}} / \Sigma P_{\text{clim}}$; values above 1.0 indicate rainfall above the climatological mean is required. (Source: 11b_03_pflood_per_well.csv).

The winter maximum depth map (Figure 39) shows that Eastern Block wells achieve the shallowest winter maxima, consistent with the flashy, fill-and-spill regime of the shallow till aquifer. Western Residual and Forest wells rarely approach the 0 m threshold required by SD15b communities. The flooding frequency map (Figure 40) gives the most ecologically direct summary: C1 wells show the highest flooding frequency across the monitoring period; C3 wells show very low frequency; C4 and C5 Forest wells show essentially zero, with the water table failing to reach the ground surface in any monitored year.

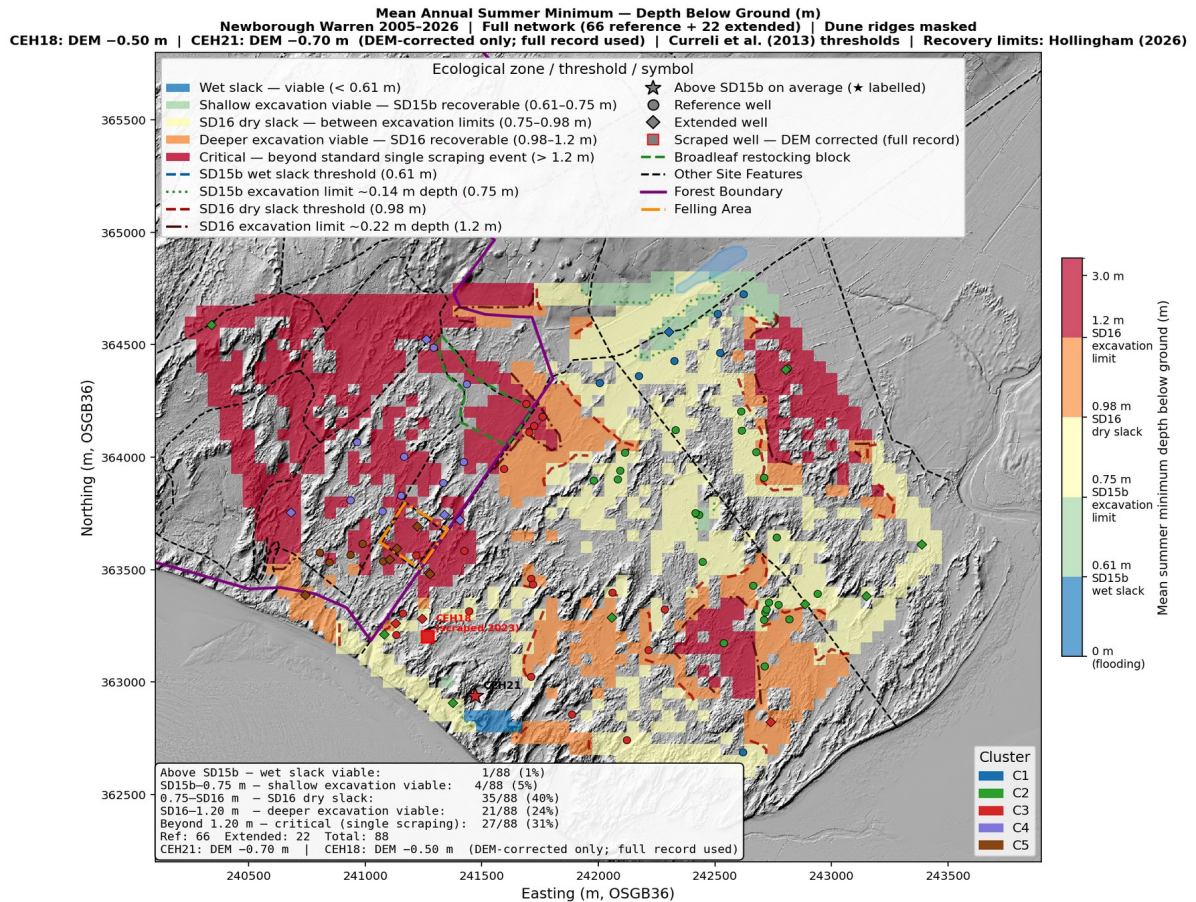


Figure 1.34: Spatial distribution of mean annual summer minimum water table depth below ground across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026. Depth is expressed as metres below the LiDAR ground surface (DEM_Ground_Elev – mean maOD), with positive values indicating the water table lies below the surface. The continuous IDW-interpolated surface is masked across dune ridges (cells where the DEM raster exceeds the interpolated well surface by more than 1 m). Ecological zone boundaries follow the eco-hydrological thresholds of Curreli et al. (2013): SD15b wet slack viability (0.61 m); SD15b excavation limit following one scraping event (0.75 m); SD16 dry slack threshold (0.98 m); SD16 excavation limit (1.20 m). Recovery limits are indicative, derived from the observed BACI scraping benefit of +0.129 m at CEH36 (Section 4.5); deeper scraping or repeat interventions may achieve greater uplift. Wells CEH18 and CEH21 use post-October 2023 data only, with DEM corrections of –0.50 m and –0.70 m respectively to account for material removed during scraping. Well symbols are coloured by cluster; circles indicate reference wells, diamonds indicate extended network wells. (LiDAR data: © NRW & OS; see Section 10 for full attribution.) A ridge-masked variant of this surface is available in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026b). (Source: 11b_01_summer_minima_depth.png).

The Eastern Block (C1 and C2) therefore remains within the operational domain of current scraping practice. C3 Western Residual is approaching the margin of operational recoverability. The two forest clusters (C4 and C5), as a whole, cannot be returned to ecological compliance by the existing excavation strategy alone under the present hydrological regime.

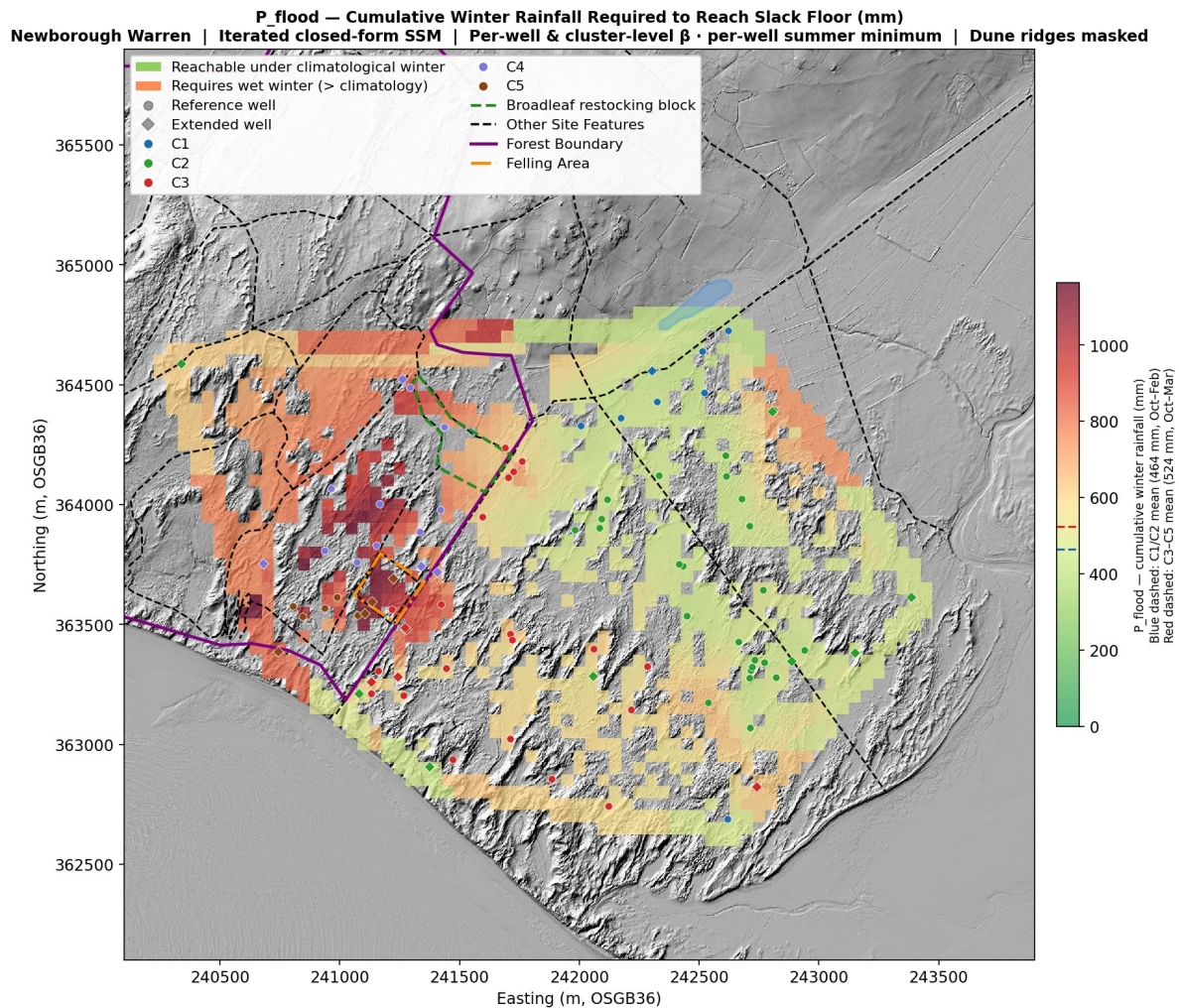


Figure 1.35: Spatial distribution of P_{flood} — the minimum cumulative winter rainfall (mm) required to raise the water table from its observed mean summer minimum to the local slack floor ($h_{\text{target}} = 0$ m) — across the 88 well classified network, Newborough Warren. P_{flood} is computed at each well using per-well SSM coefficients where available (66 reference wells) or cluster-level centroid coefficients assigned via Pearson affinity (22 extended wells), applying the iterated closed-form threshold equation of Section 3.6.3 to that well's observed mean summer minimum. The IDW-interpolated surface is masked across dune ridges. Dashed reference lines on the colourbar indicate the cluster-specific climatological rainfall totals: 401 mm for C1/C2 (Oct–Jan, blue) and 464 mm for C3–C5 (Oct–Feb, red). Wells whose P_{flood} falls below their cluster's reference line are reachable under average winter rainfall. (LiDAR data: © NRW & OS; see Section 10 for full attribution.) (Source: 11b_03_pflood.png).

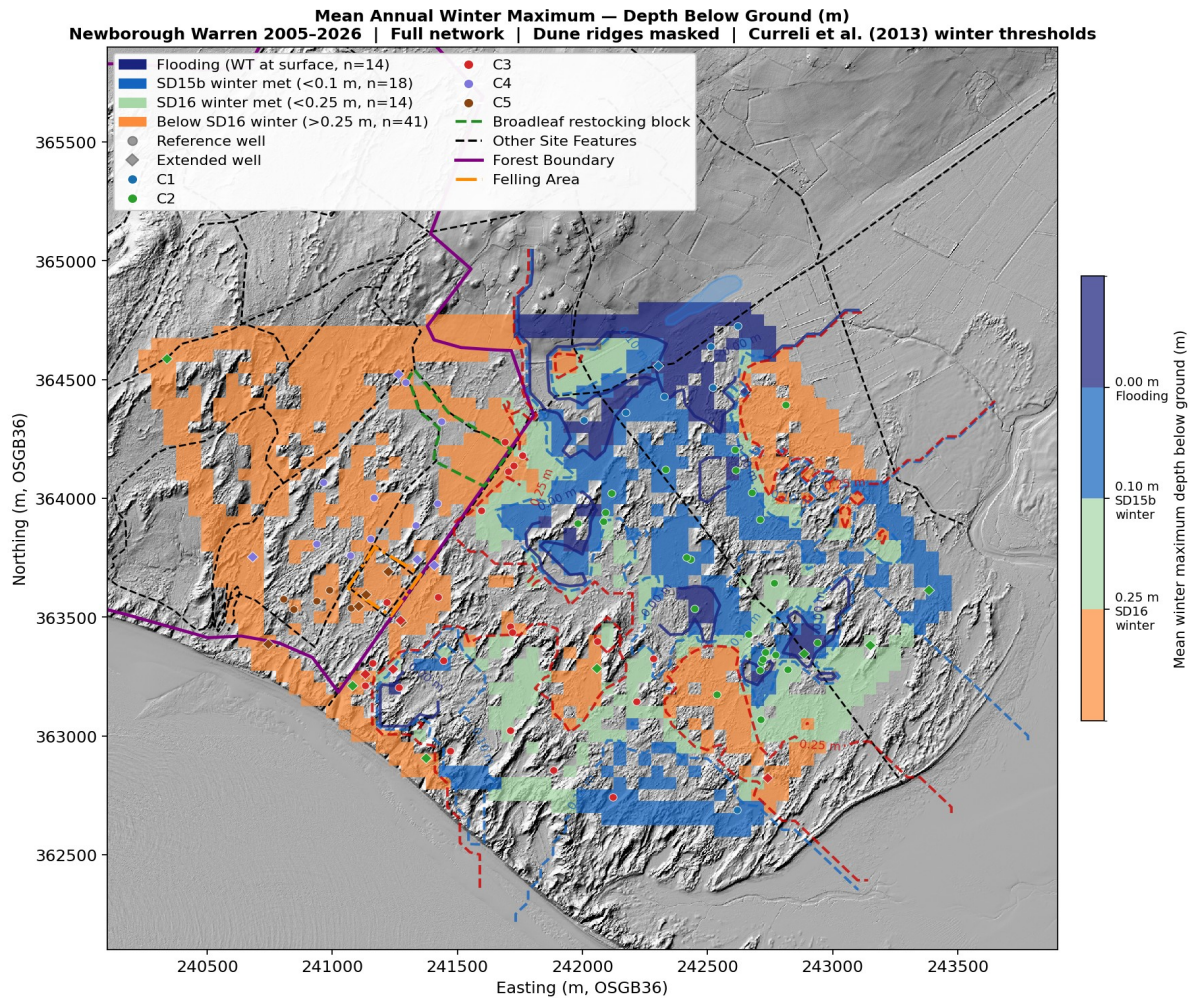


Figure 1.36: Spatial distribution of mean annual winter maximum water table depth below ground across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026. Winter maximum is defined as the highest monthly maOD value in October–March of each hydrological year (October start); the mean is taken across all years with at least three October–March readings. Depth is expressed using the same DEM-referenced convention as Figure 37. The IDW-interpolated surface is masked across dune ridges. Contours mark the eco-hydrological thresholds of Curreli et al. (2013) for winter water table requirements: 0 m (ground surface — required for SD15b wet slack communities); 0.10 m below ground (minimum winter maximum for SD15b viability); 0.25 m below ground (SD16 dry slack winter threshold). The spatial contrast between the C1 Lake Edge — where winter maxima frequently reach or approach the surface — and the C3 Western Residual, C4 Main Forest and C5 Coastal Forest, where winter maxima rarely approach 0 m, reflects the contrasting aquifer storage architectures identified in Section 4.2. (LiDAR data: © NRW & OS; see Section 10 for full attribution.) (Source: 11b_02_winter_maxima_depth.png).

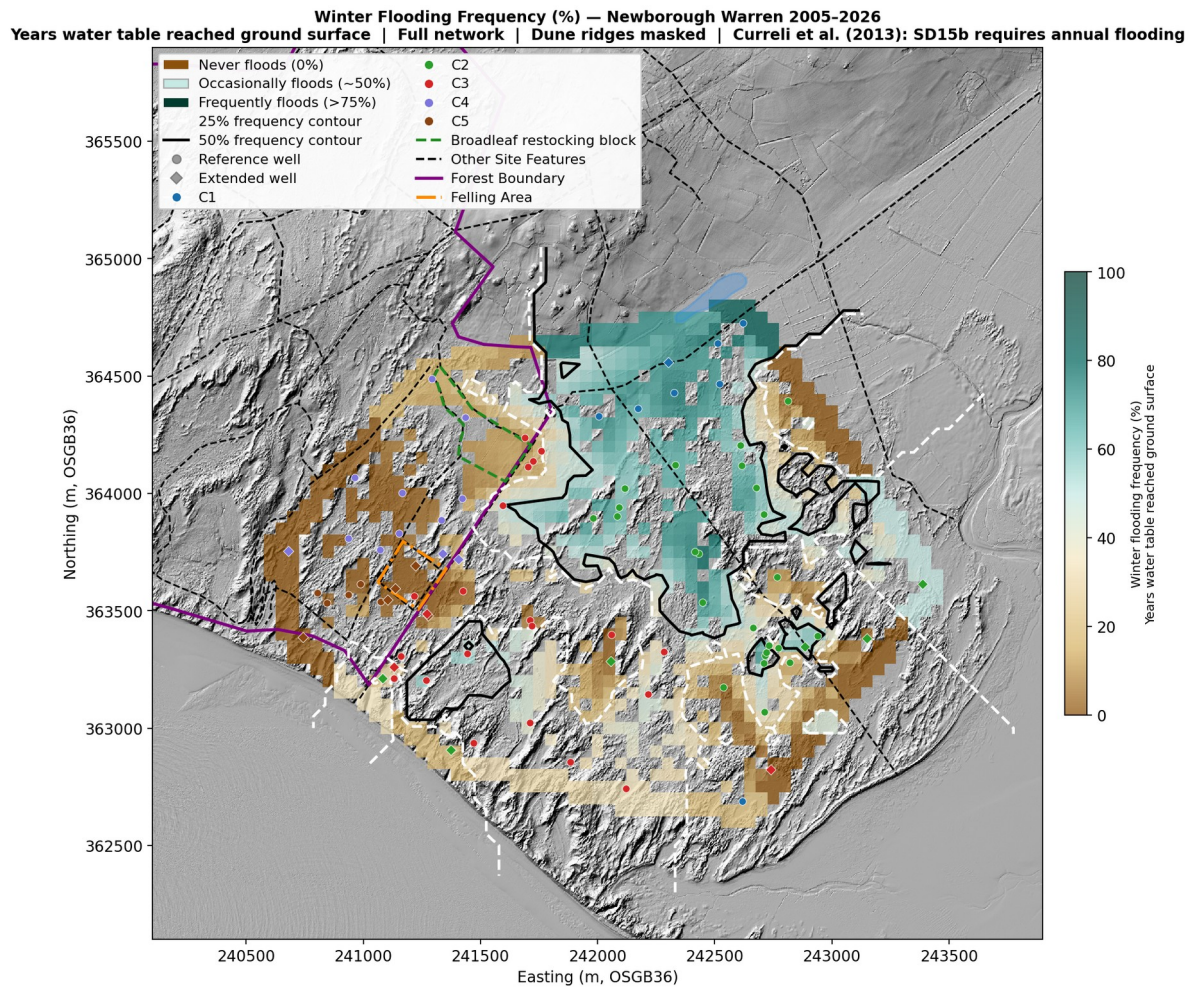


Figure 1.37: Winter flooding frequency across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026, expressed as the percentage of monitored hydrological years in which the winter maximum water table reached or exceeded the local ground surface (DEM ground elevation used as ground surface proxy at each dipwell location). Only wells with at least five years of winter data are included. The IDW-interpolated surface is masked across dune ridges. Contours mark the 25% and 50% frequency thresholds. The C1 Lake Edge shows the highest flooding frequency, consistent with the flashy, fill-and-spill dynamics of the shallow till substrate. C3 Western Residual wells show very low flooding frequency, and C4 and C5 Forest wells record essentially zero — the water table fails to reach the ground surface in any monitored year. This map provides the most ecologically direct assessment of habitat viability: the SD15b wet slack community identified by Curreli et al. (2013) requires the winter water table to rise within 0.10 m of the ground surface, a condition approached regularly only by the C1 Lake Edge under current conditions. (LiDAR data: © NRW & OS; see Section 10 for full attribution.) (Source: 11b_04_flood_frequency.png).

1.7.5 Critical Rainfall Threshold Equations (P_{flood})

The structural difference between clusters is immediately apparent from the threshold equations (Table 15; Section 3.6.3). The coefficients from Table 3, together with the cluster-

specific recharge horizons from October to each cluster's historical peak month (four months for C1 and C2, five months for C3, C4 and C5) and the long-term monthly RAF Valley rainfall and PET climatology, propagate through the iterated closed form of Section 3.6.3 to produce a linear relationship between observed summer-minimum depth and required winter rainfall. The C4 recharge horizon endpoint is marginal — the cluster's historical mean peak falls between February and March depending on the treatment of partial monitoring years — and the shorter February horizon is adopted as the conservative choice; the C1, C2, C3 and C5 horizons are unambiguous.

The iterated derivation reduces to the form $P_{\text{flood}} = A \cdot d + B$ for each cluster, where d is the observed summer-minimum depth below ground (m, positive) and P_{flood} is the cumulative winter rainfall required over the cluster's recharge horizon (mm). The slope A represents the marginal rainfall demand per additional metre of summer drawdown, and the intercept B represents the rainfall demand required even for a well already at the surface — non-zero because accumulated winter PET continues to draw down the water table through the recharge season. Both terms express the cluster gradient directly: A rises from 173 mm per metre at C1 through 213 (C2) and 237 (C3) to 364 (C5) and 375 (C4), and B ranges from 287 mm (C4) to 424 mm (C5).

As a worked example, at a typical C3 well with a mean summer minimum of 1.0 m below ground, $P_{\text{flood}} = 236.56 \times 1.0 + 375.26 = 611.8$ mm of cumulative winter rainfall over the October–February recharge horizon, corresponding to a rainfall multiplier $\lambda = 612/464 = 1.32$ times the long-term climatological total. The same depth at a C2 well produces $P_{\text{flood}} = 213.04 \times 1.0 + 296.15 = 509.2$ mm ($\lambda = 509/401 = 1.27$), a 103 mm difference reflecting the stronger antecedent-depth dependence and longer recharge horizon of C3. These differences translate directly into operational recoverability status: the C2 well lies within a marginally-wet-winter-reachable range, while the C3 well requires slightly wetter conditions over a longer accumulation of months.

Cluster	Label	Horizon	P_flood equation (mm)	ΣP_{clim} (mm)
C1	Lake Edge	Oct–Jan (4 mo)	$173.48 \cdot d + 315.97$	401
C2	Dune	Oct–Jan (4 mo)	$213.04 \cdot d + 296.15$	401
C3	Western Residual	Oct–Feb (5 mo)	$236.56 \cdot d + 375.26$	464
C4	Main Forest	Oct–Feb (5 mo)	$374.97 \cdot d + 287.43$	464
C5	Coastal Forest	Oct–Feb (5 mo)	$364.29 \cdot d + 424.36$	464

Table 1.15: Cluster-specific linear forms derived from the iterated closed-form P_{flood} solution of Section 3.6.3, reduced by substituting fitted SSM coefficients ($\beta_1, \beta_2, \beta_3$), cluster recharge horizons ($n = 4$ for C1/C2, 5 for C3/C4/C5), and RAF Valley long-term monthly rainfall and PET climatology (2005–2026). Input variable d is the observed mean summer minimum water table depth below ground in metres, entered as a positive number (e.g. $d = 1.26$ for a water table 1.26 m below the surface). Output P_{flood} is the cumulative rainfall (mm) required over the cluster's recharge horizon (October to the historical mean peak month) to raise the water table to the slack floor ($h_{target} = 0$ m), assuming monthly rainfall scales uniformly by a multiplier λ and monthly PET follows climatological means. The rainfall multiplier is $\lambda = P_{flood} / \Sigma \bar{P}_i$; values above 1.0 indicate rainfall above the climatological mean is required. (Source: 11_forecast_pflood_summary.csv).